MACHINE LEARNING METHODS

Data

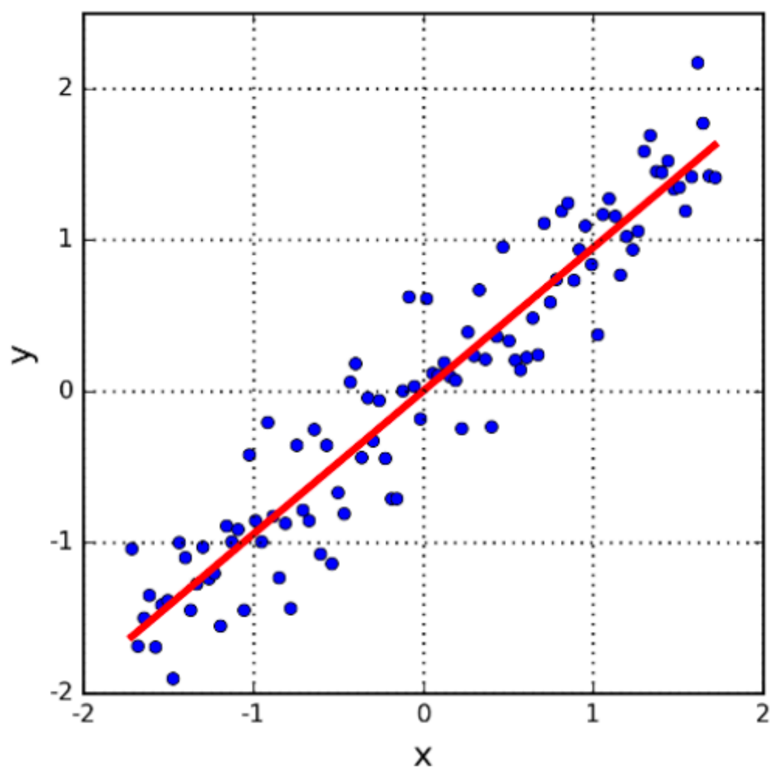
	“interdisciplinary”	“Statistics”	…
“Data science”	5	10	
…			

Labeled

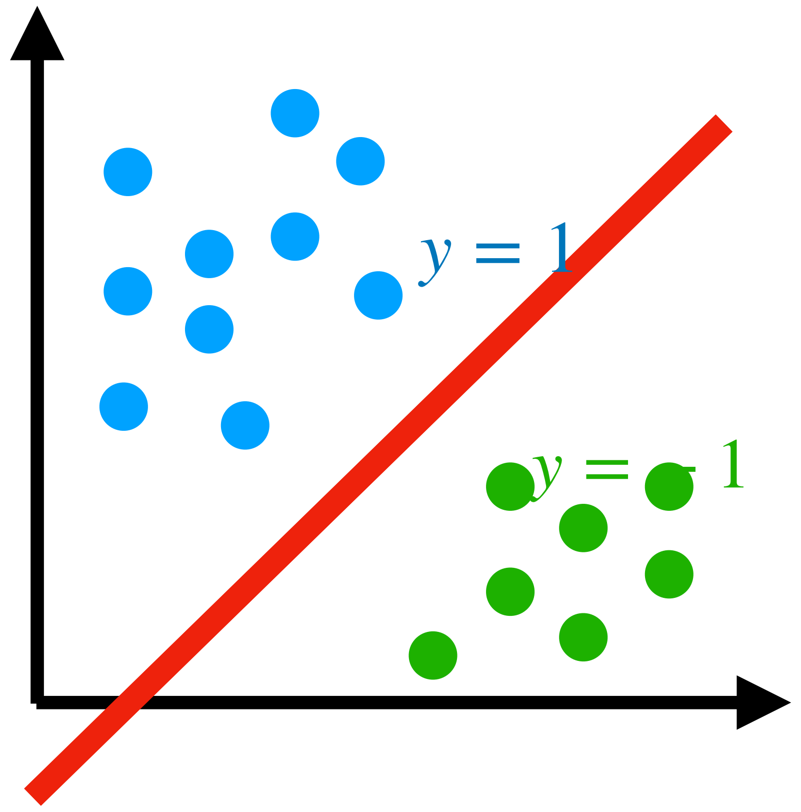
Supervised Learning

Quantitative

Categorical



Regression



Classification

MACHINE LEARNING METHODS

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Unlabeled

Supervised Learning

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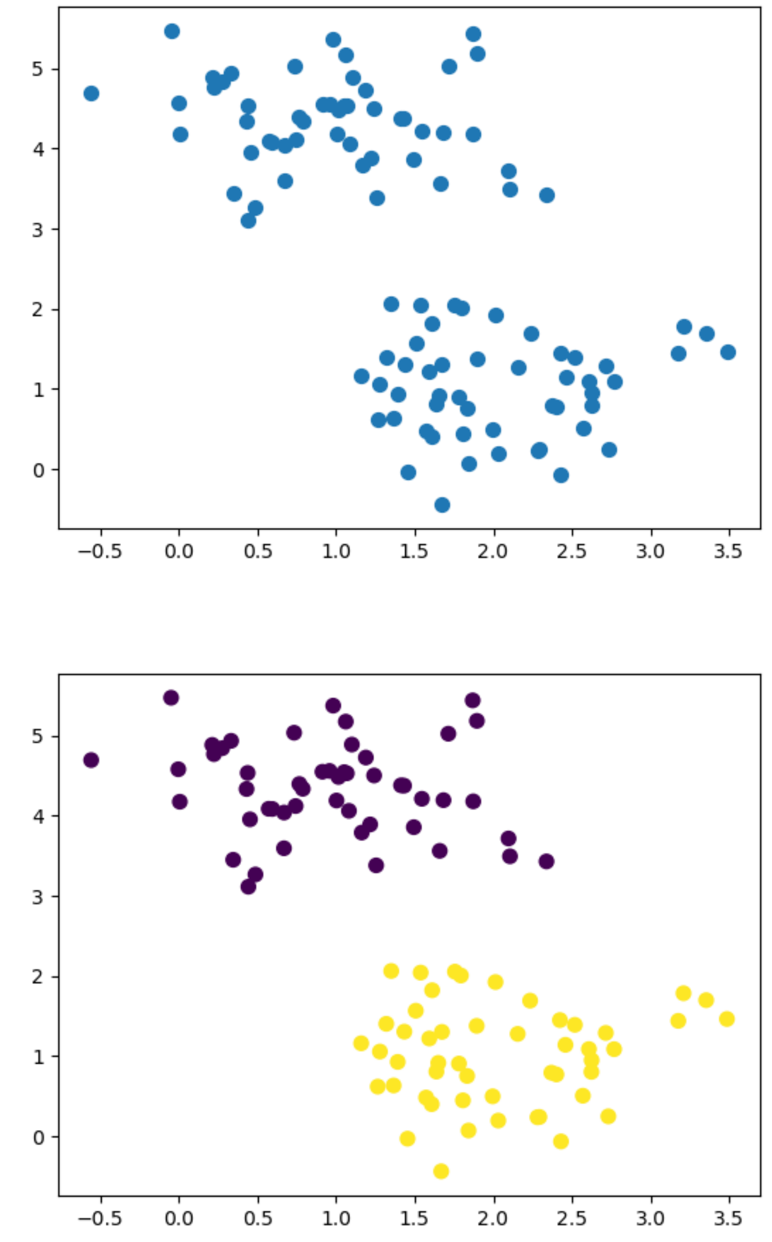
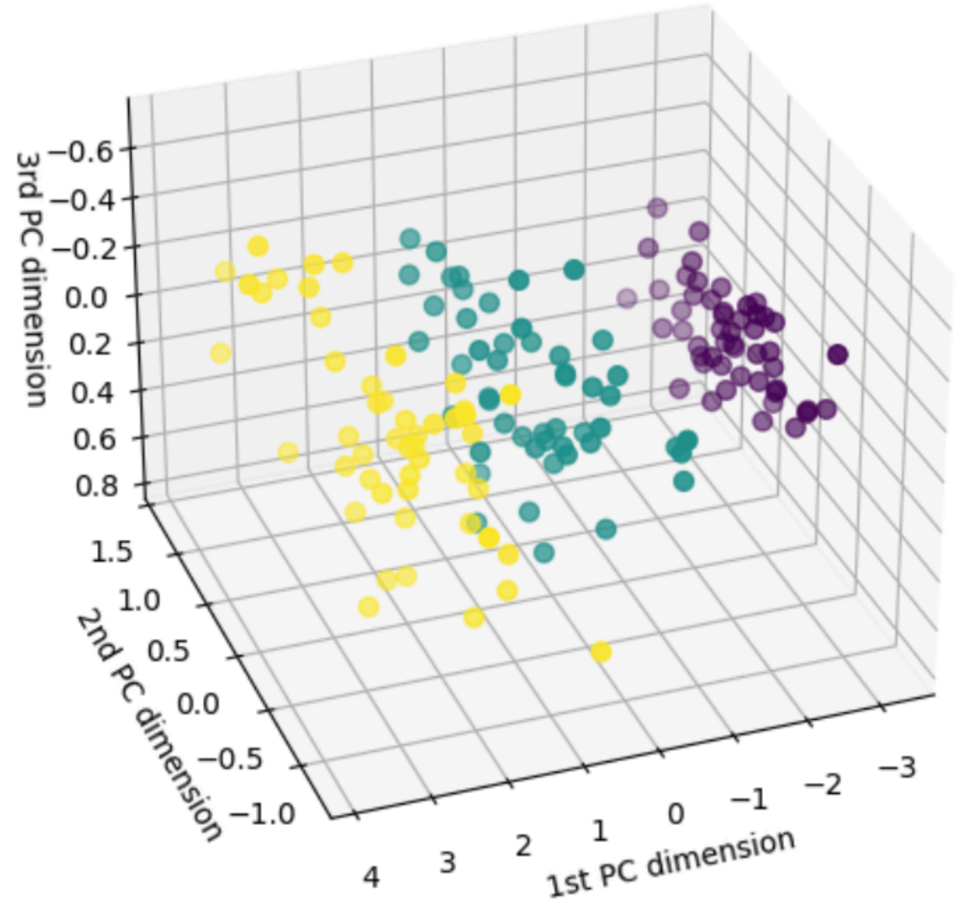


Clustering



Dimension reduction

MACHINE LEARNING METHODS

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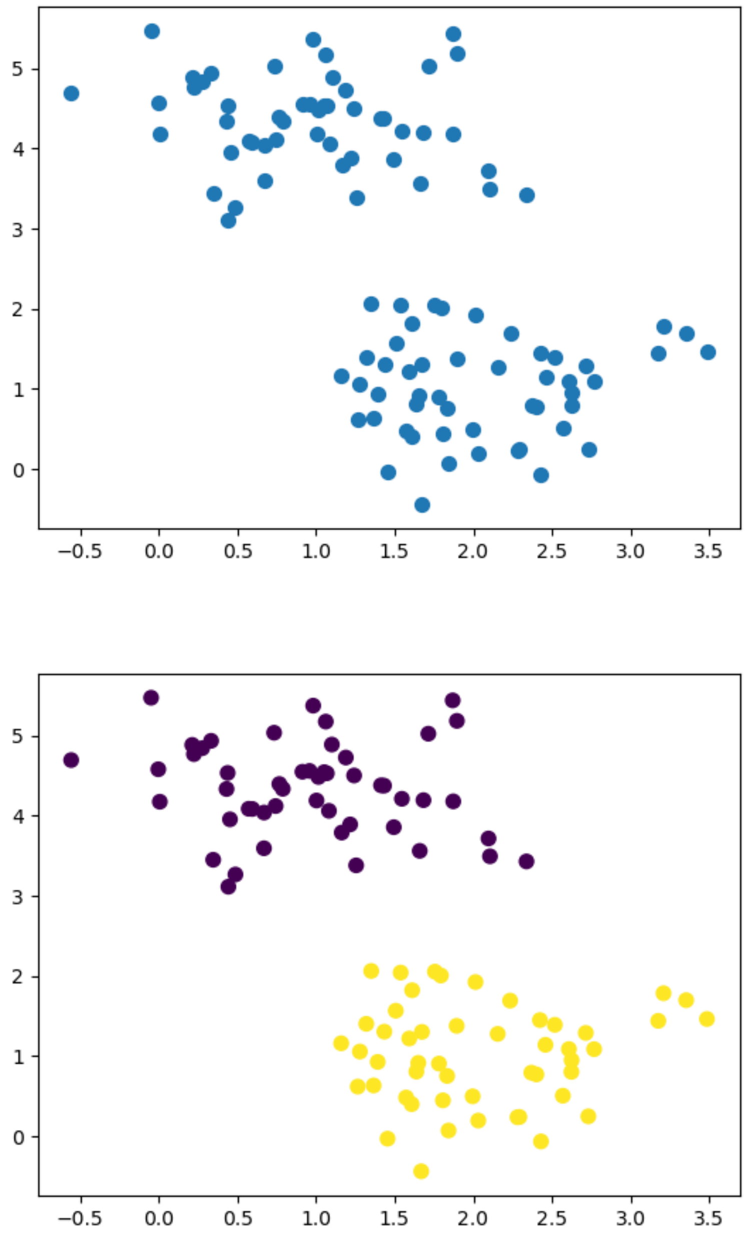


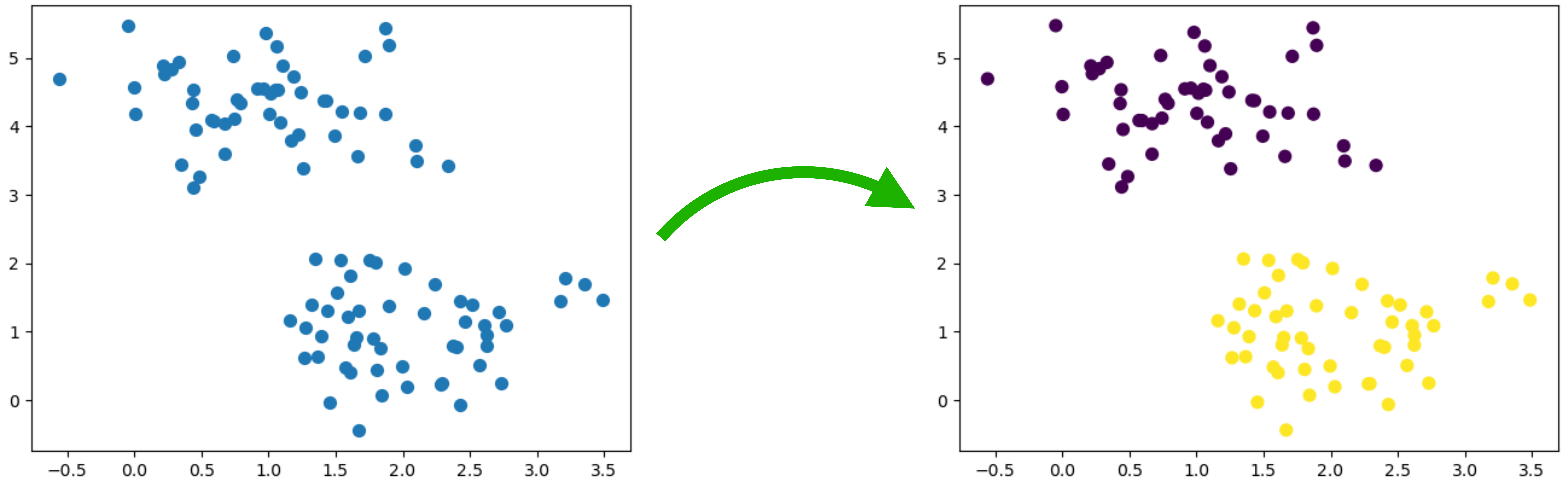
Clustering

Dimension reduction

CLUSTERING

Given a set of data points, construct groups of objects such that:

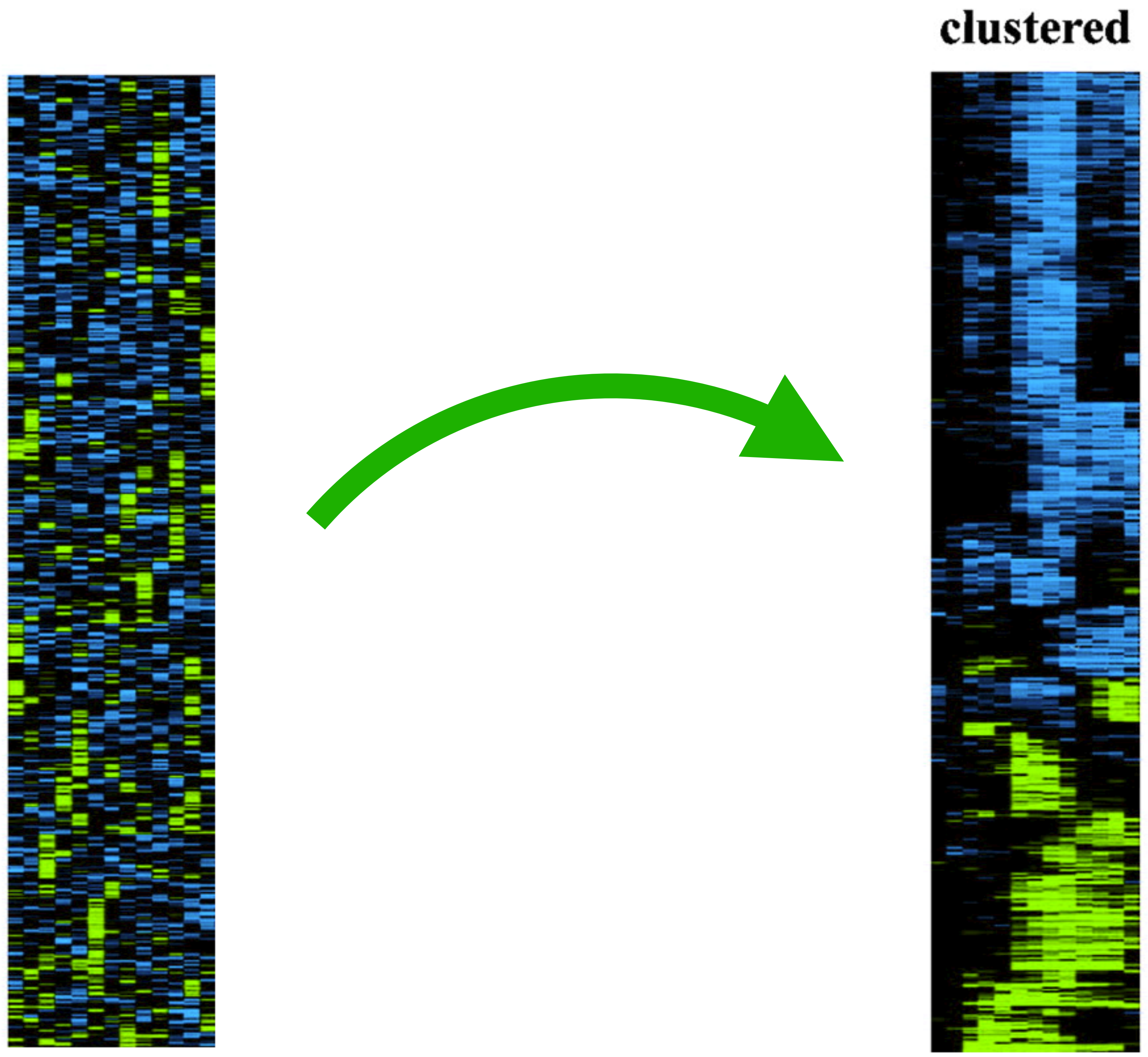
- Objects in the same group are similar
- Objects in the different group are dissimilar



CLUSTERING

Some Motivating Application Areas:

- Biology/Genomics: Find groups of genes with similar functionality



- Information Retrieval: Web searches, movie searches etc.
- Disease studies: Detect patterns in the spatial or temporal distribution of a disease

CLUSTERING

- Can be used for hypothesis development in EDA

What do points in the same cluster have in common?

CLUSTERING

- Can be used for hypothesis development in EDA

  What do points in the same cluster have in common?

- Modeling over smaller subsets of data

  Dataset with large number of observations:

  Train a model for each cluster separately.

  Forecast for a new point:

  Find its cluster, use the model trained for that cluster.

CLUSTERING

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  Train a model for each cluster separately.

  Forecast for a new point:

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- Outlier detection:

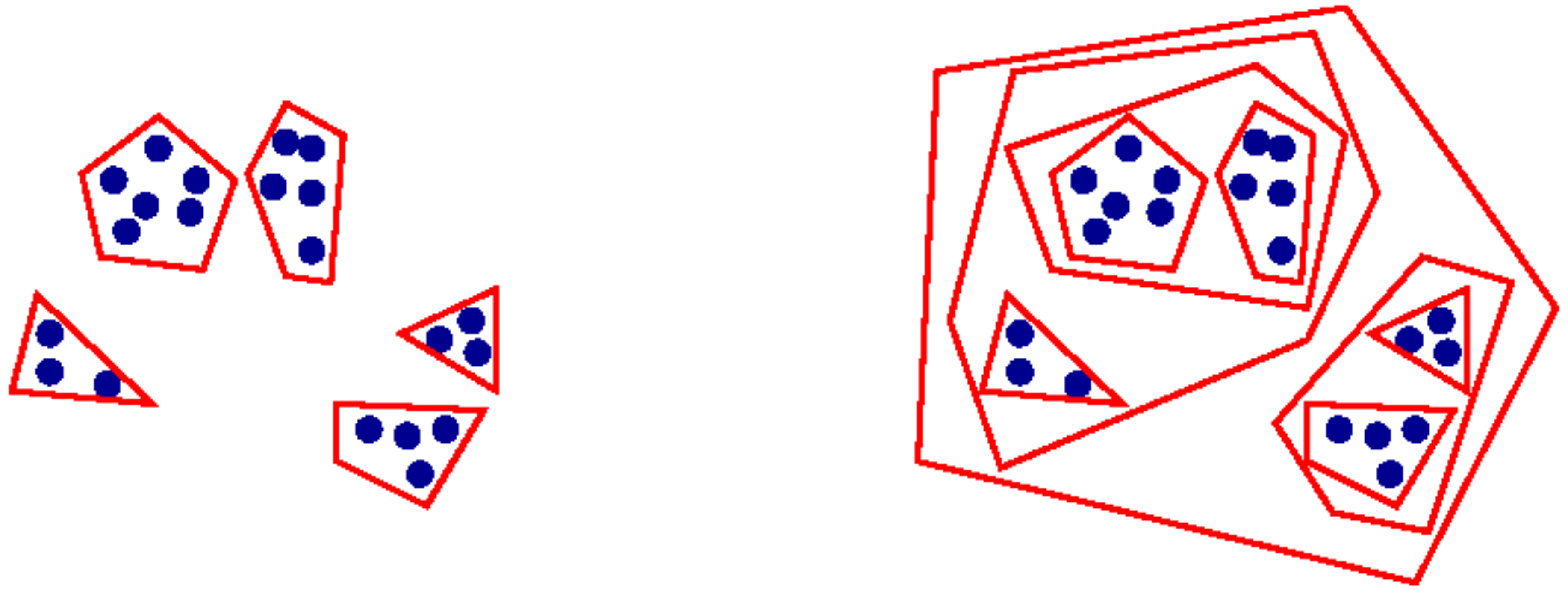
  Points far away from the center of cluster.

  Small clusters with centers far away from other centers.

CLUSTERING

Difficulty 1: How to decide on the properties of a clustering?

Types of clusterings:



Partitional vs Hierarchical

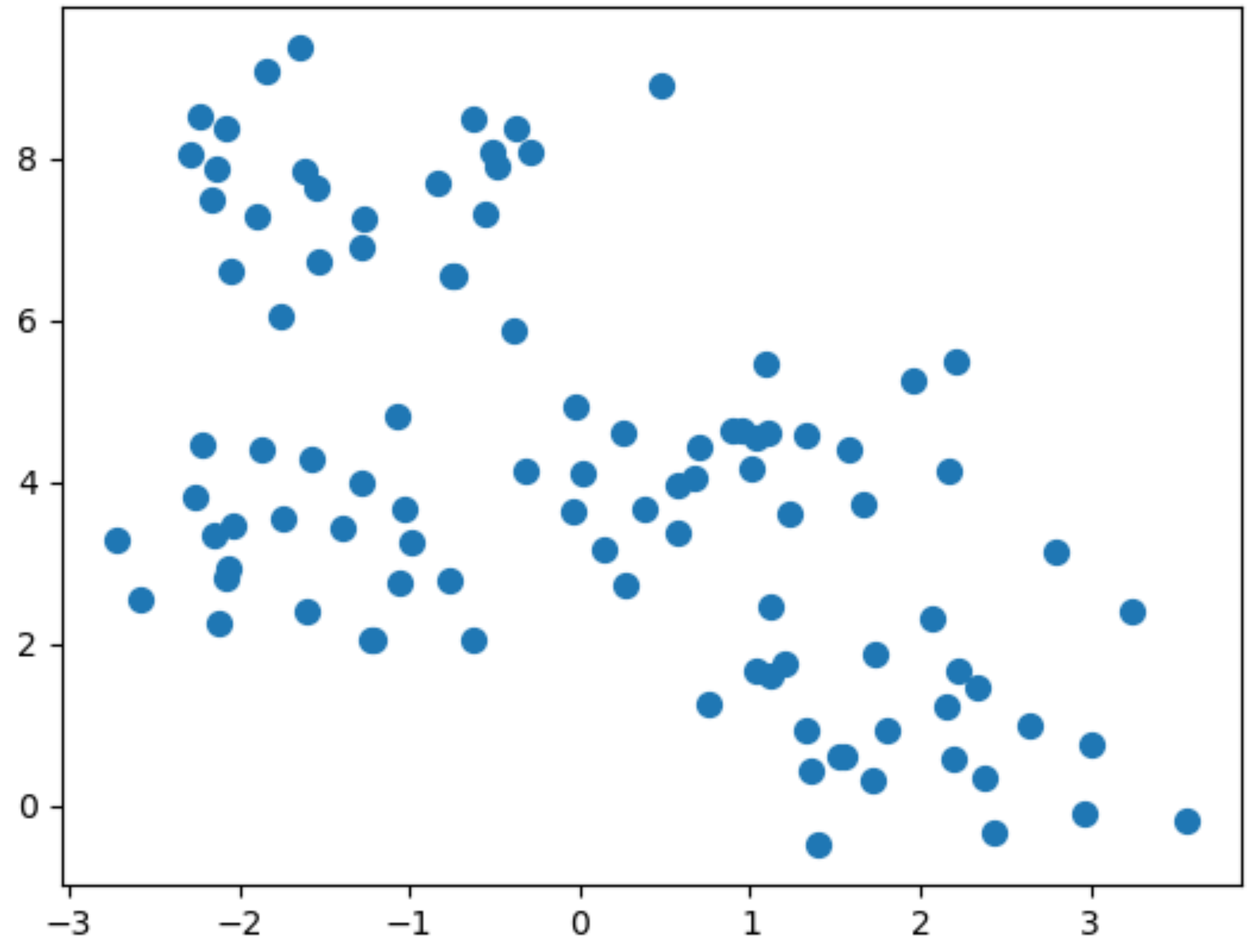
Exclusive vs Overlapping

Complete vs Partial

CLUSTERING

Difficulty 2: How to define a good clustering?

‘Clustering beauty lies in the eye of the beholder.’



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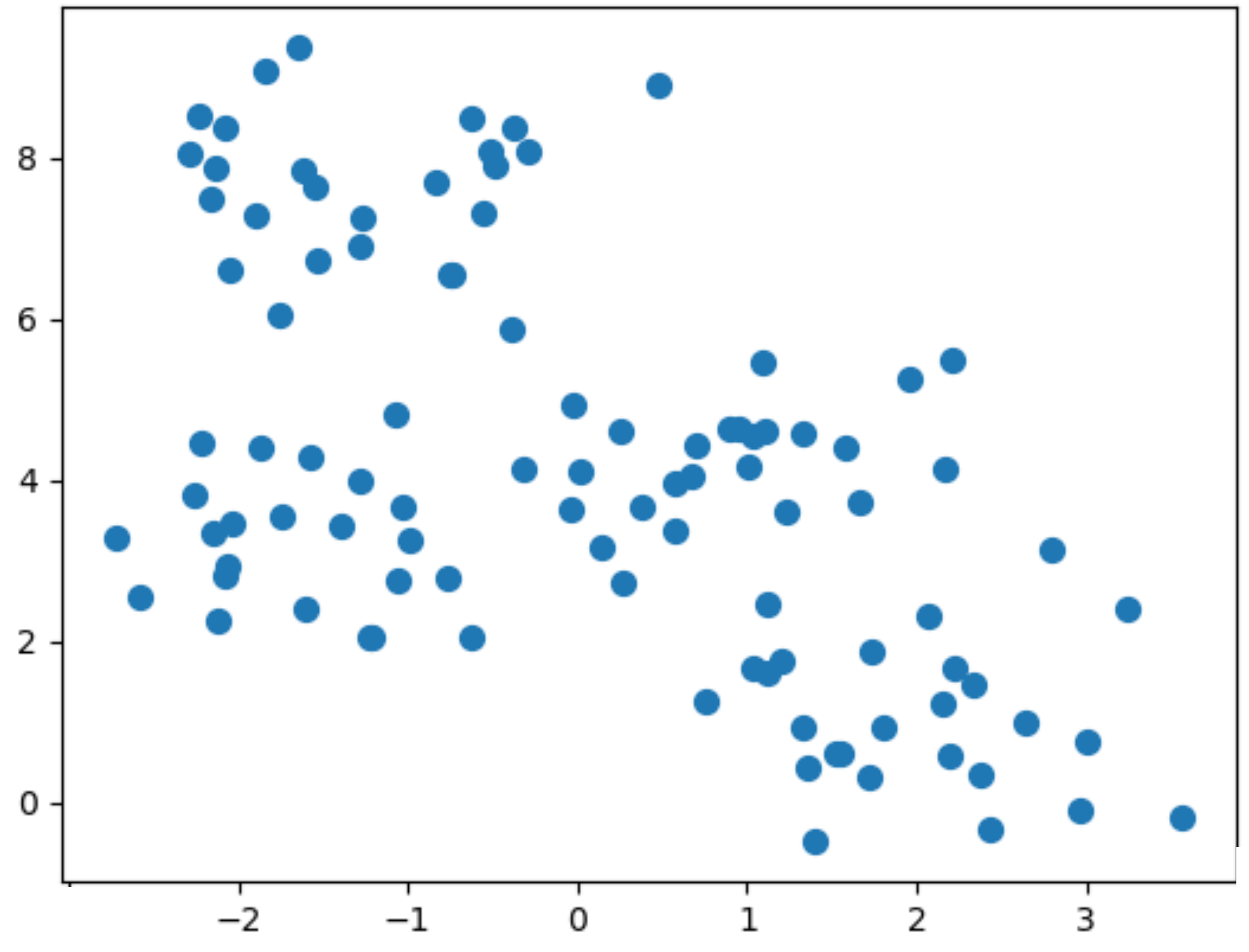

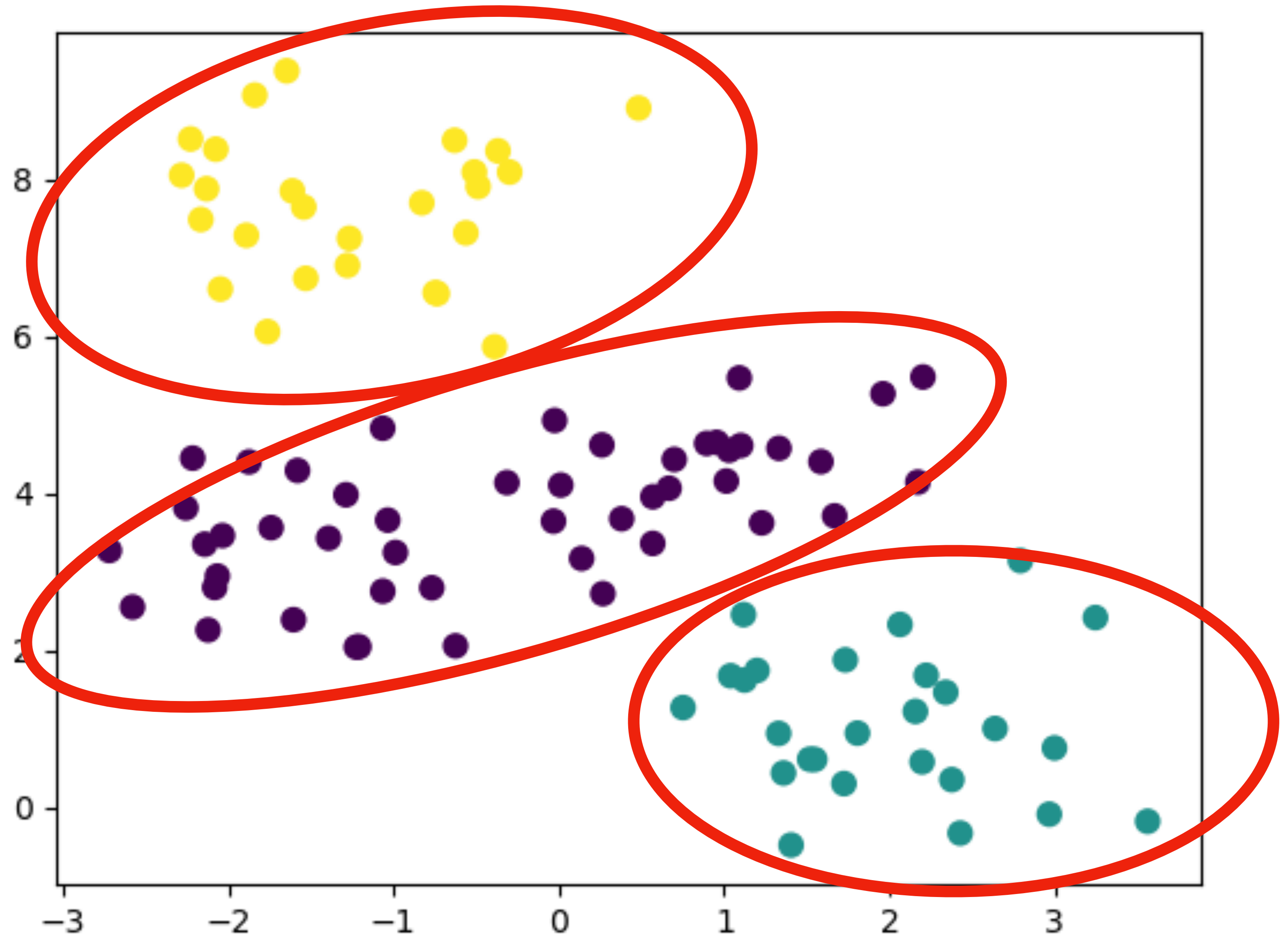

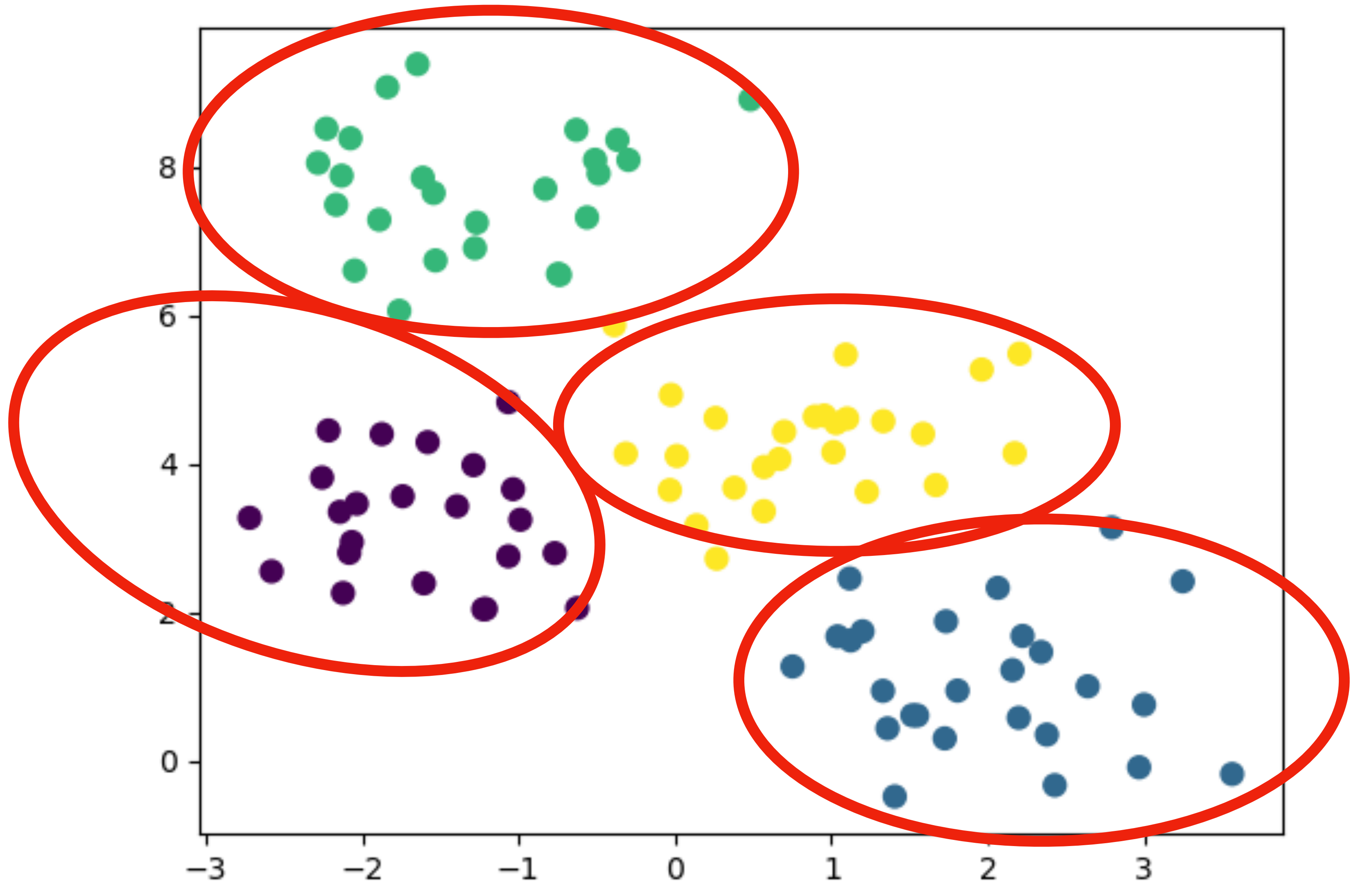

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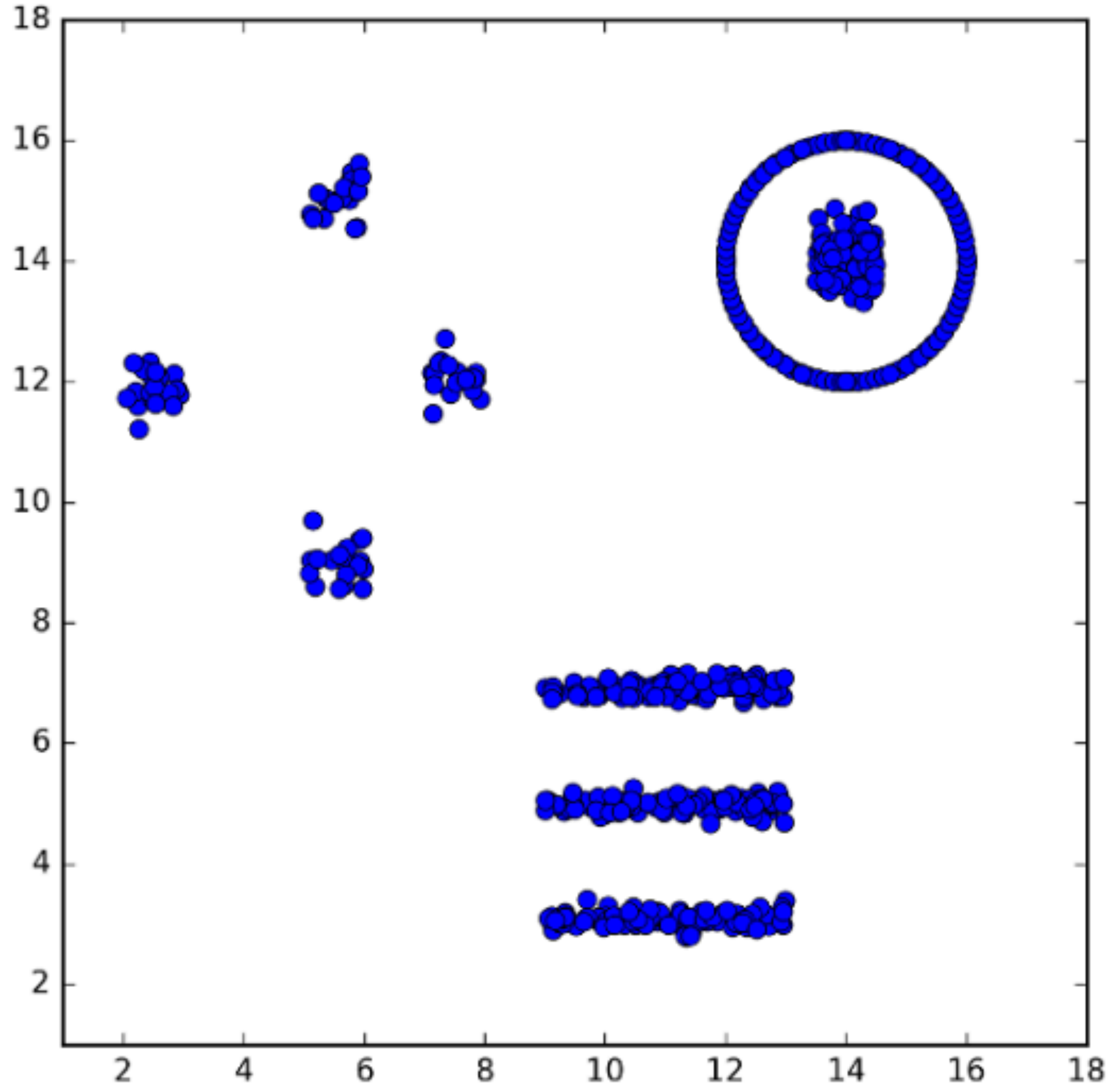


How many clusters are there?

K-MEANS CLUSTERING

Algorithm:

Select k random points as centers

Repeat until convergence

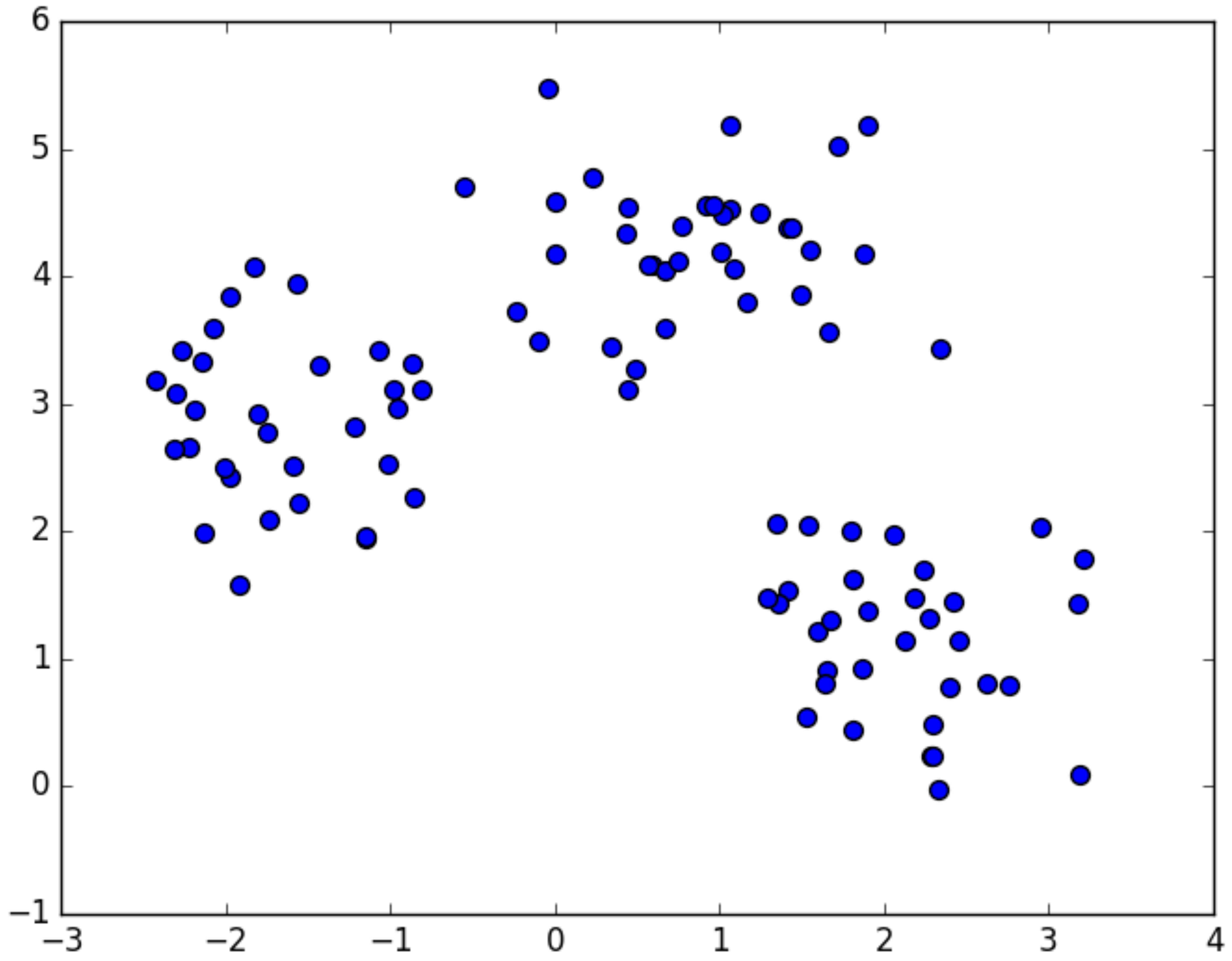
- Assign label of each point to its closest center
- Recompute centers

Postprocessing if necessary

At this point take centers as ‘centroids’:
average value of each dimension over all points in cluster.

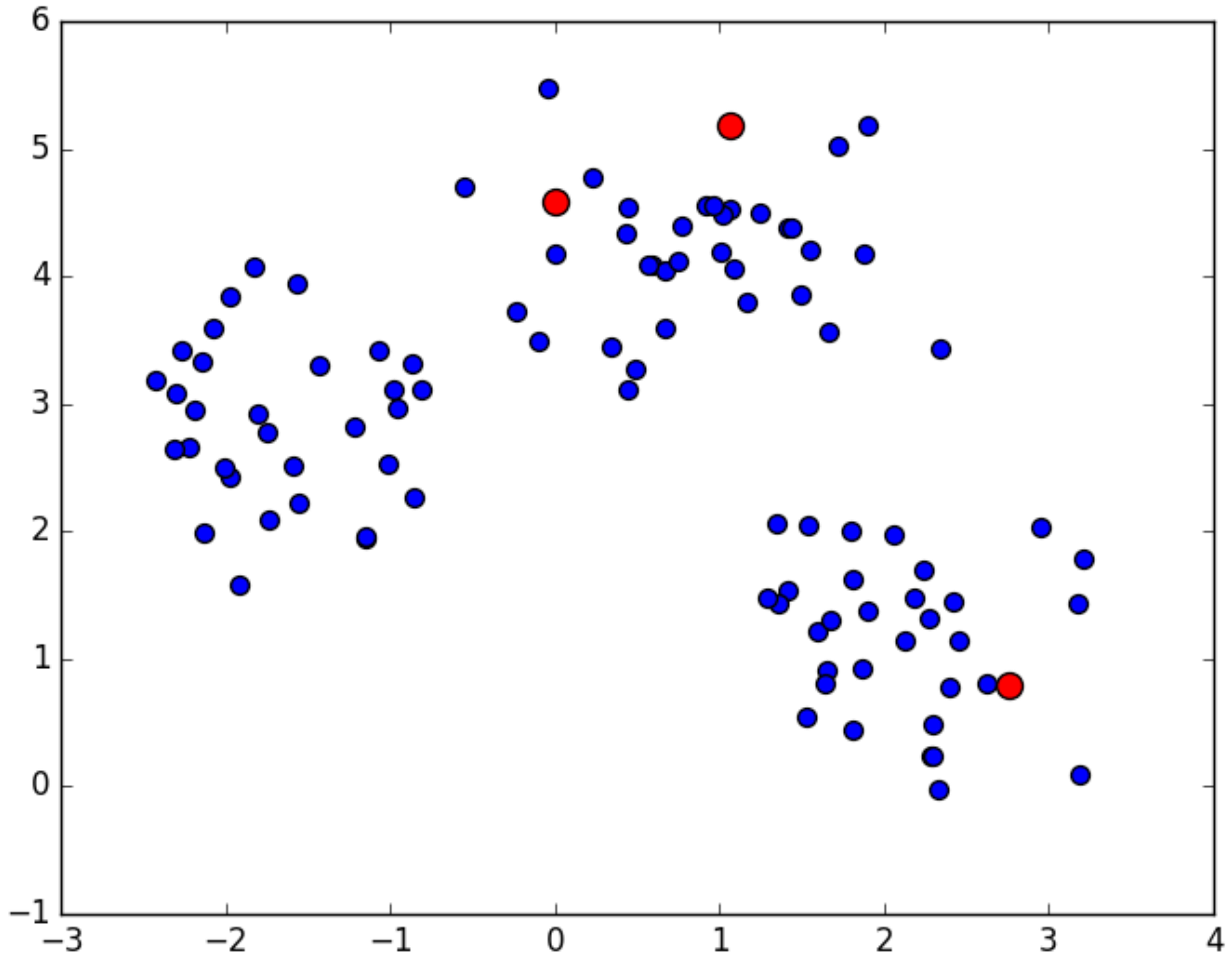
K-MEANS CLUSTERING

$k = 3$. Input configuration.



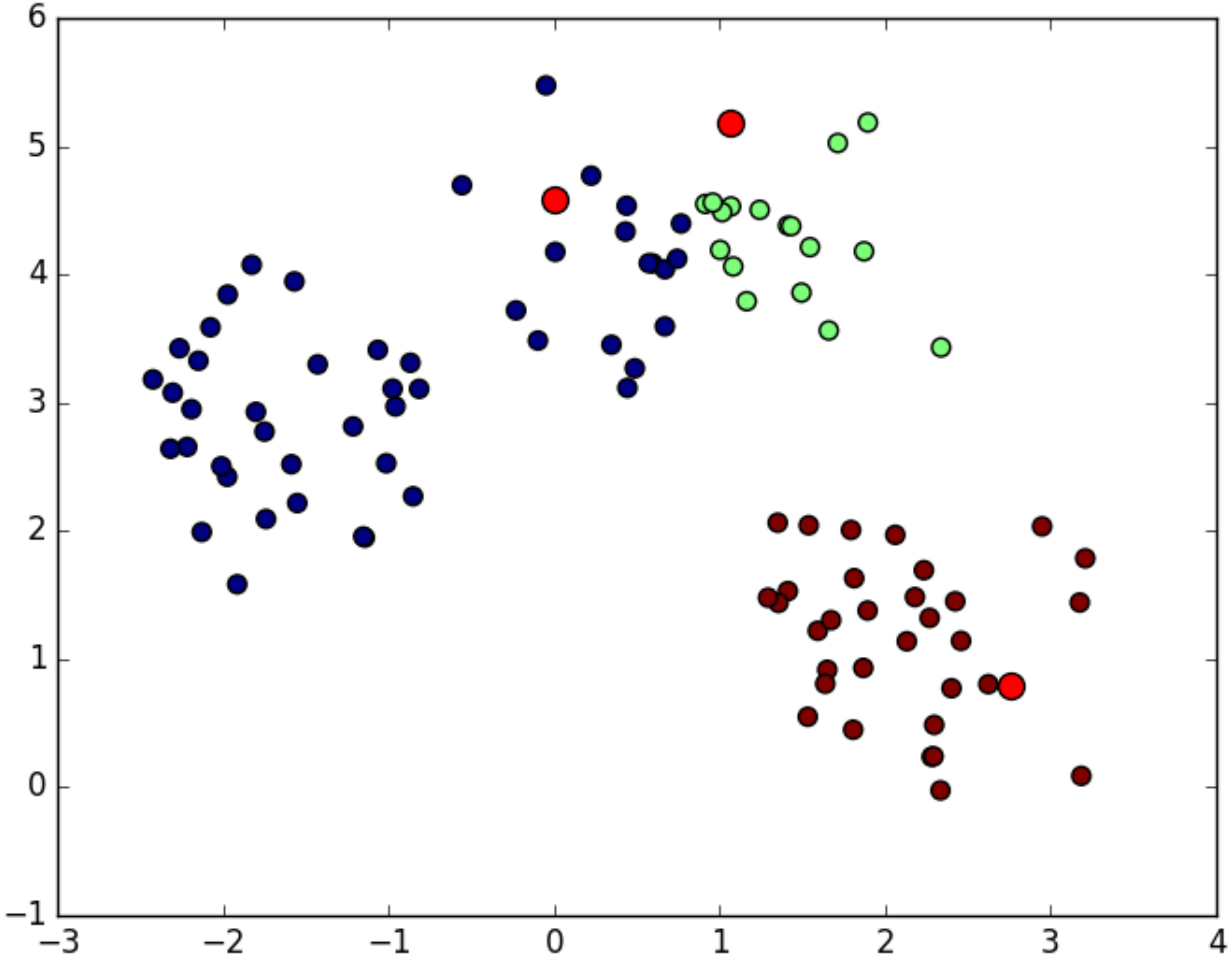
K-MEANS CLUSTERING

Random selection of centers.



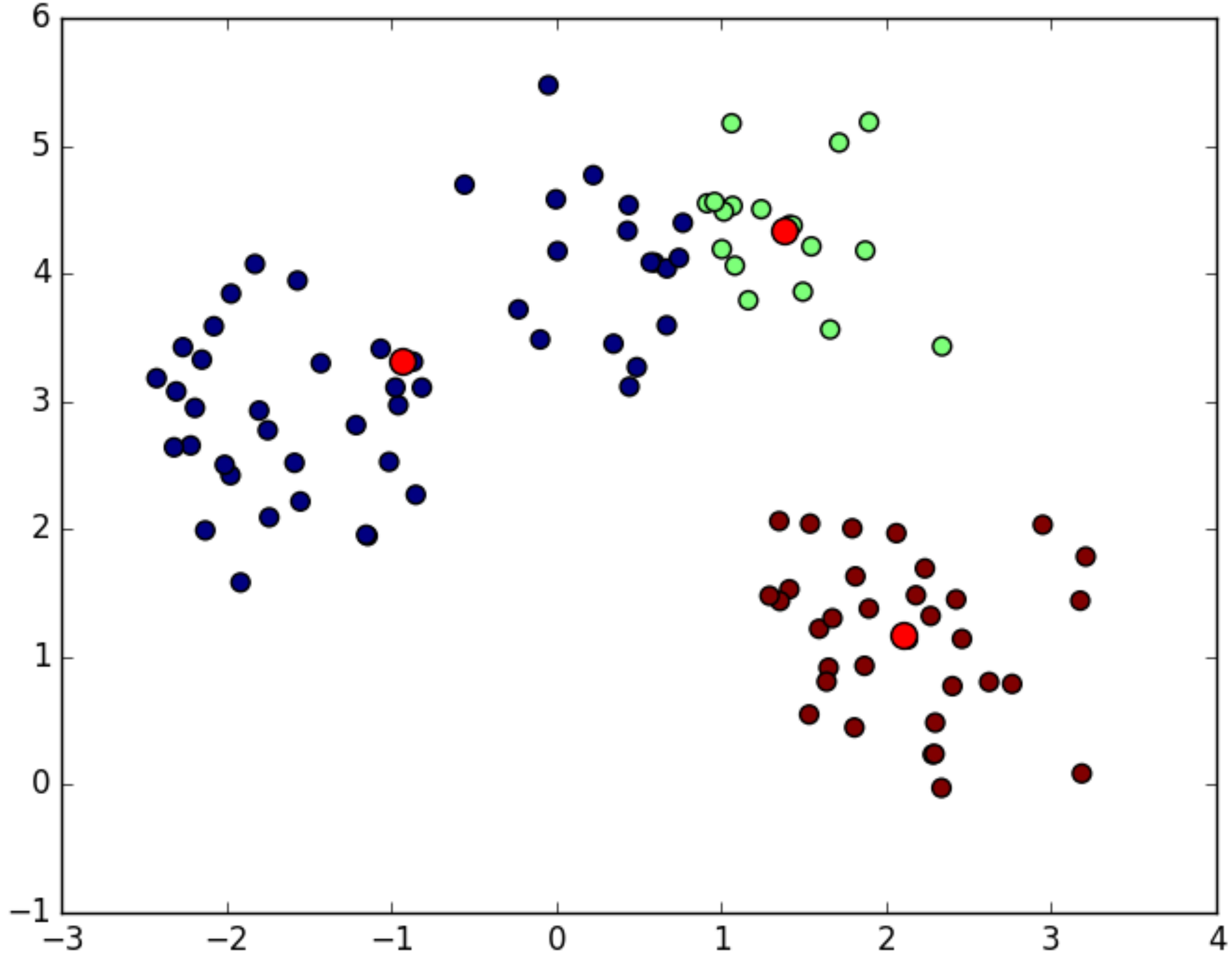
K-MEANS CLUSTERING

Iteration 1. Label each point with respect to its closest center



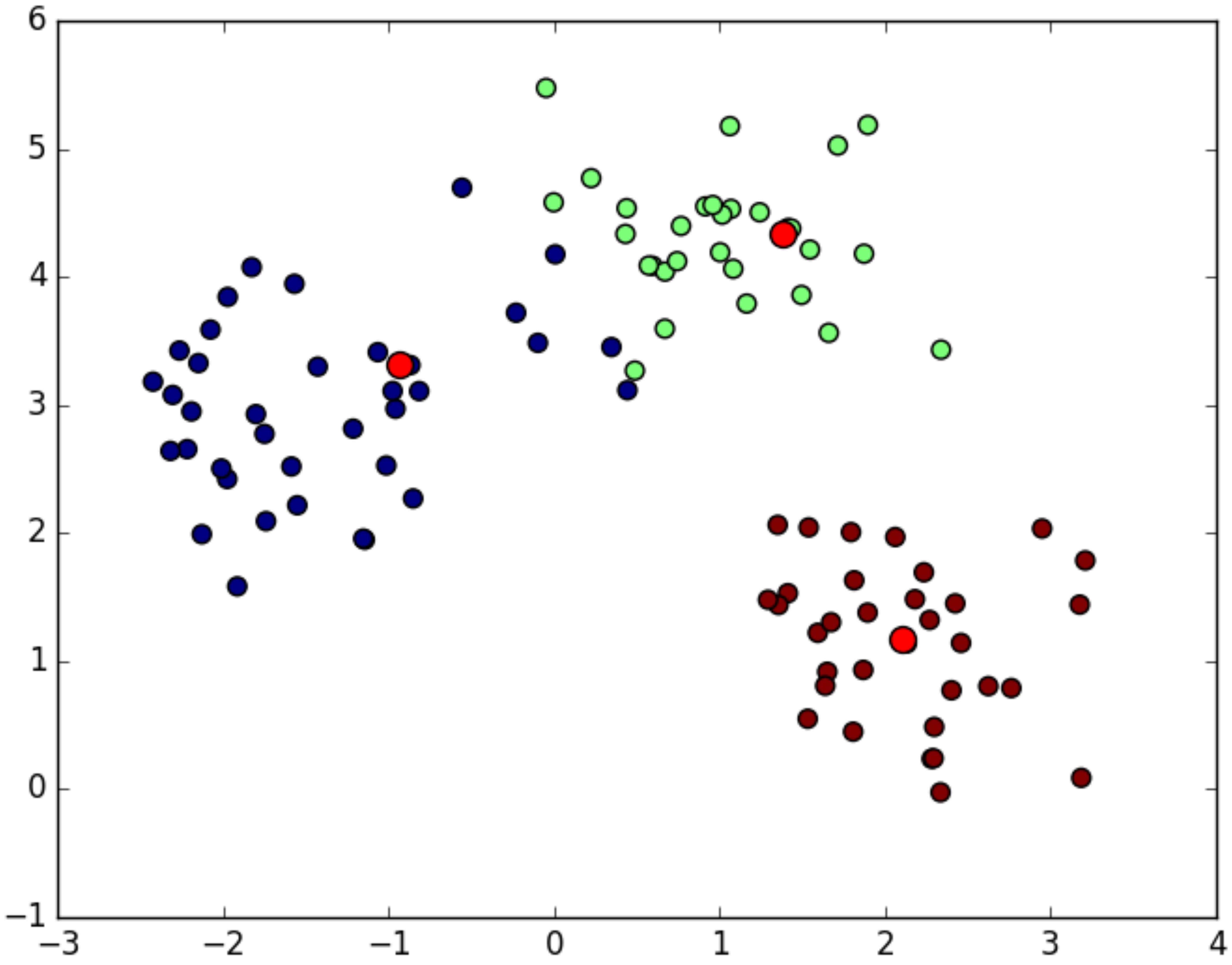
K-MEANS CLUSTERING

Iteration 1. Recompute centers



K-MEANS CLUSTERING

Iteration 2. Label each point with respect to its closest center



K-MEANS CLUSTERING

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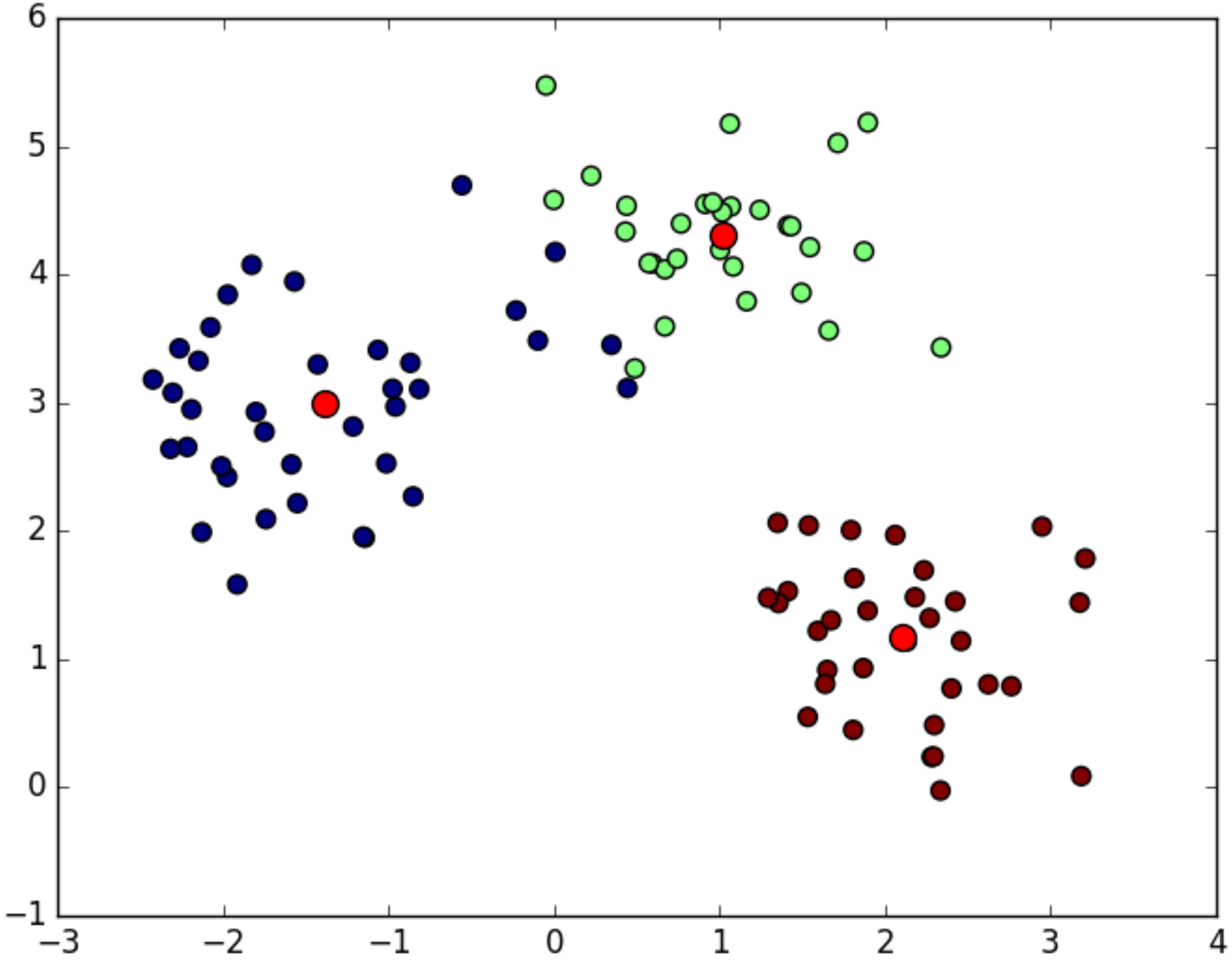
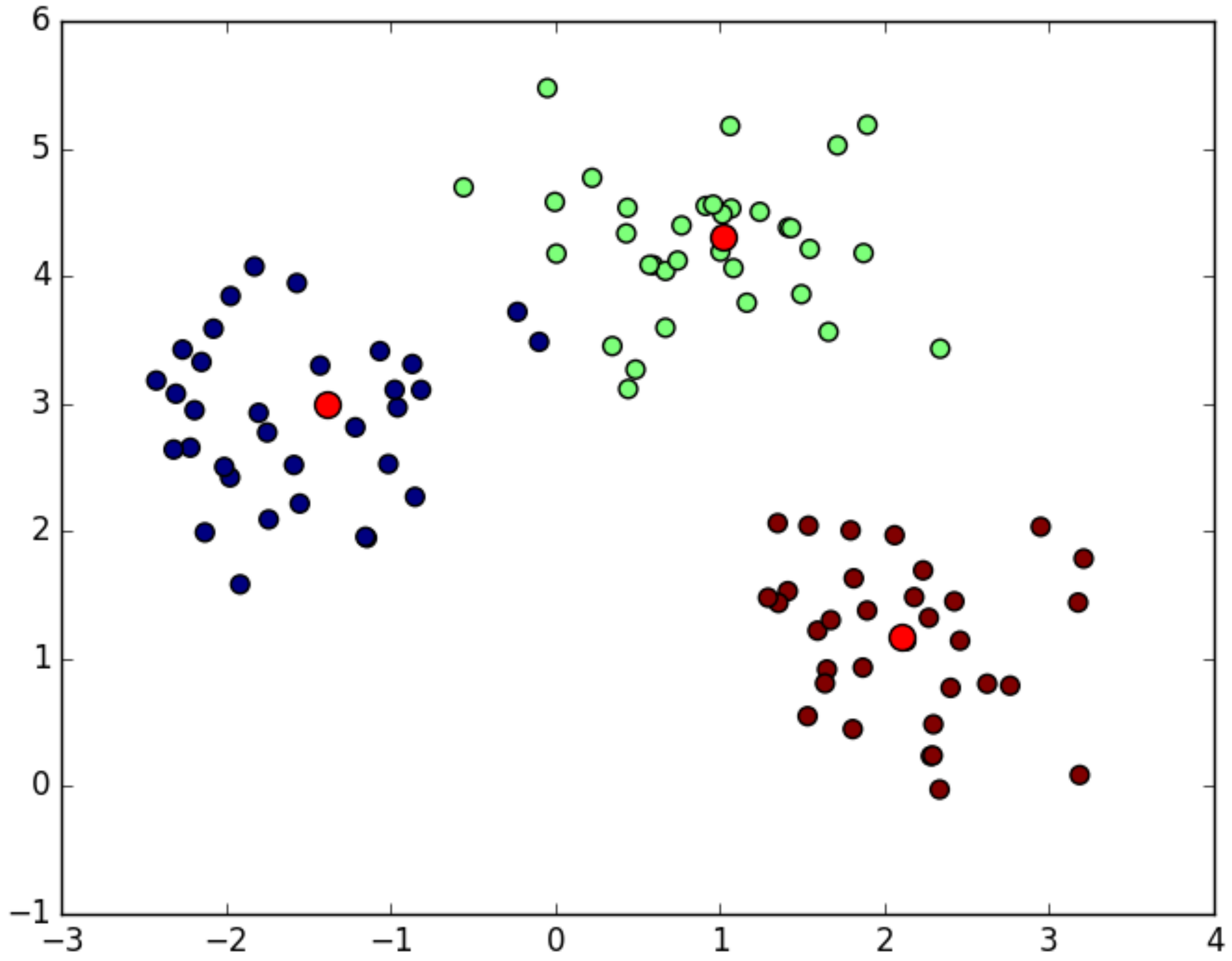


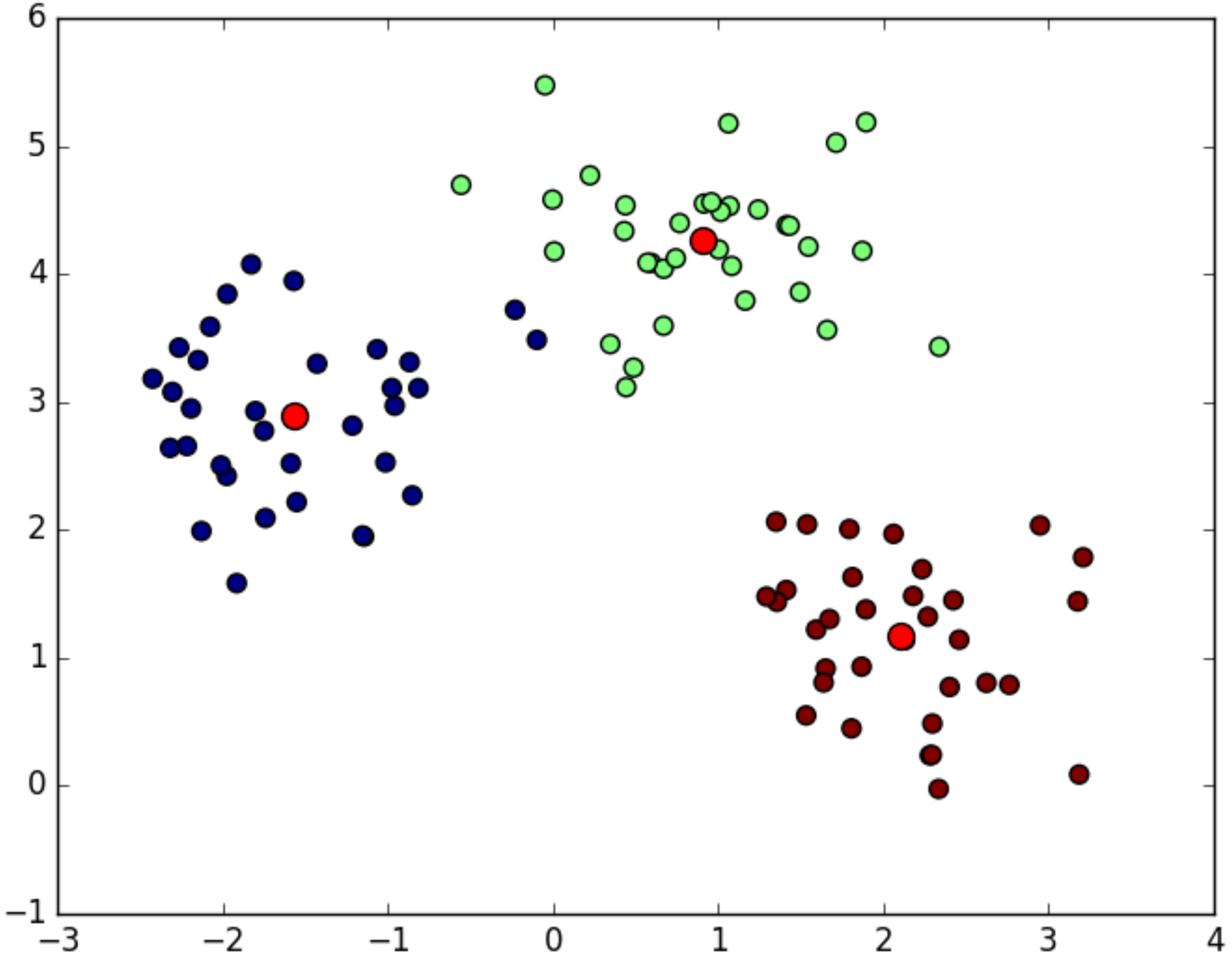
K-MEANS CLUSTERING

Iteration 3. Label each point with respect to its closest center



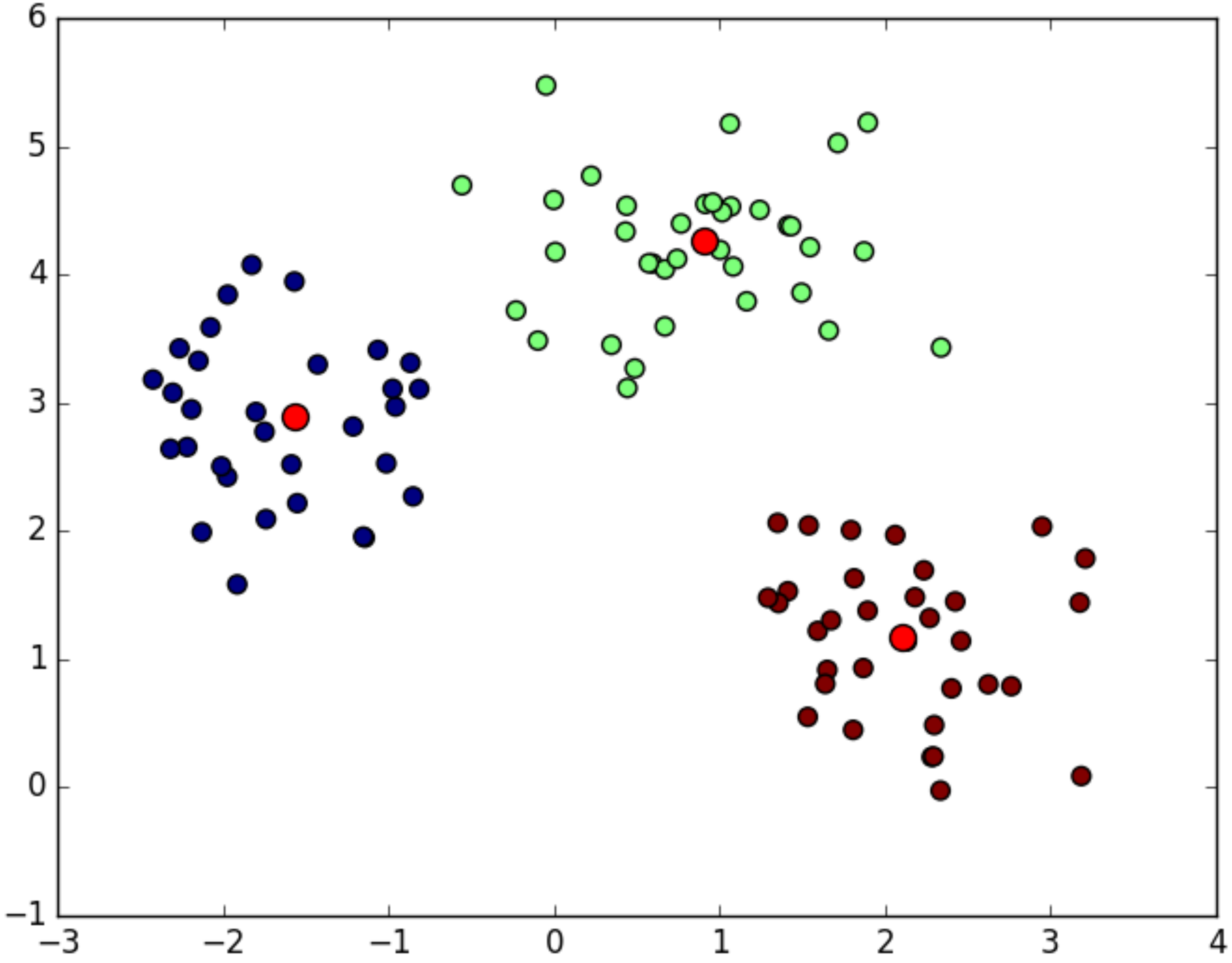
K-MEANS CLUSTERING

Iteration 3. Recompute centers



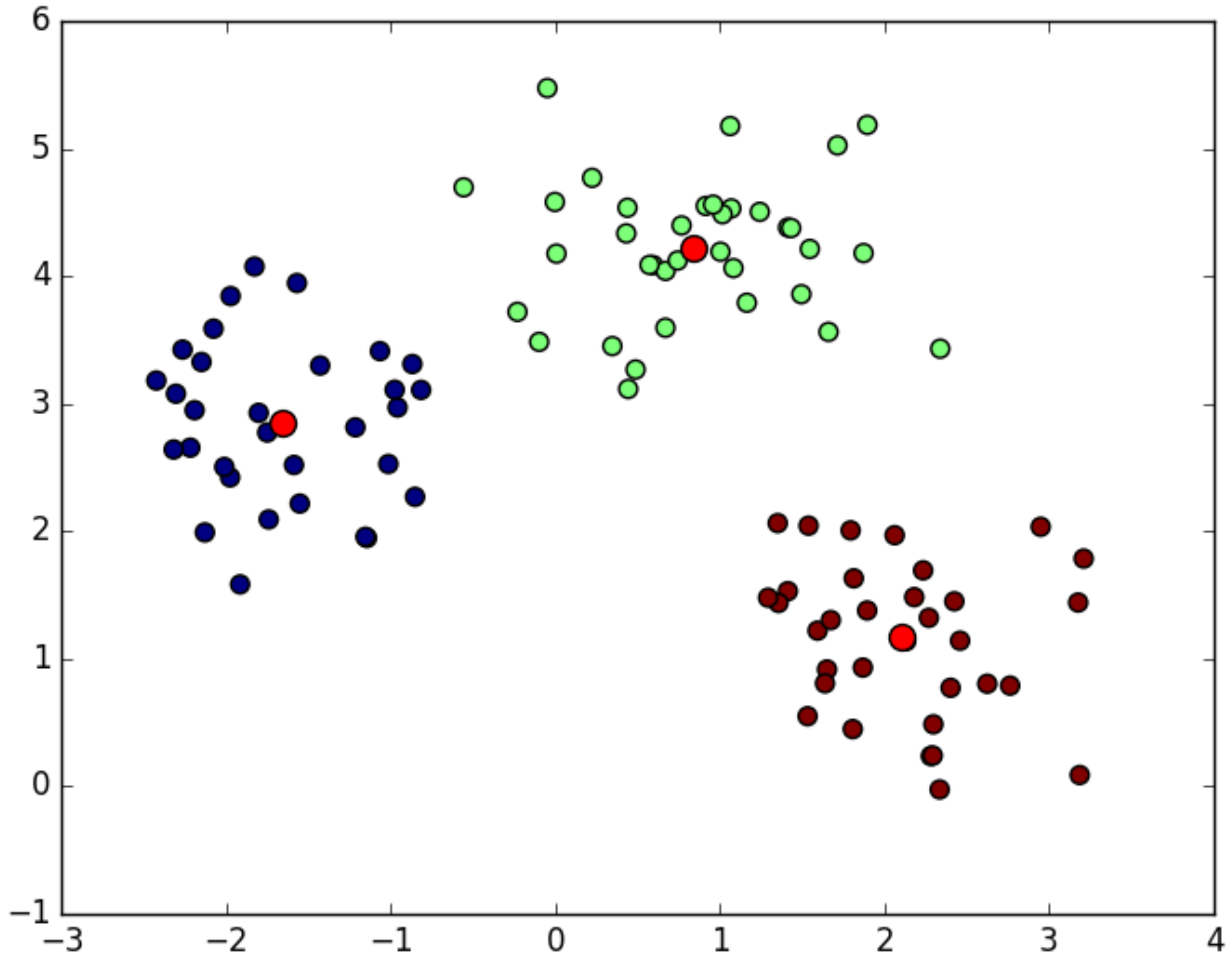
K-MEANS CLUSTERING

Iteration 4. Label each point with respect to its closest center



K-MEANS CLUSTERING

Iteration 4. Recompute centers



K-MEANS CLUSTERING: DETAILS

Several issues need to be addressed.

- Does it always converge? Is it optimal?
- How to find good starting centers?
- Are there problematic cases?
- How to define the centers?
- How to find a good value of k?

K-MEANS CLUSTERING: DETAILS

Convergence? YES (we need formal definition)

Optimality? NOT ALWAYS (we need formal definition)

K-MEANS CLUSTERING: DETAILS

Convergence? YES (we need formal definition)

Optimality? NOT ALWAYS (we need formal definition)

Formal definition: Partition S into $C_1, C_2, \cdots, C_k$ such that

$$\sum_{i=1}^{k} \sum_{p \in C_i} ||p - centroid_i||^2$$

is minimized ($centroid_i$ is the centroid of C_i).

Hard problem!

K-means tries to minimize it, but doesn't not always succeed.

K-MEANS CLUSTERING: DETAILS

Convergence? YES

Optimality? NOT ALWAYS   Let’s see an example with $k = 4$.

K-MEANS CLUSTERING: DETAILS

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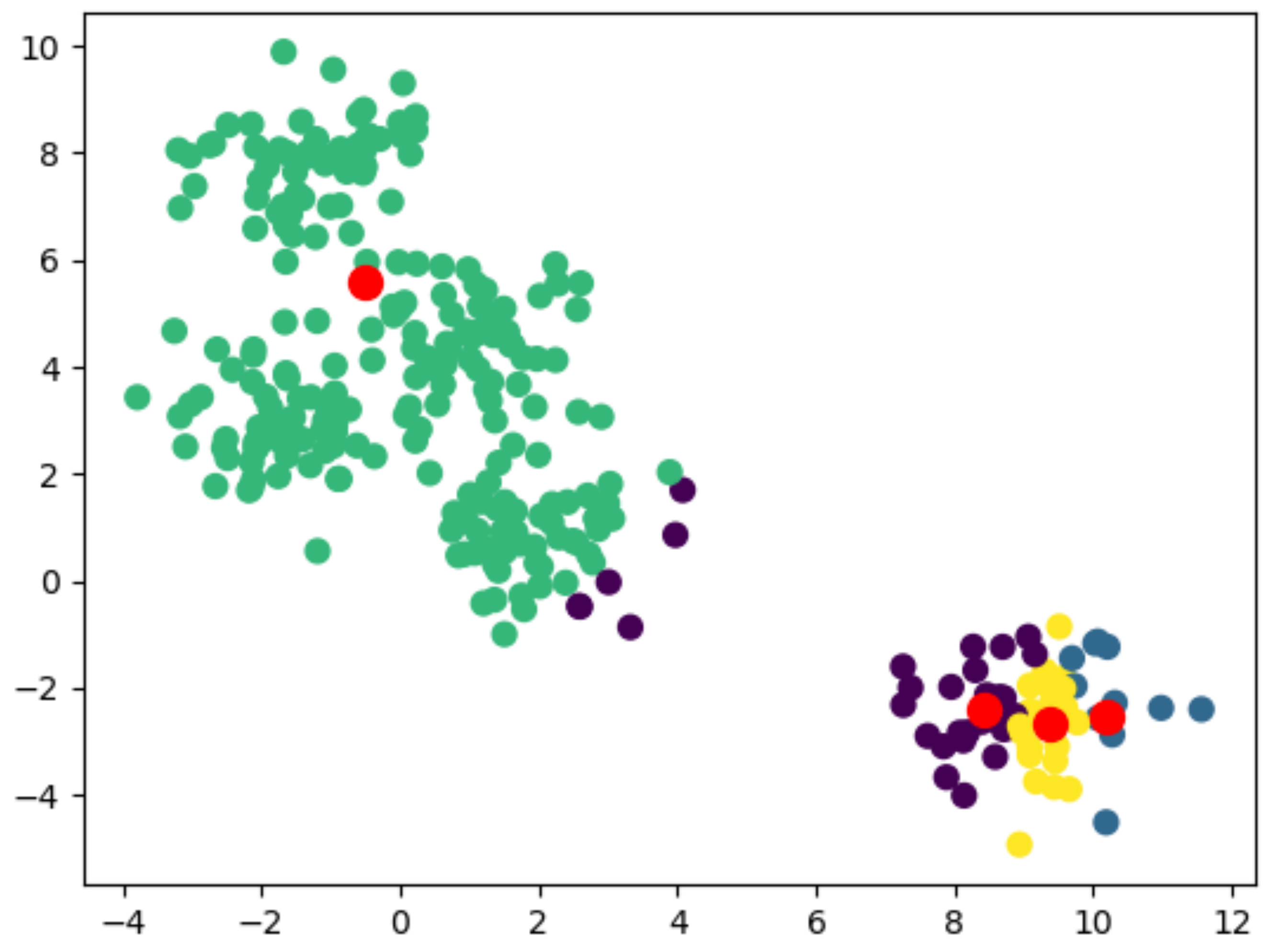
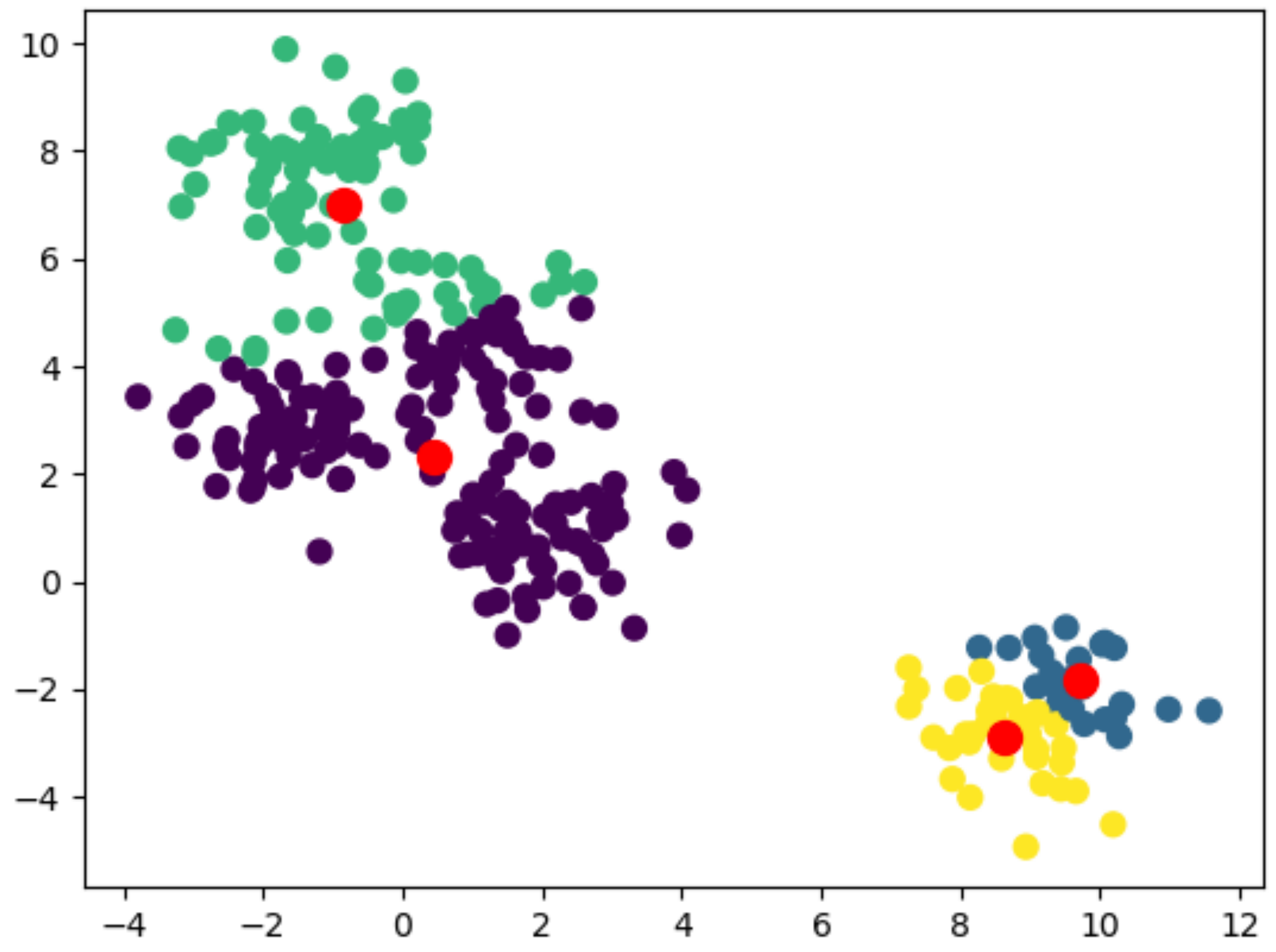
Initial



Clustering depends on initial centers

Final



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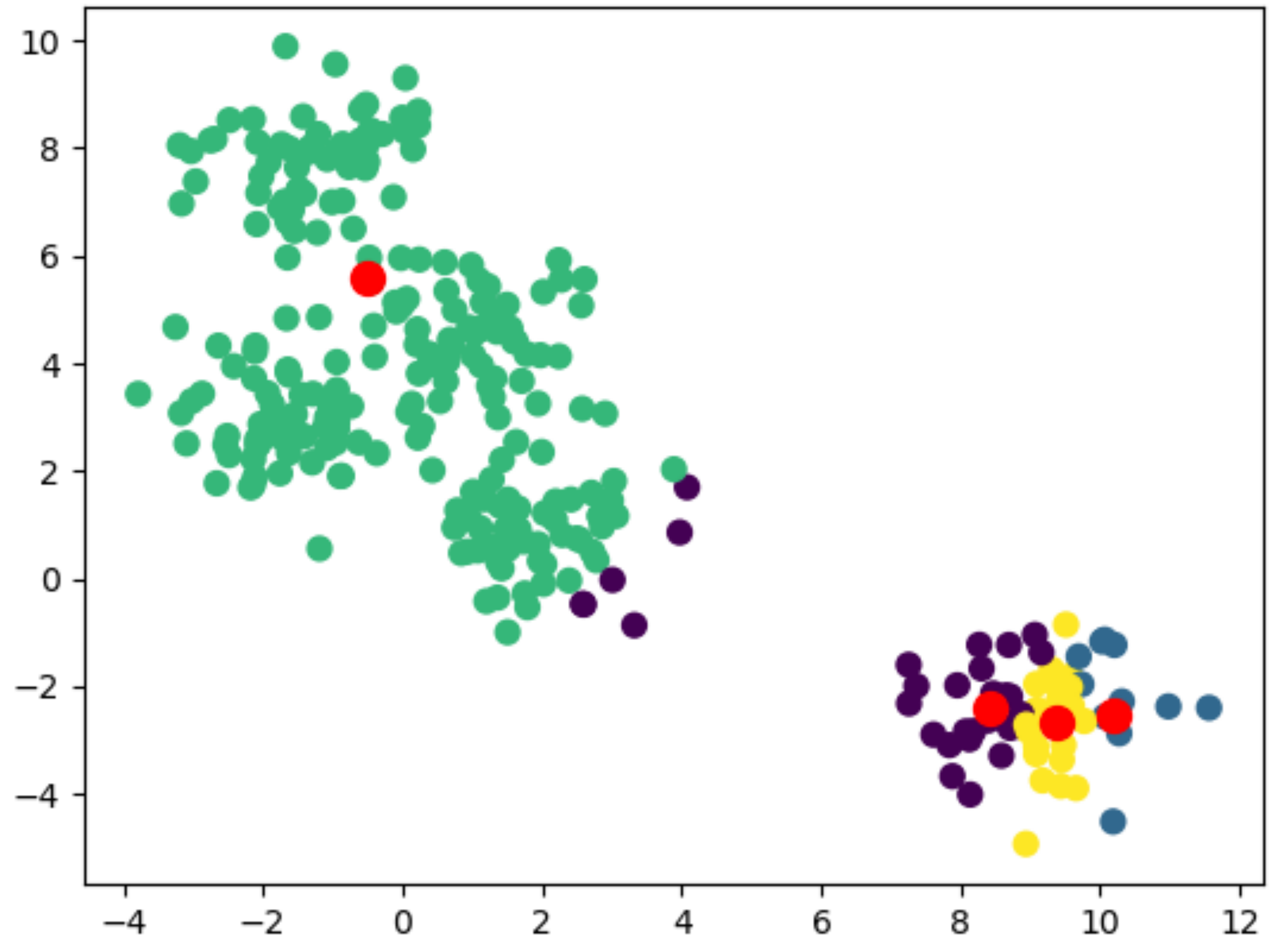

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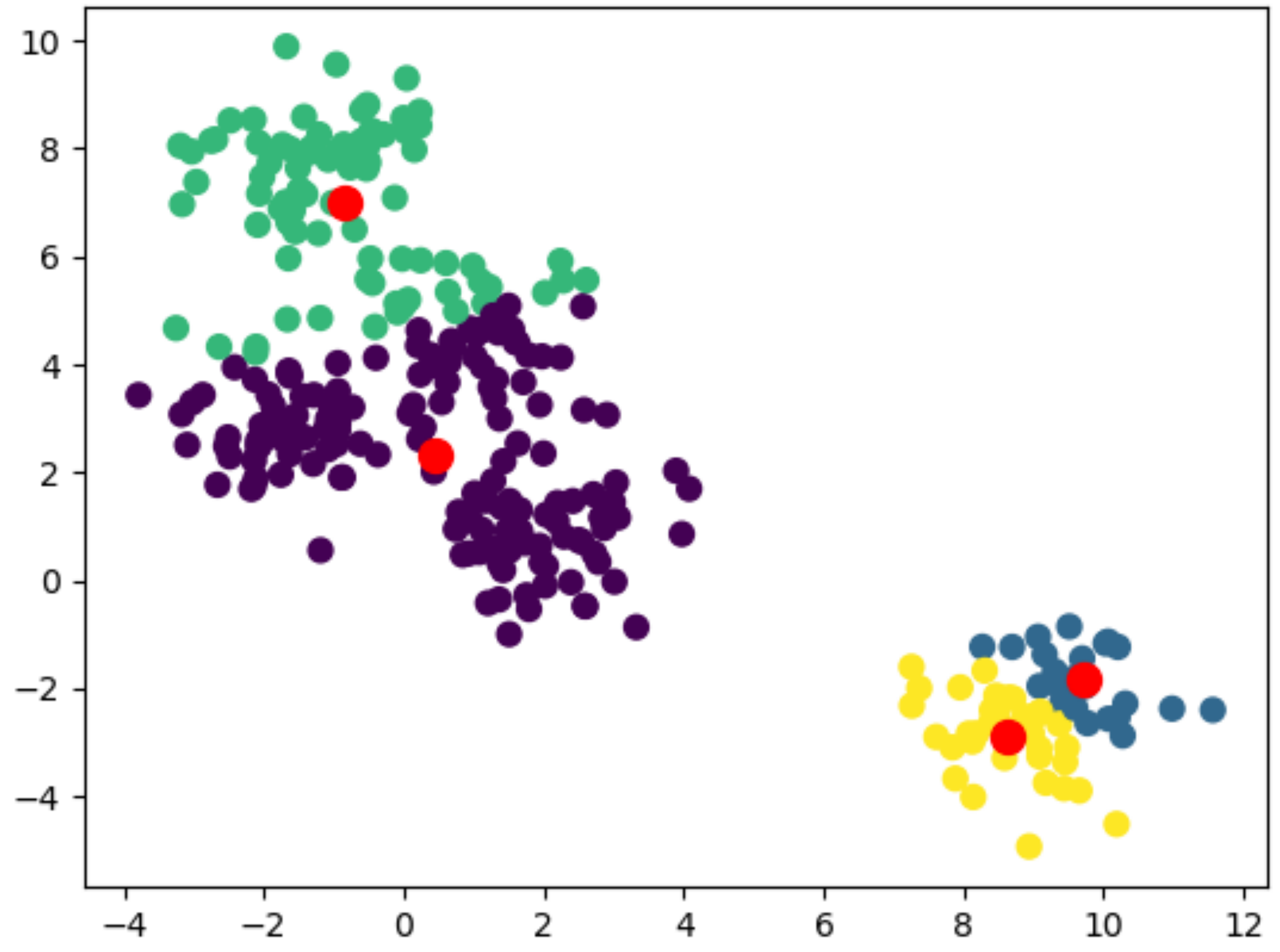

Doesn't minimize sum of squared distances.

K-MEANS CLUSTERING: DETAILS

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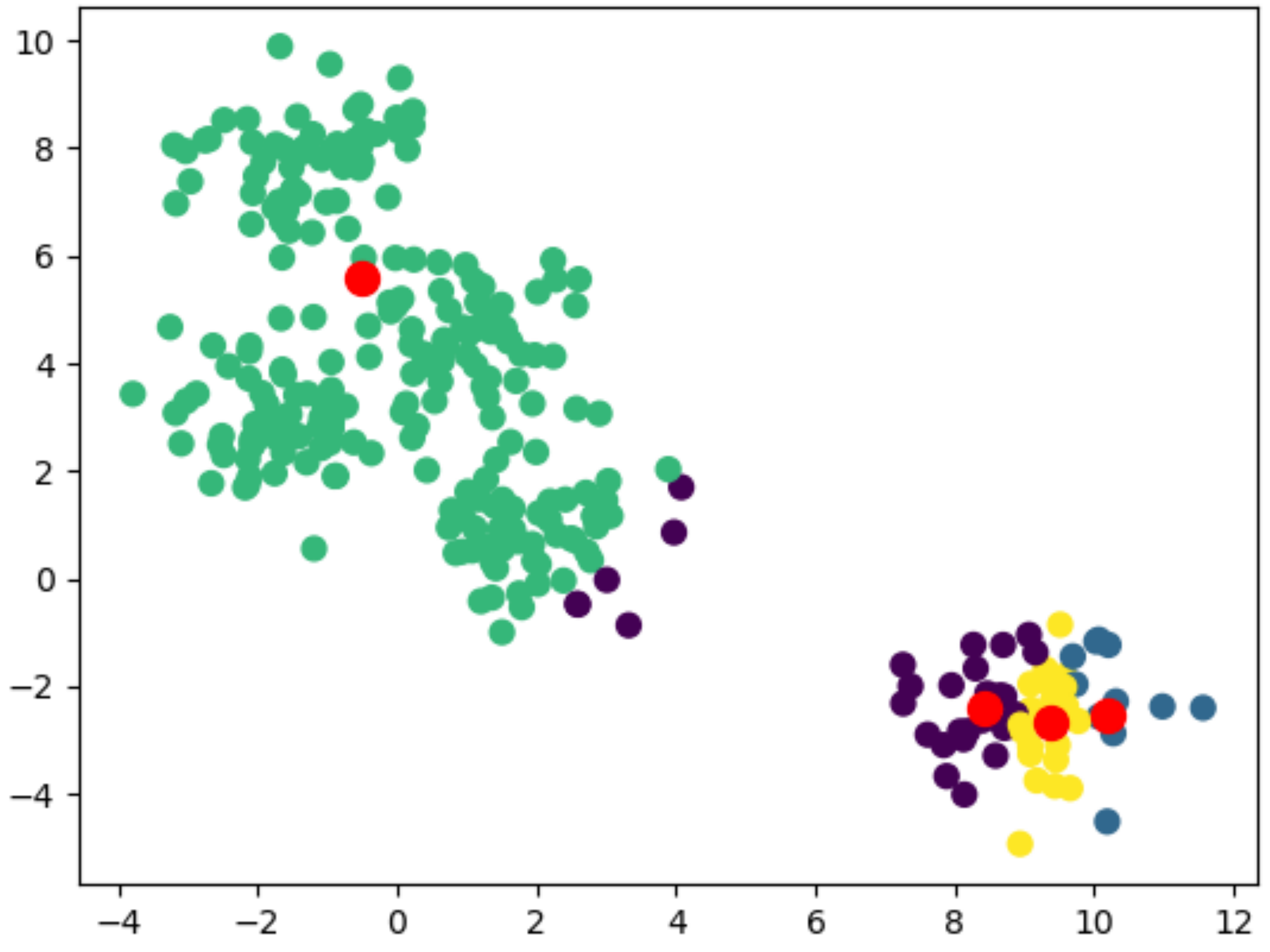

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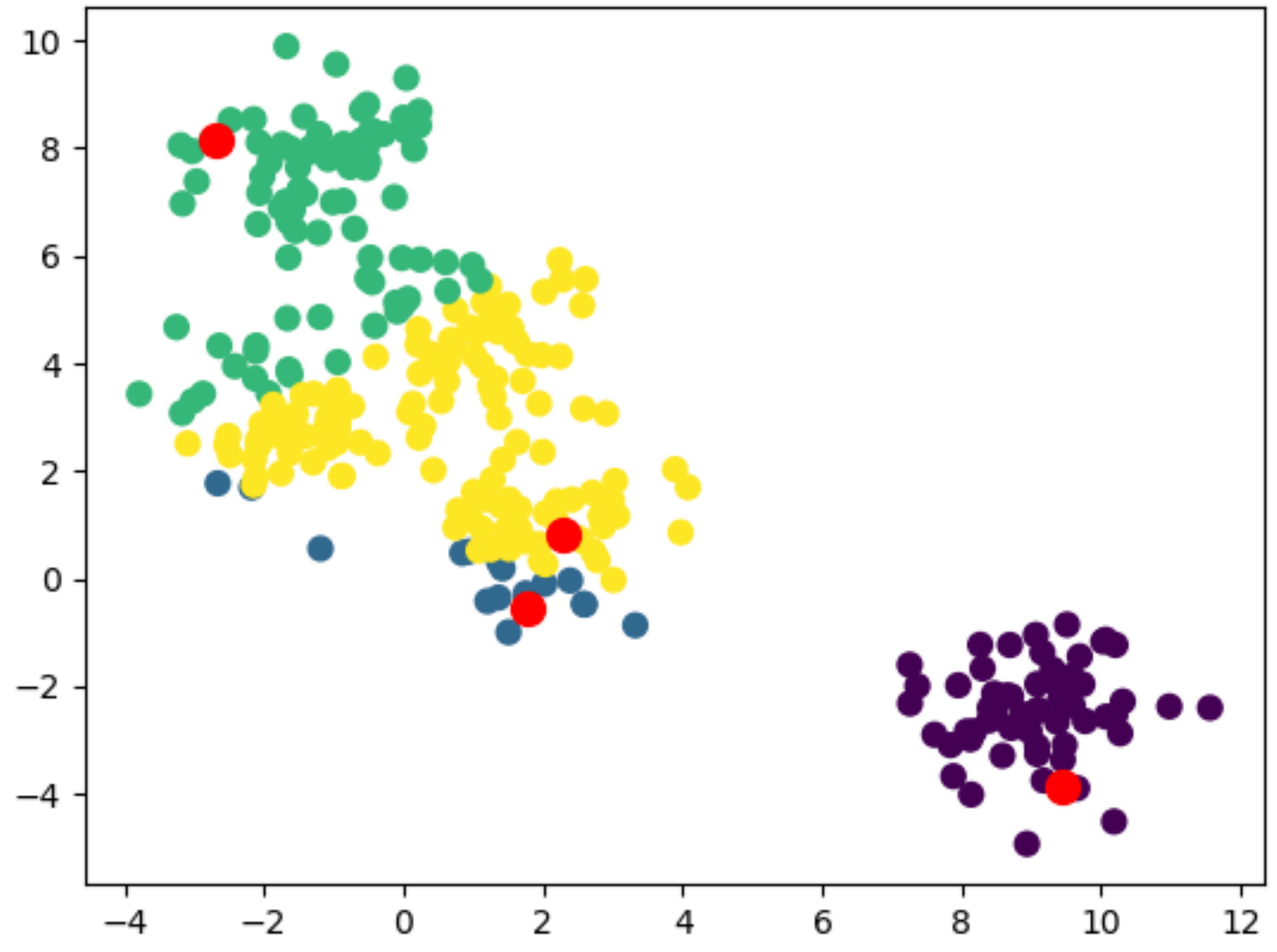

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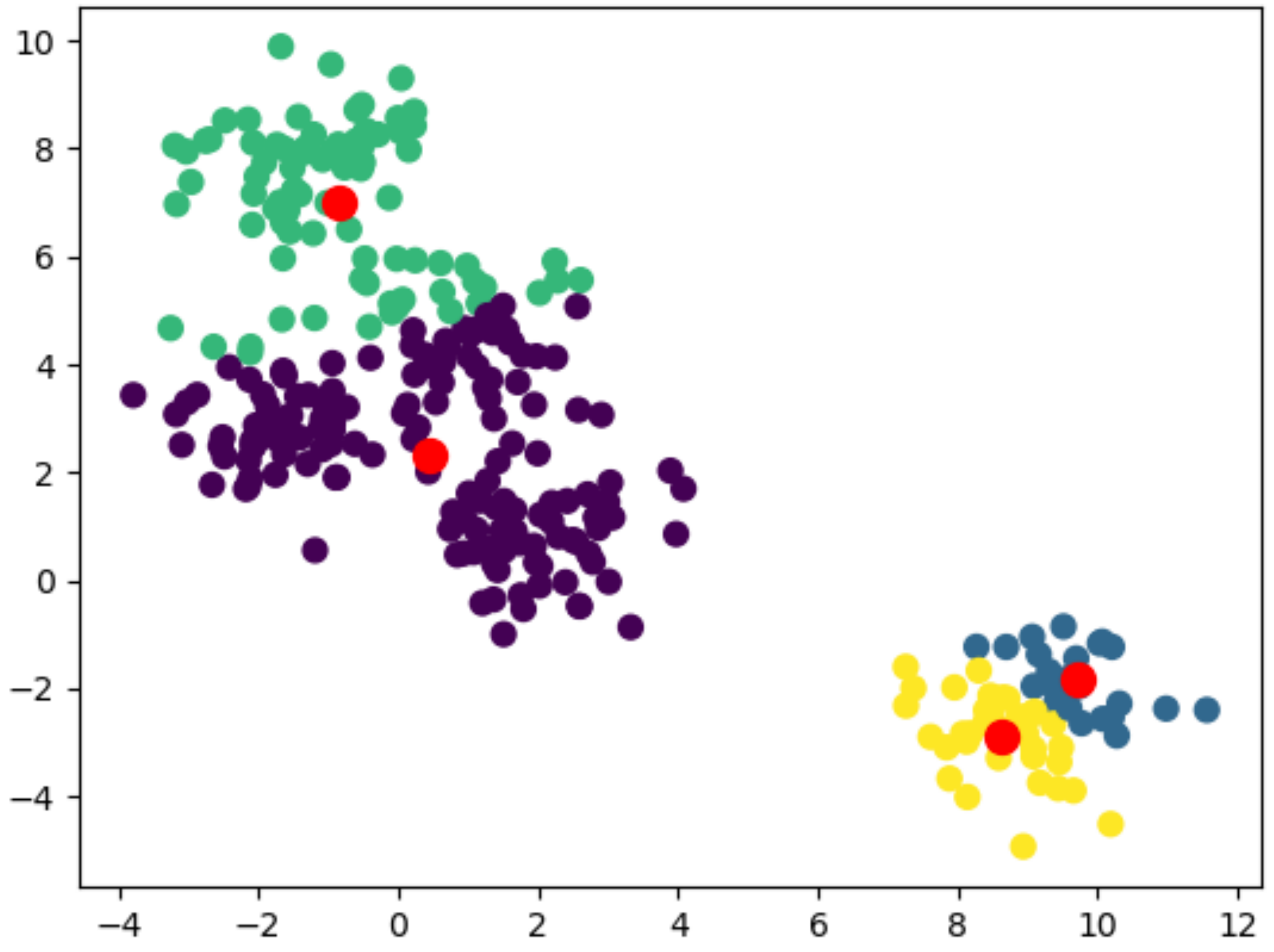

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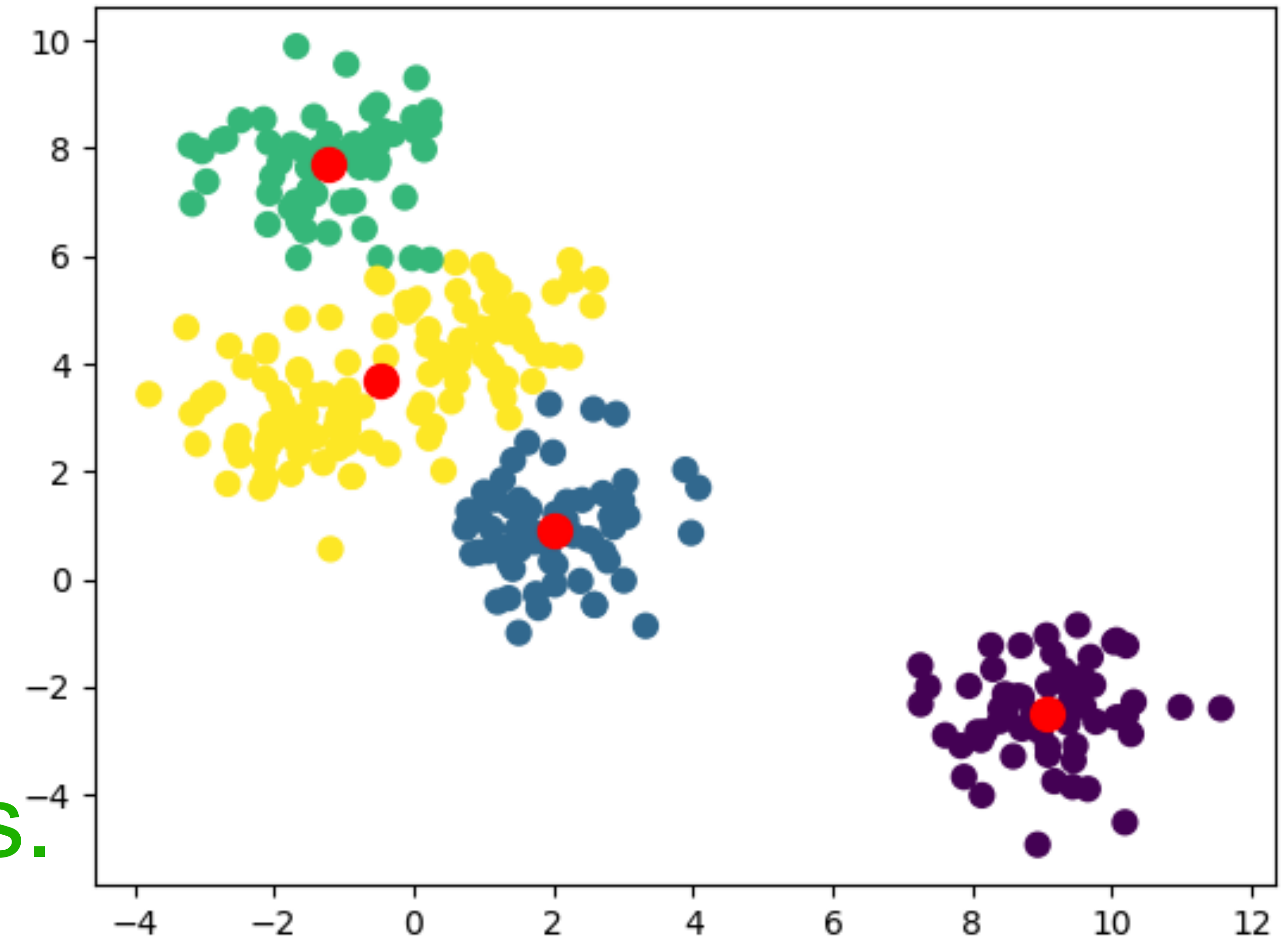


Final

K-MEANS CLUSTERING: DETAILS

How to pick good starting centers? Try repeated runs AND/OR

K-MEANS CLUSTERING: DETAILS

How to pick good starting centers? Try repeated runs AND/OR

Rather than selecting at random:

- Select first center randomly
- Each successive center: the farthest point from any center

Possible further problems:

- If both n, k are large: execution speed
- May select outliers at distant locations

Possible further fix:

- Apply on a random sample of points

K-MEANS CLUSTERING: DETAILS

Problematic cases even if we were to solve optimally:

Outliers: Influence centers towards them $\Rightarrow$ large cost

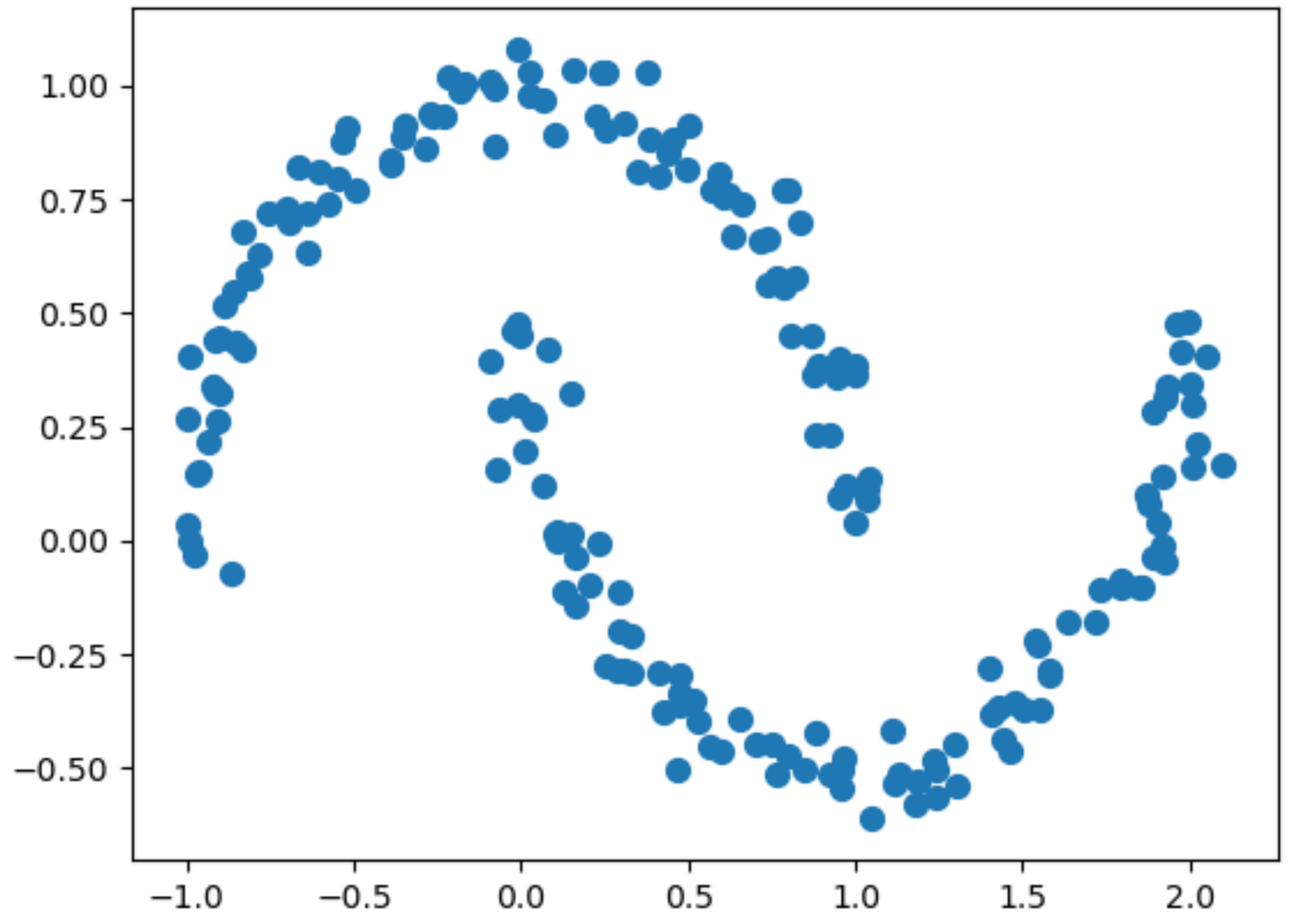
Solution: Depending on the application, if outliers not important:

- Remove points contributing to cost the most for centers
- Remove small clusters

K-MEANS CLUSTERING: DETAILS

Problematic cases even if we were to solve optimally:

The natural clusters not convex:



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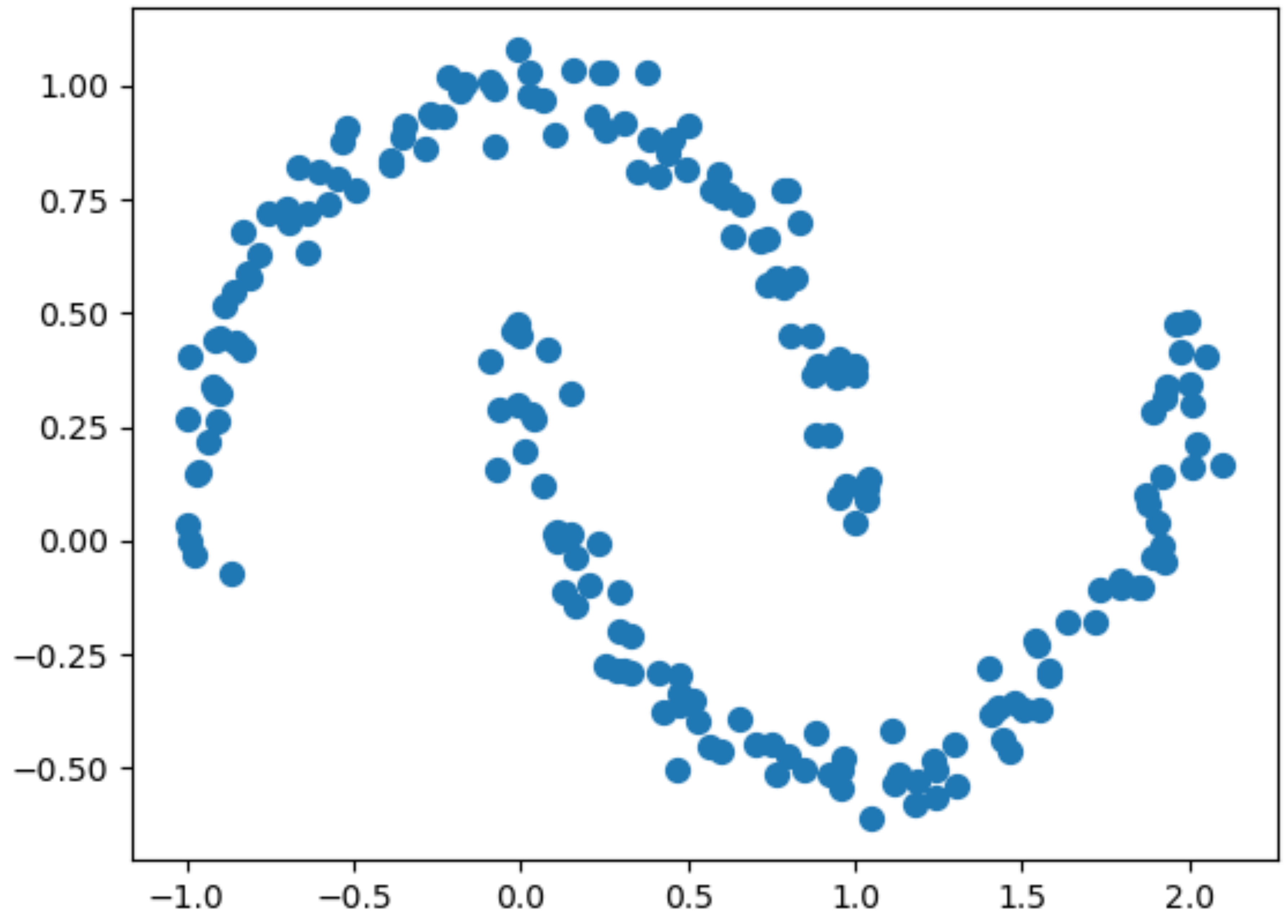

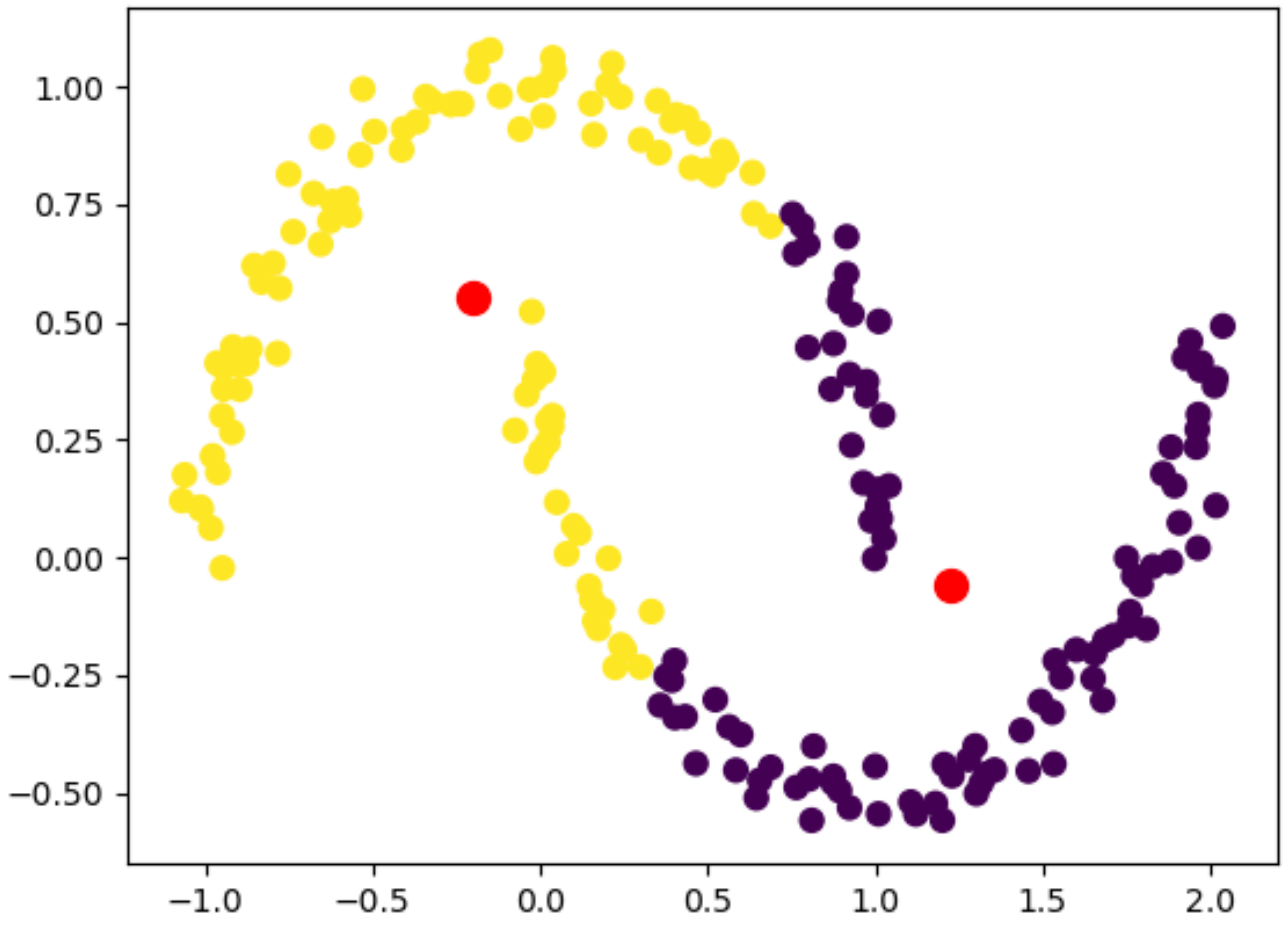

- Can be resolved if it is ok to have more clusters than necessary.

K-MEANS CLUSTERING: DETAILS

How to define the centers?

So far we defined the center to be the centroid of the cluster, i.e.,

For a cluster i:

$$centroid_{i,d} = \frac{1}{|C_i|} \sum_{p \in C_i} p_d$$

$centroid_{i,d}$: d^{th} dimension of $centroid_i$

p_d: d^{th} dimension of point p.

$centroid$ doesn't make sense if non-numeric attributes:

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$centroid$ doesn't make sense if non-numeric attributes:

$\Rightarrow$ use medoid to define the center of cluster (k-medoids):

$$medoid_i = \min_{p' \in C_i} \sum_{p \in C_i} d(p, p')$$

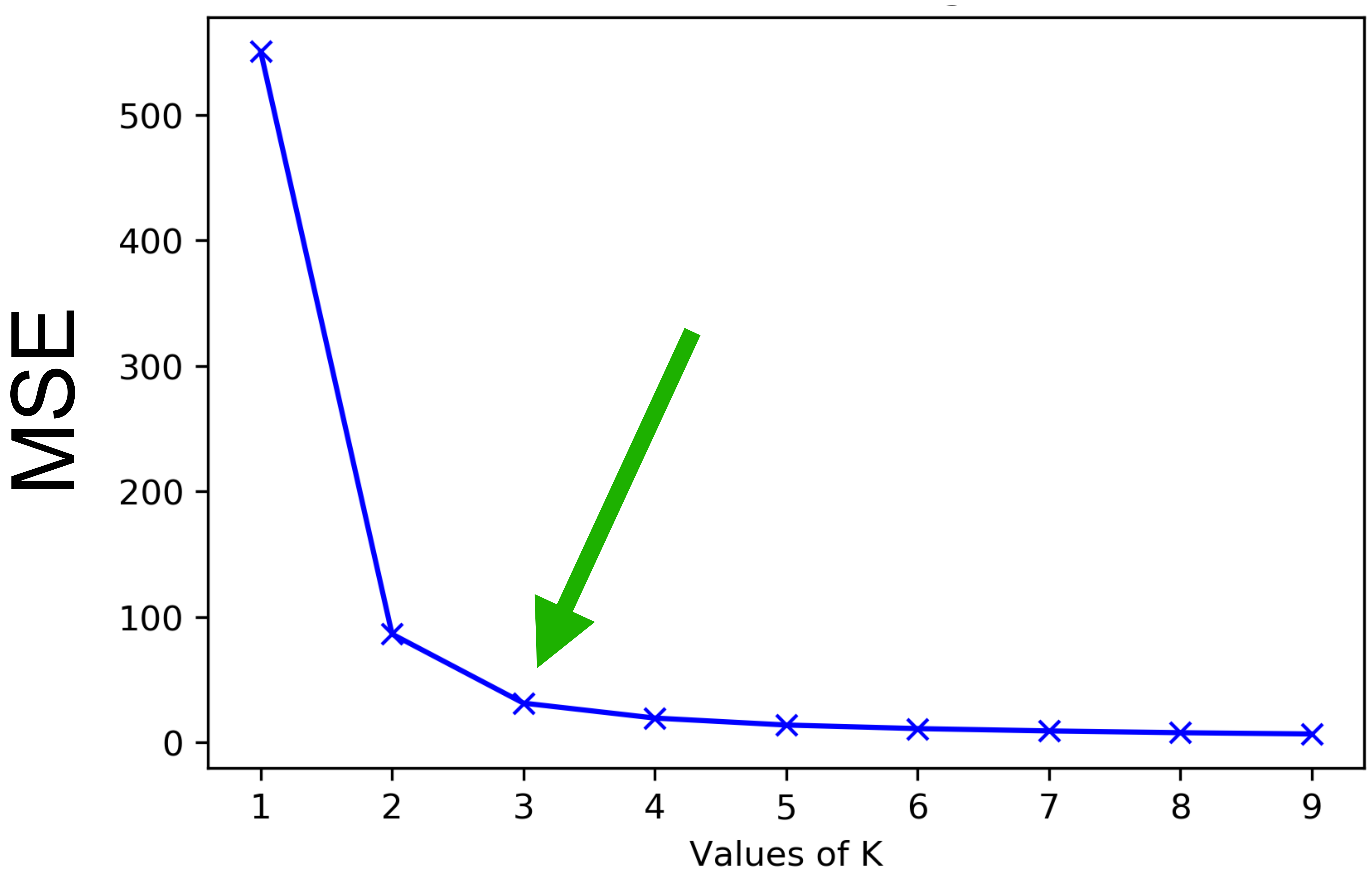
Center is always a point in cluster.

Need a proper definition of distance.

K-MEANS CLUSTERING: DETAILS

How to find a good value of k?

Elbow Method:



Pick k with diminishing returns.

This is not quantitative.

Just observational.

K-MEANS CLUSTERING: DETAILS

How to find a good value of k?

We can quantify it with formal scores such as Average Silhouette.

K-MEANS CLUSTERING: DETAILS

How to find a good value of k?

We can quantify it with formal scores such as Average Silhouette.

Silhouette score for a point p:

$$\frac{d_{closest} - d_{own}}{max(d_{closest}, d_{own})}$$

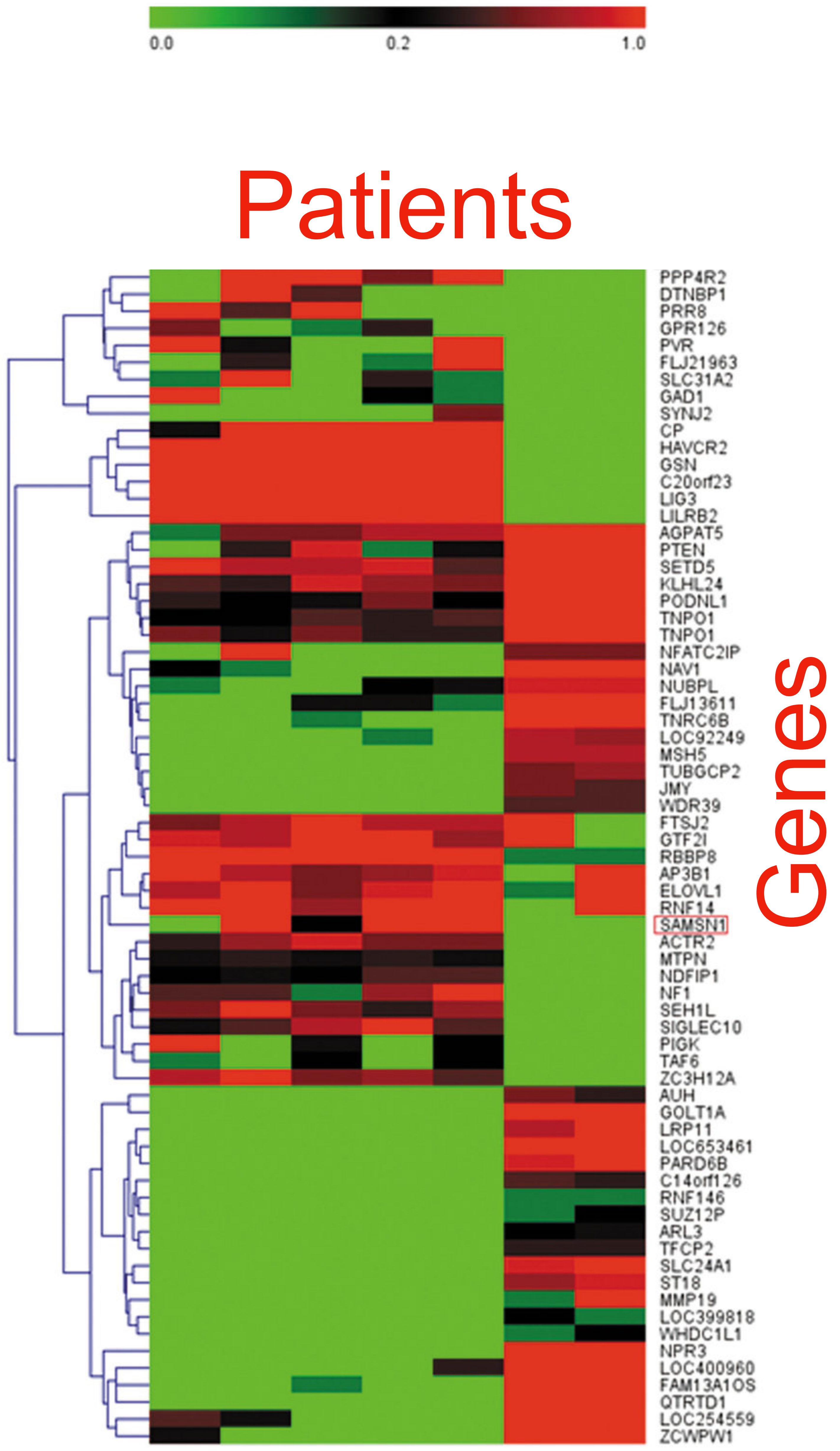
$d_{closest}$: avg dist of p to points in closest cluster

d_{own}: avg dist of p to points in its own cluster

Ranges between -1 and 1.

Can average over all points to get an overall scoring of a clustering.

HIERARCHICAL AGGLOMERATIVE CLUSTERING



Bottom-up method for clustering:

- Start with every item in its own cluster.
- Repeatedly,
  - Merge the most similar pair of clusters
  - Merging gives a new super-cluster

The whole process defines a tree:

- Leaves are the individual items
- Root defines the universe.

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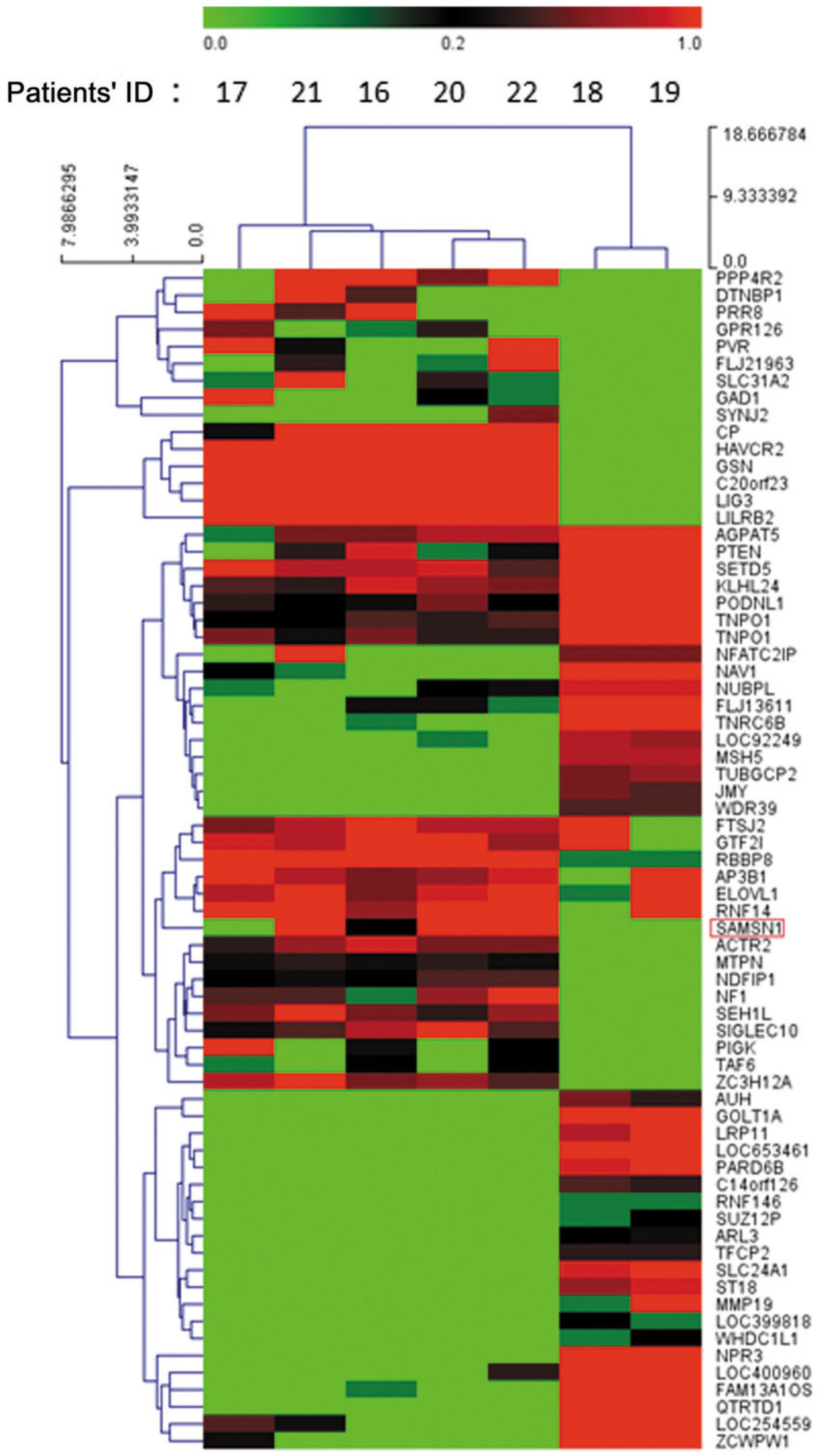

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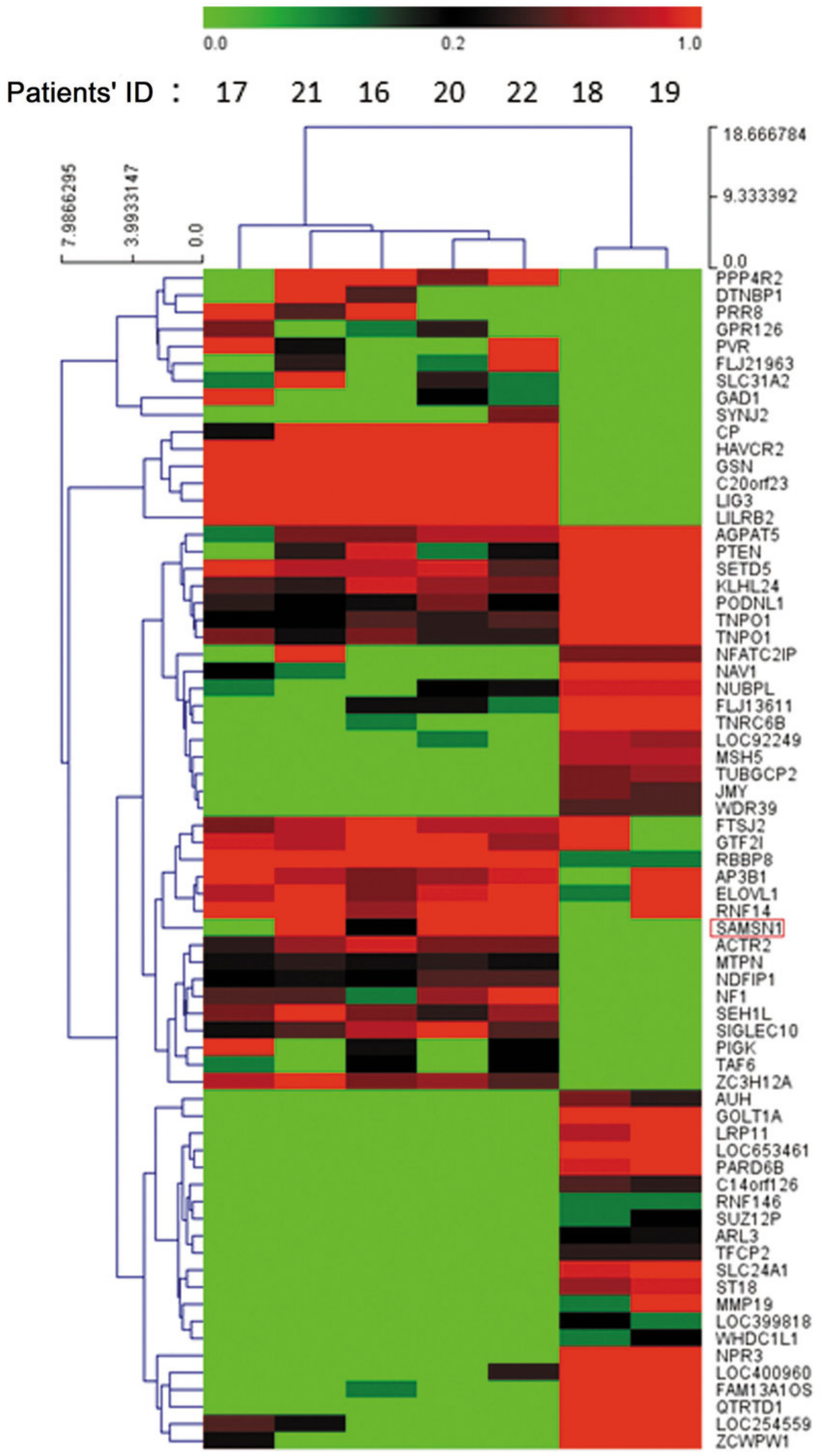
Can do it column-wise as well.

HIERARCHICAL AGGLOMERATIVE CLUSTERING



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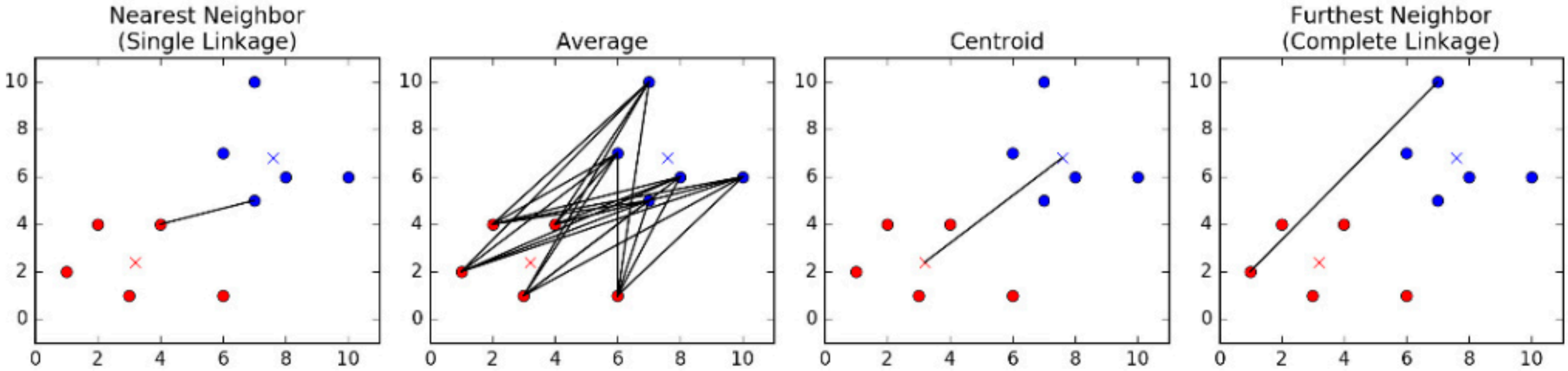
How to define?

HIERARCHICAL AGGLOMERATIVE CLUSTERING

4 different ways of defining the distance between a pair of clusters:



HIERARCHICAL AGGLOMERATIVE CLUSTERING

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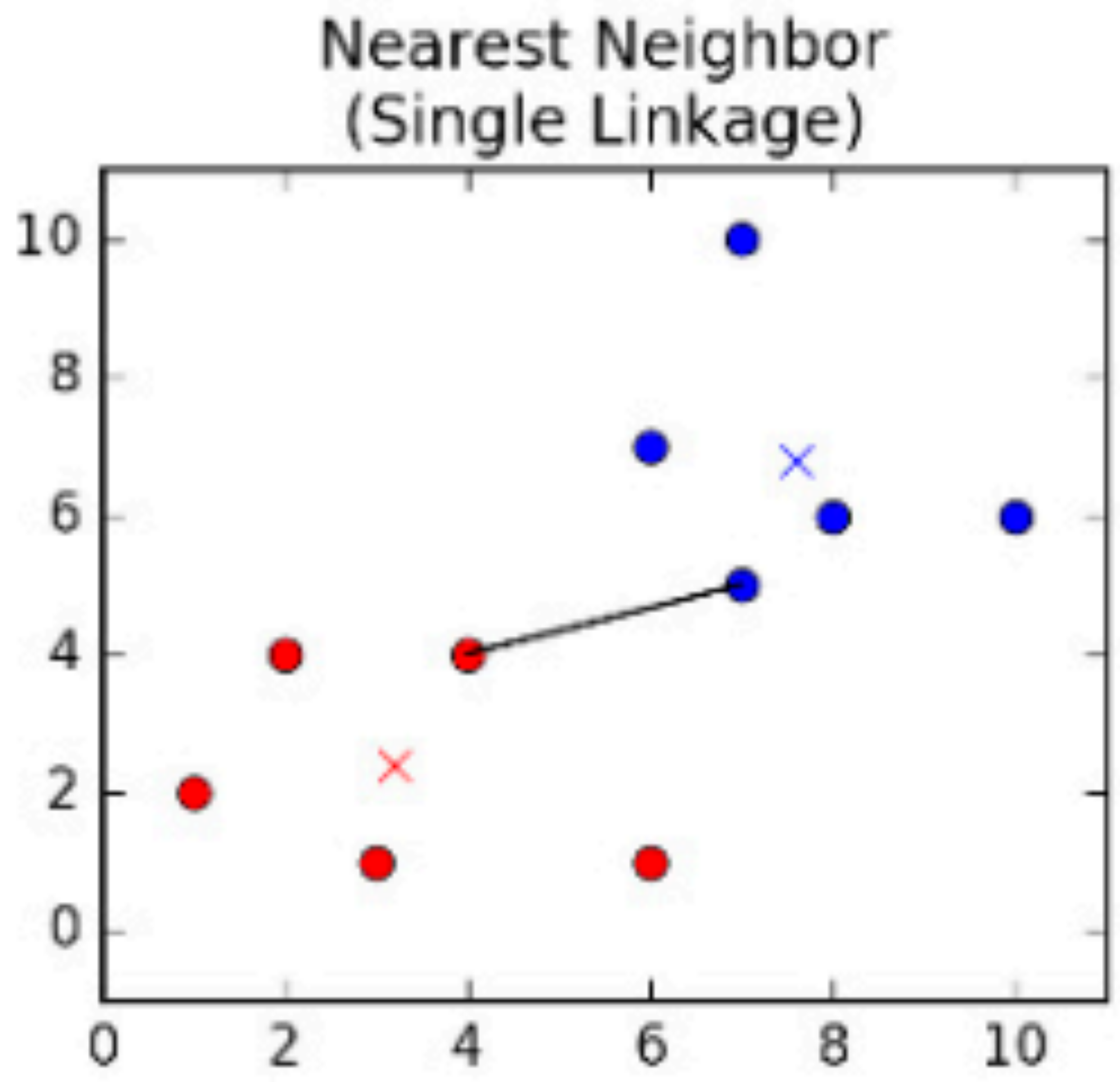

Single Linkage:

$$D(C_1, C_2) = \min_{p \in C_1, p' \in C_2} d(p, p')$$

HIERARCHICAL AGGLOMERATIVE CLUSTERING

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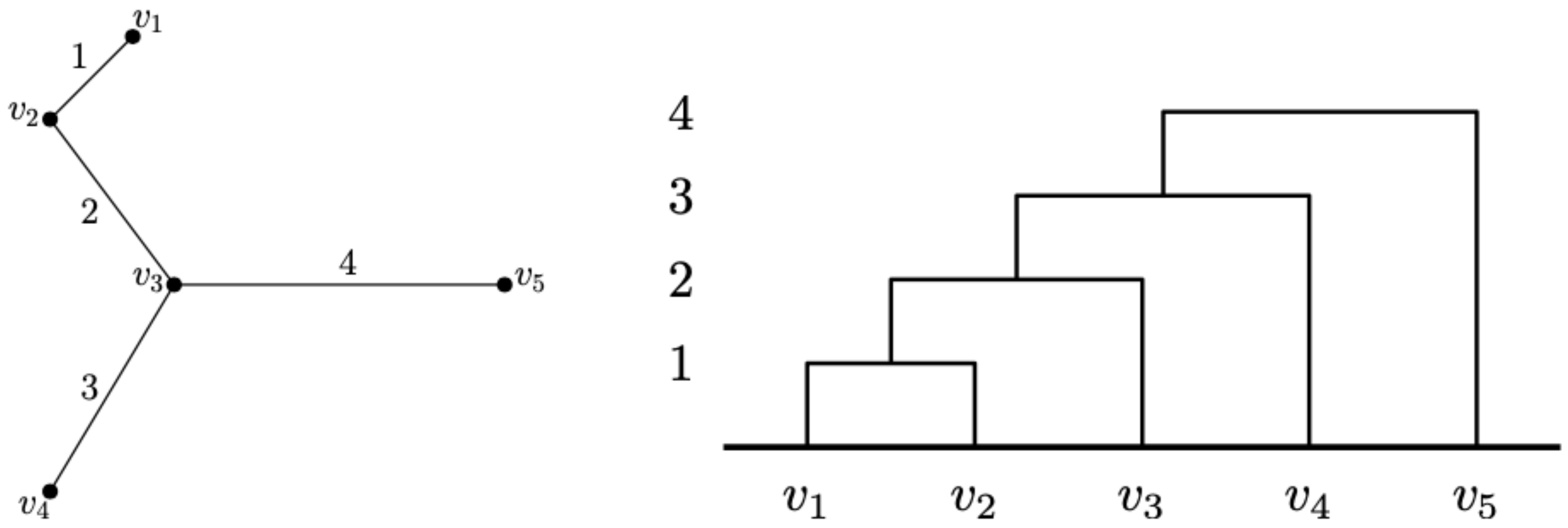

Single Linkage:

Same as MST

Problem:

May create long skinny clusters

HIERARCHICAL AGGLOMERATIVE CLUSTERING

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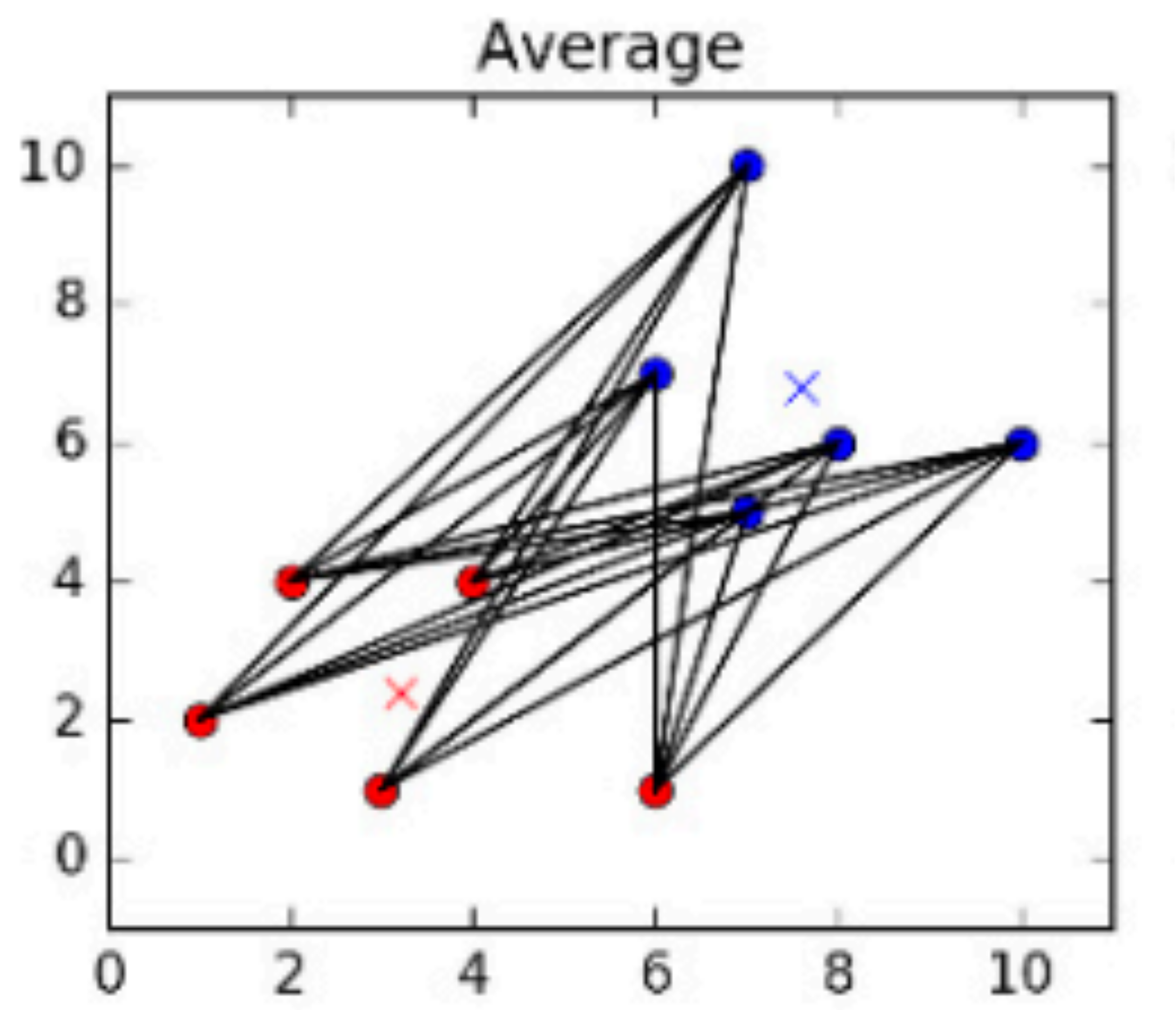
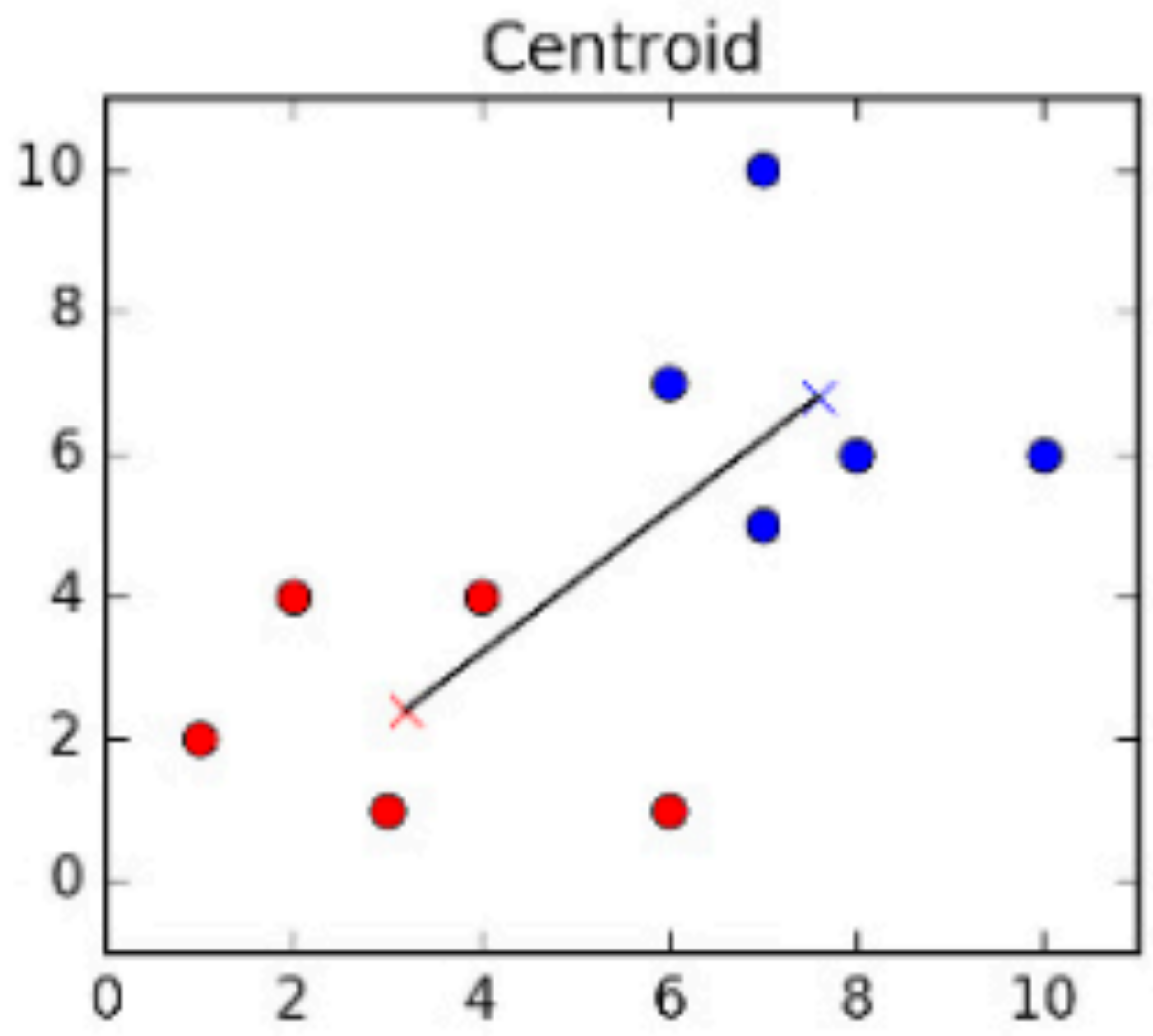


Average:

$$D(C_1, C_2) = \frac{1}{|C_1||C_2|} \sum_{p \in C_1} \sum_{p' \in C_2} d(p, p')$$

HIERARCHICAL AGGLOMERATIVE CLUSTERING

4 different ways of defining the distance between a pair of clusters:



Centroid:

$$D(C_1, C_2) = d(centroid_1, centroid_2)$$

HIERARCHICAL AGGLOMERATIVE CLUSTERING

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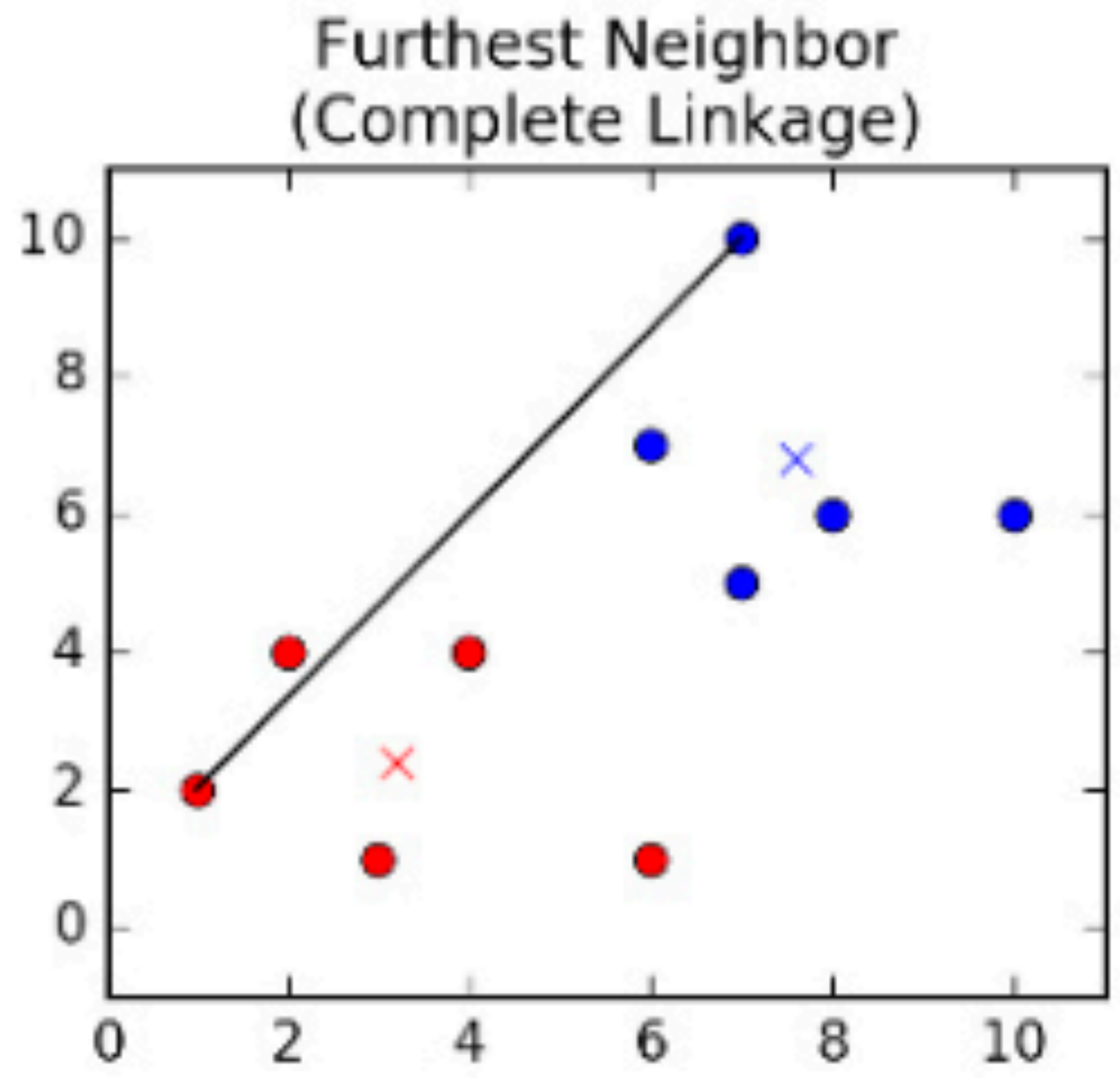
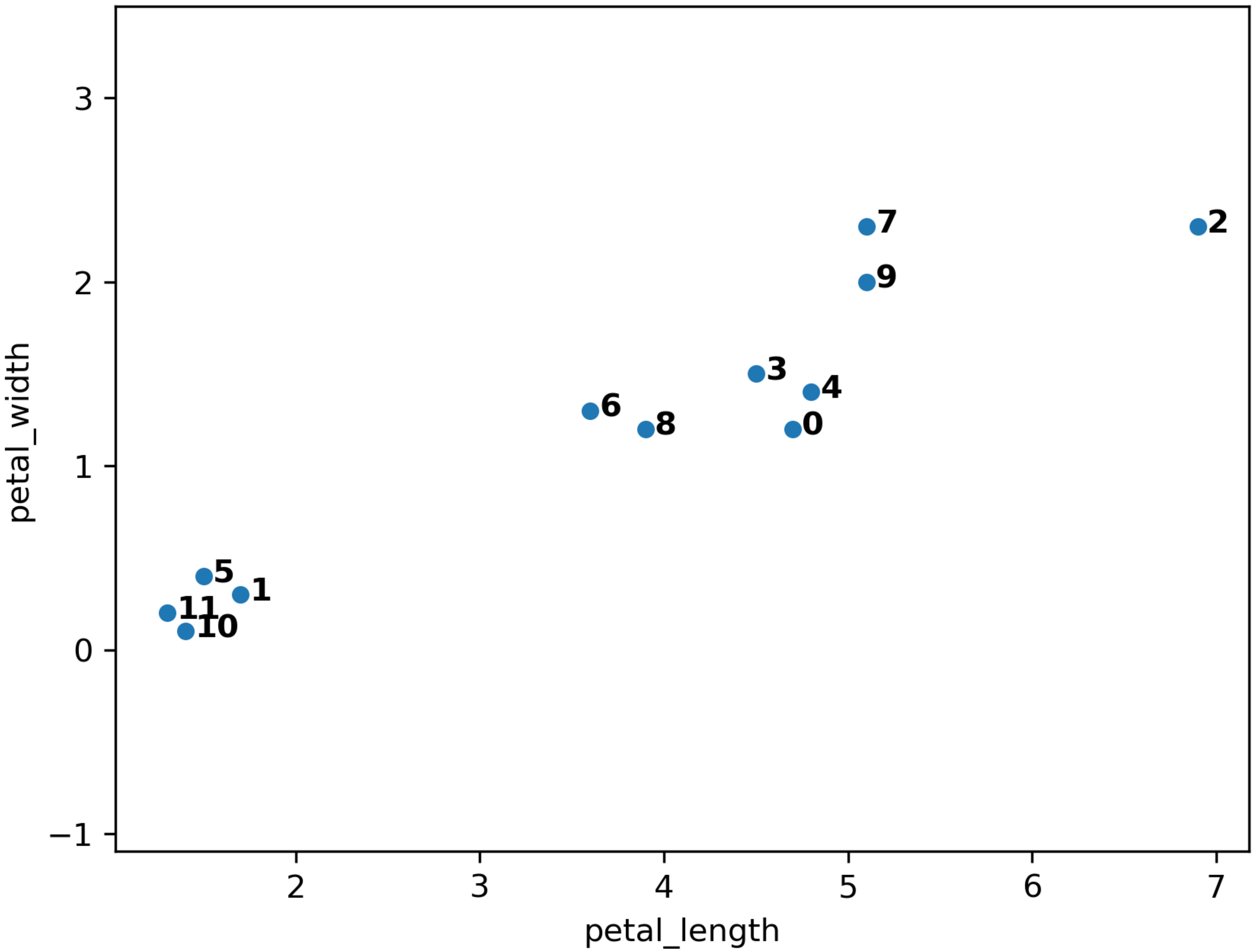


Complete Linkage:

$$D(C_1, C_2) = \max_{p \in C_1, p' \in C_2} d(p, p')$$

HIERARCHICAL AGGLOMERATIVE CLUSTERING

Example with complete linkage:



HIERARCHICAL AGGLOMERATIVE CLUSTERING

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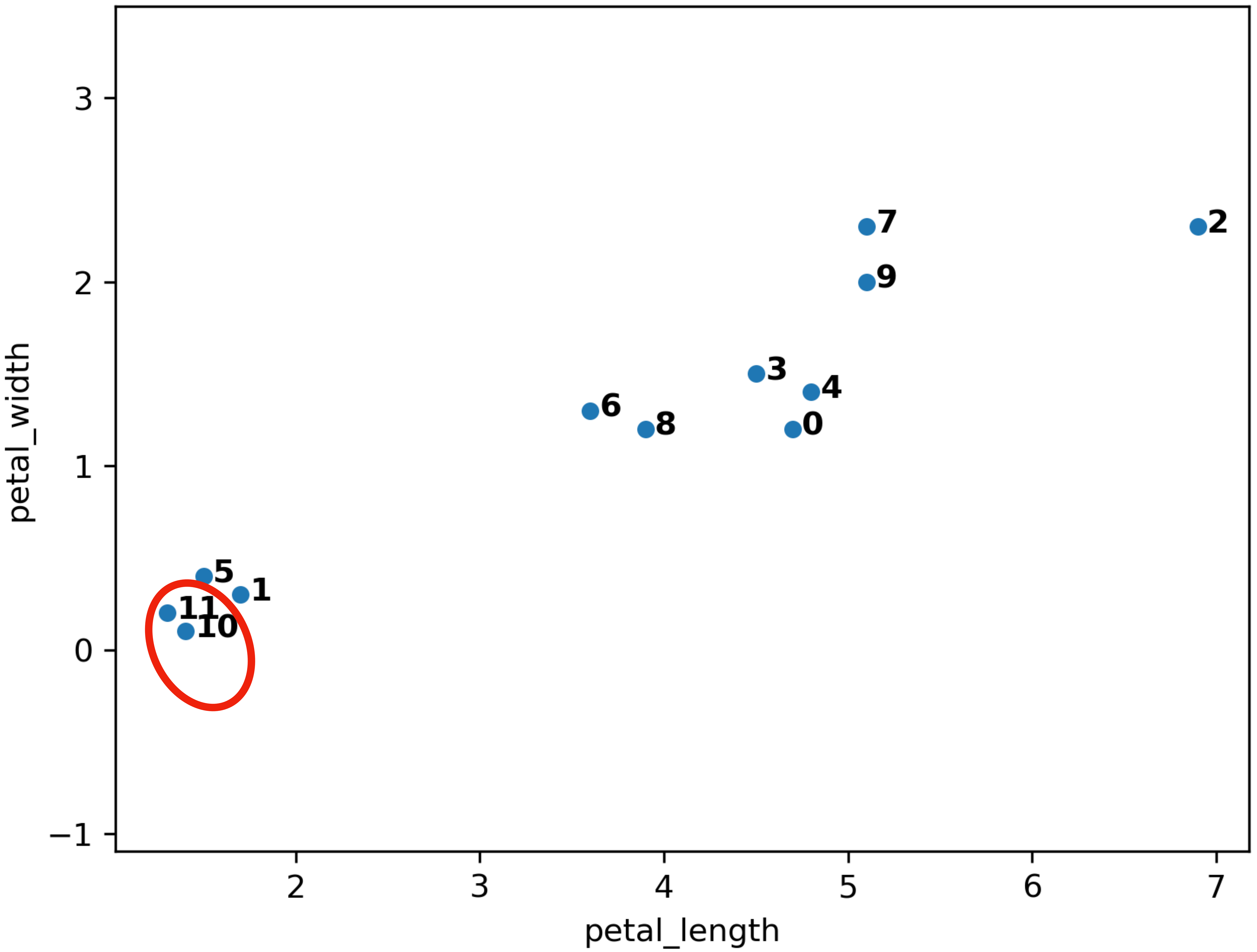

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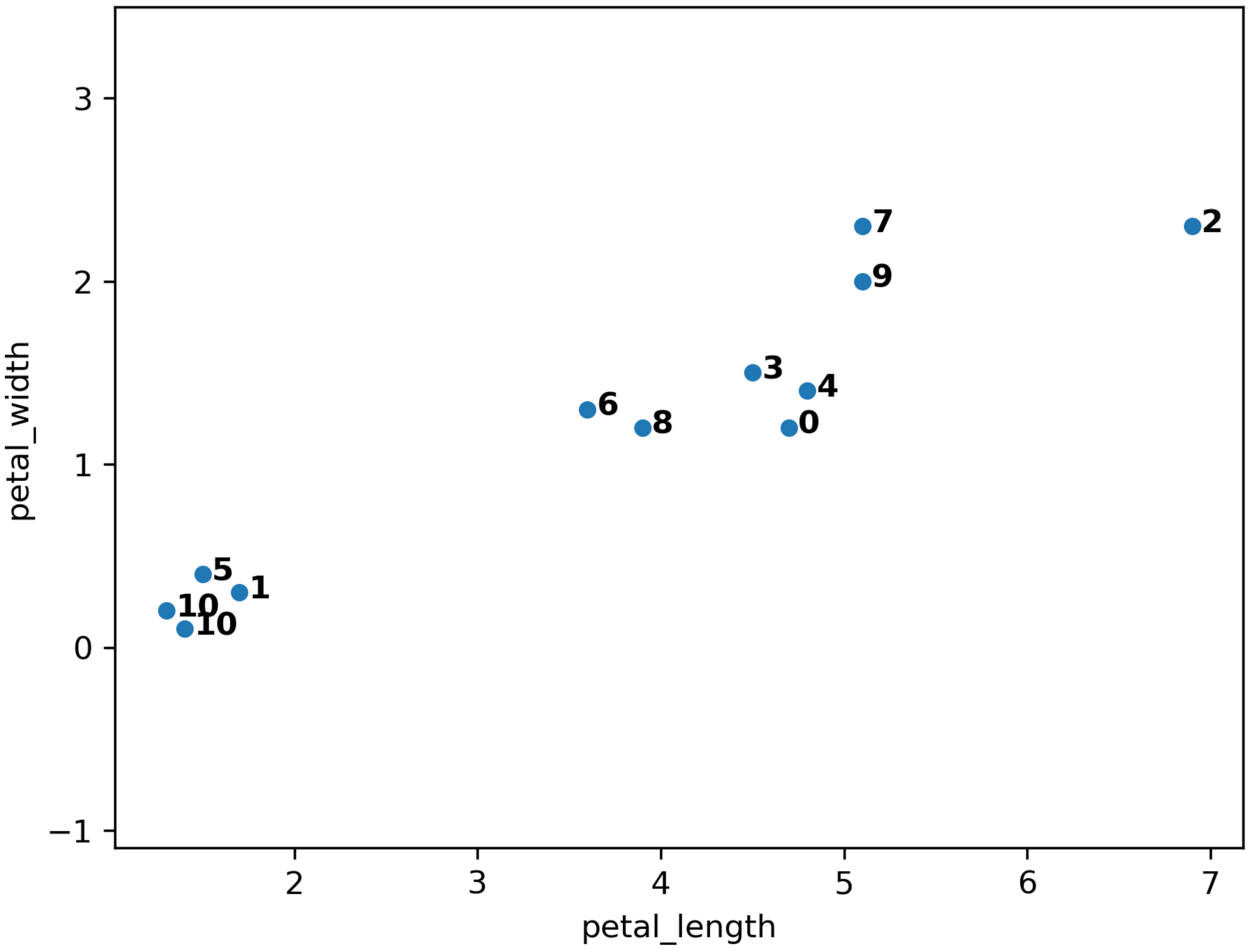

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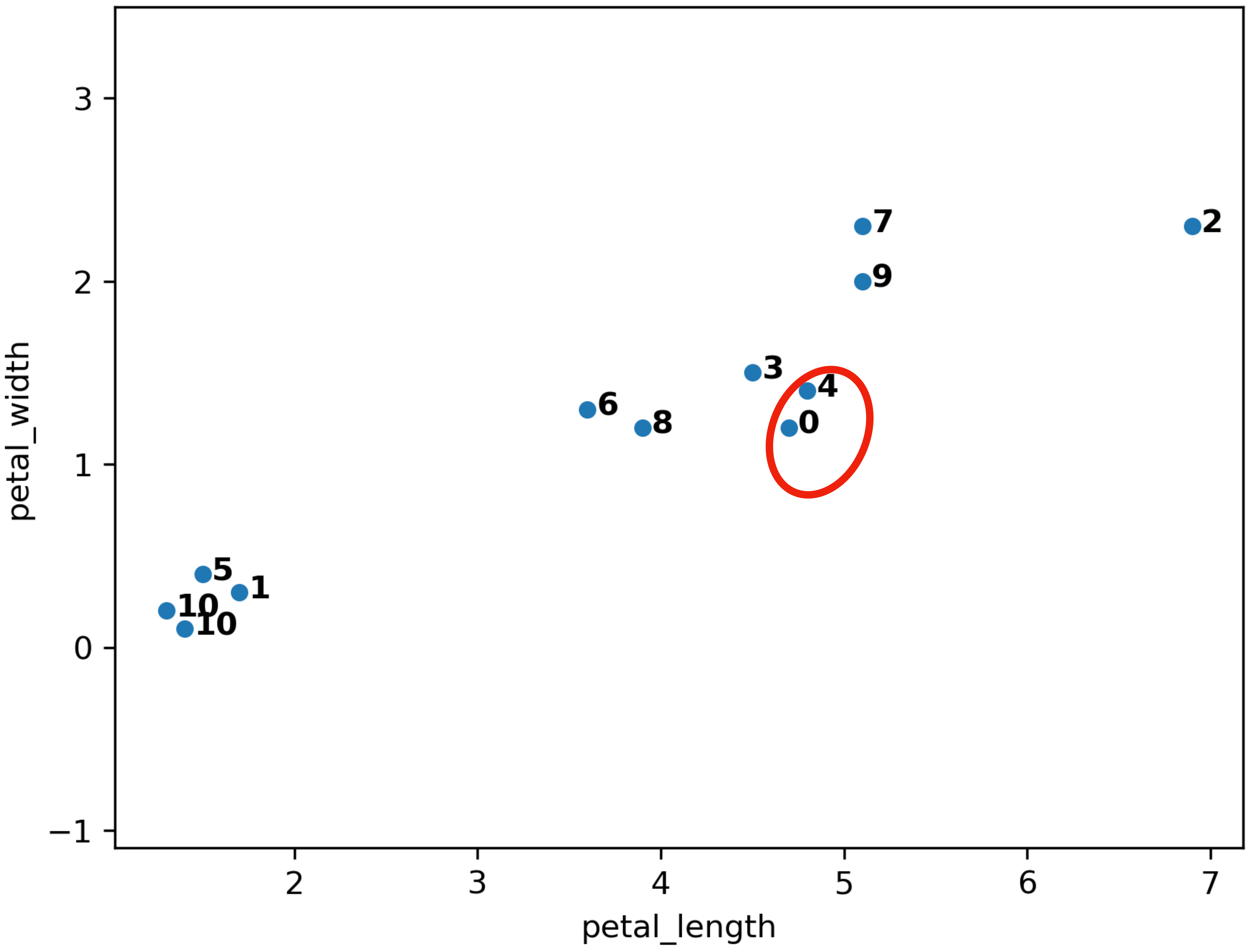
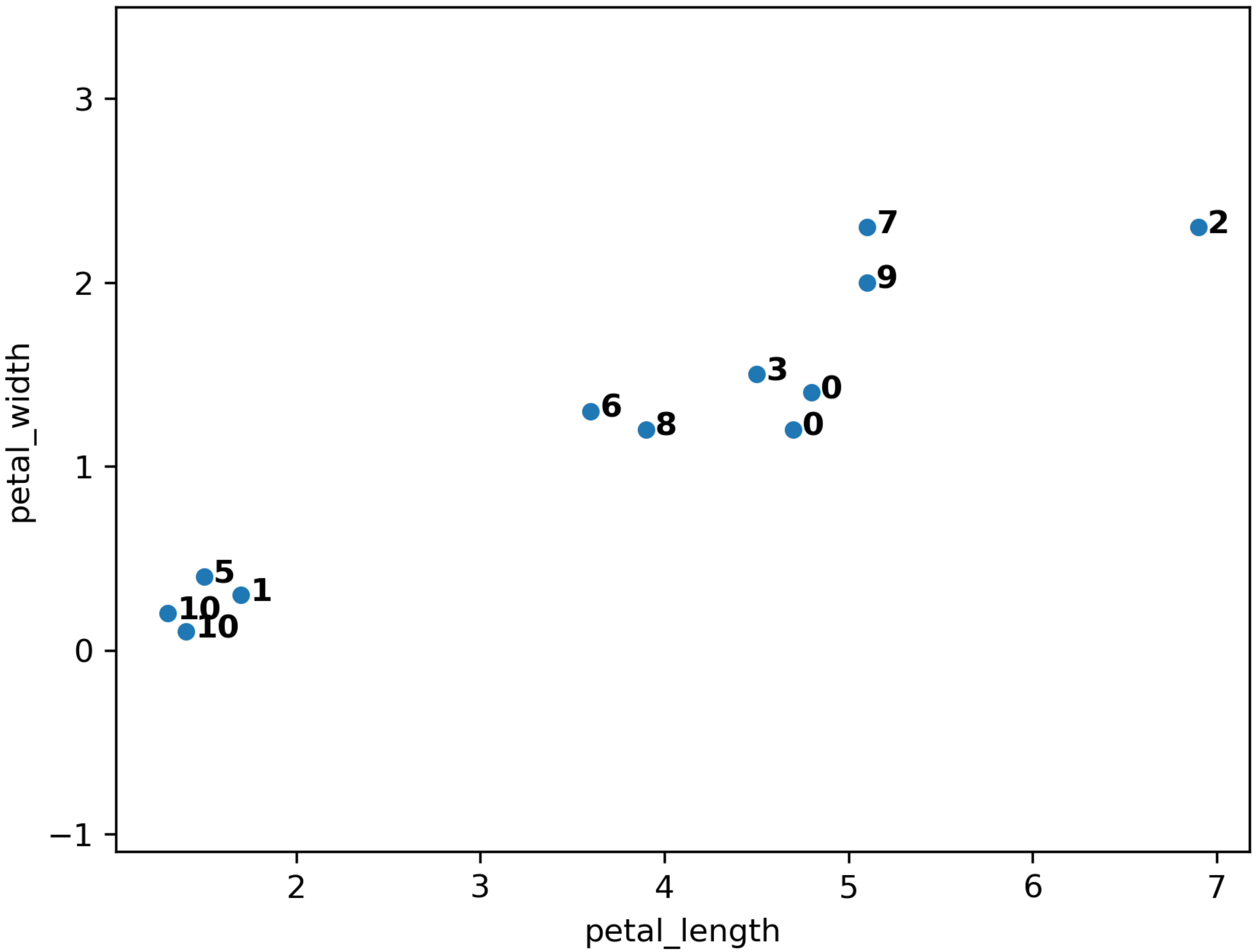


HIERARCHICAL AGGLOMERATIVE CLUSTERING

Example with complete linkage:



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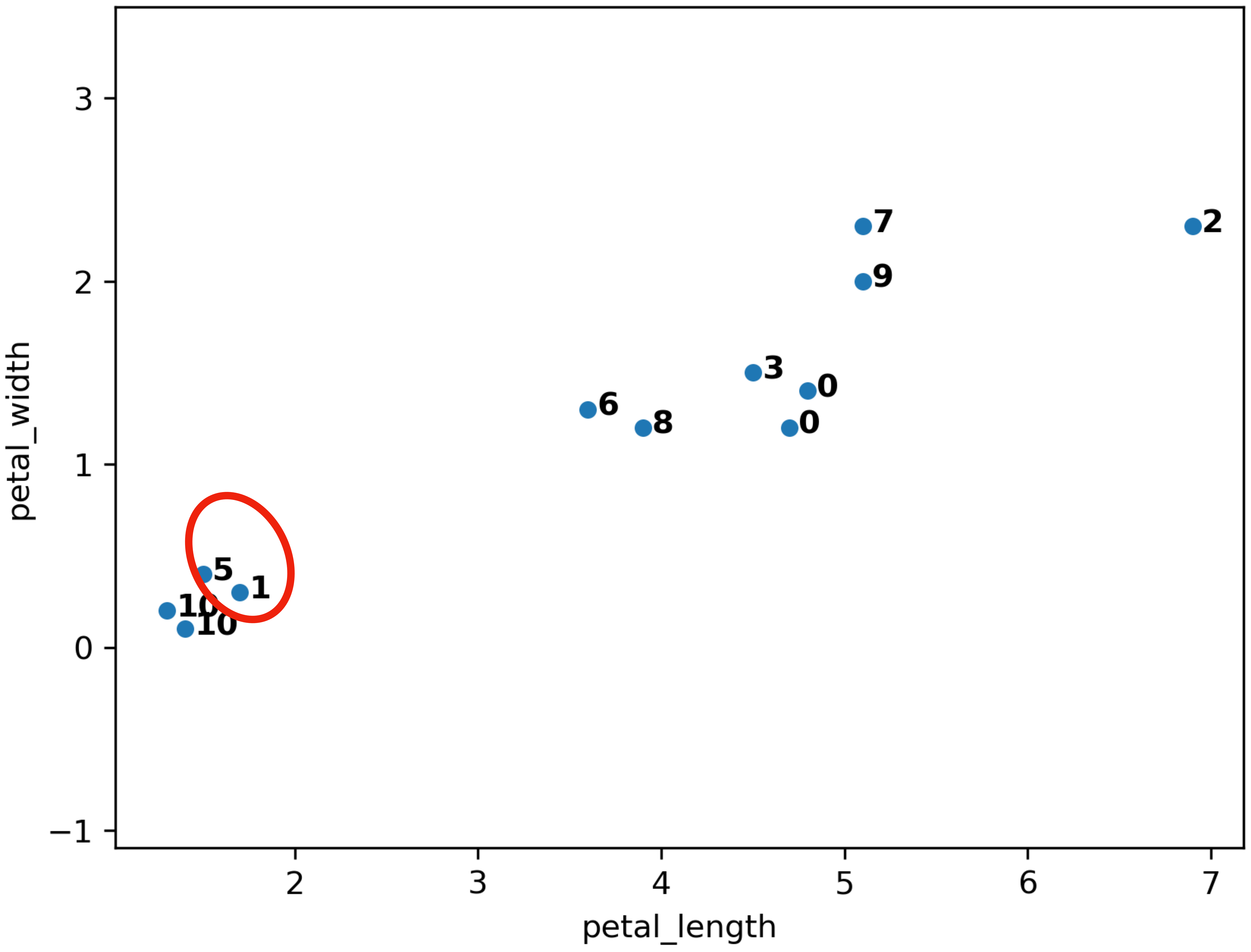

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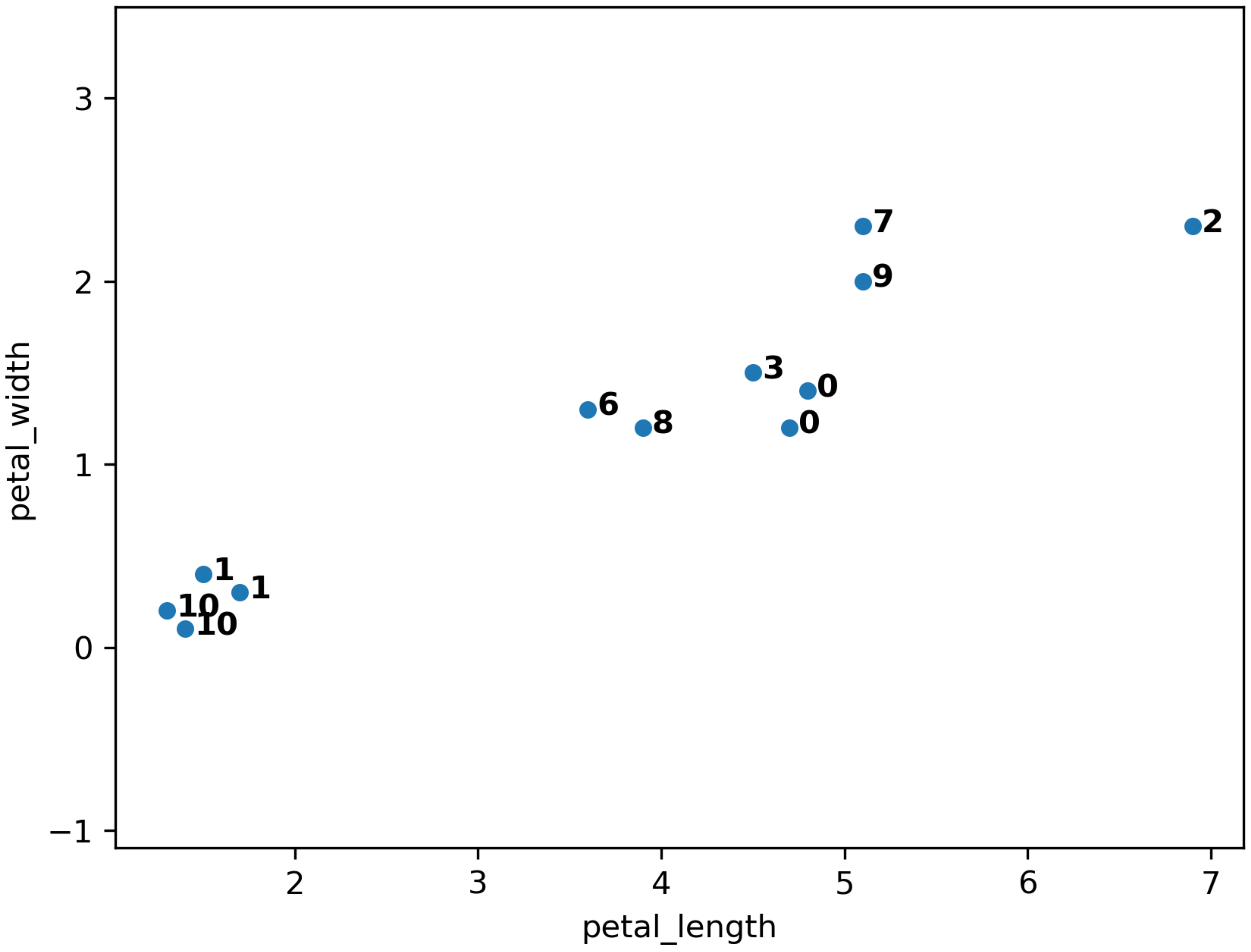

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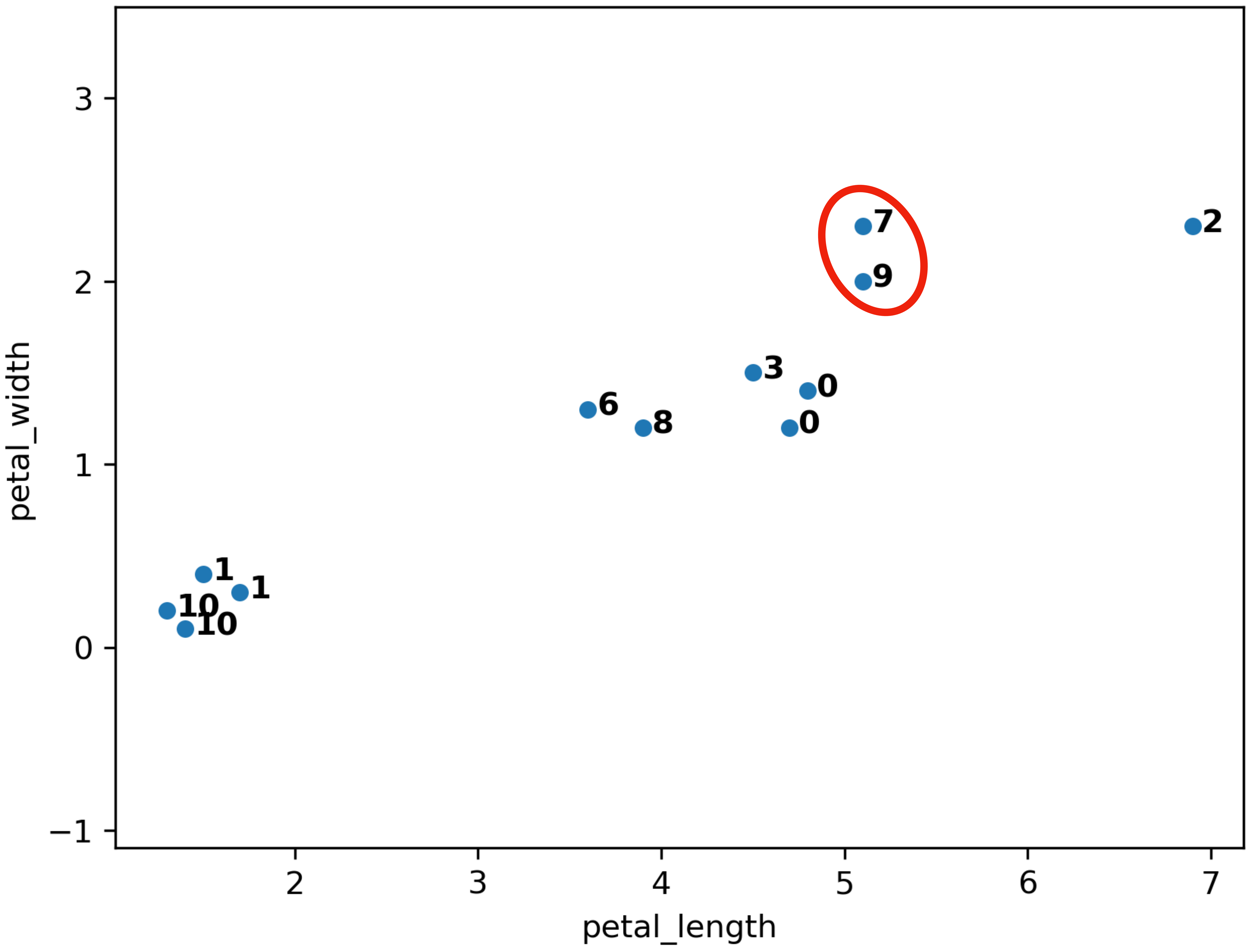

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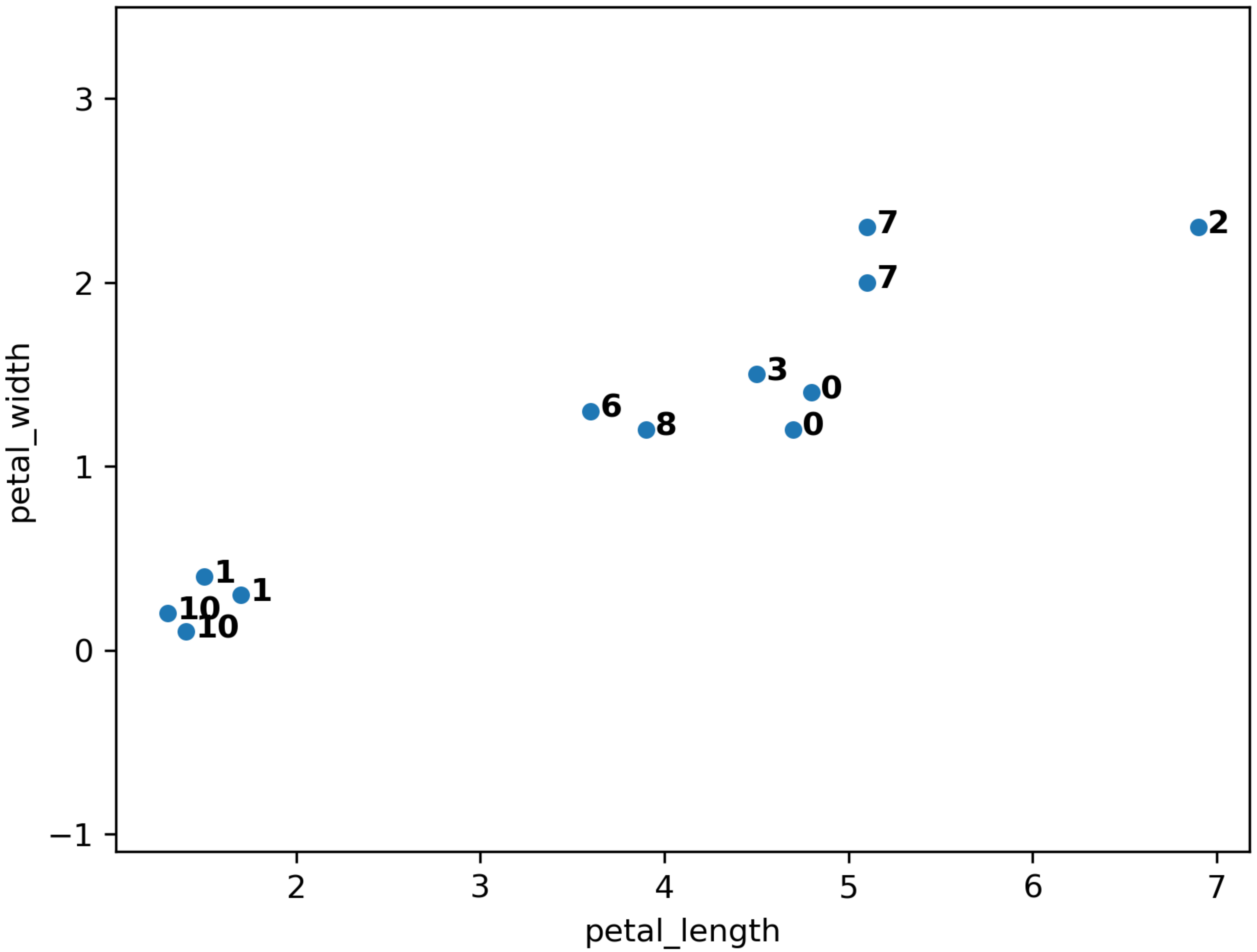

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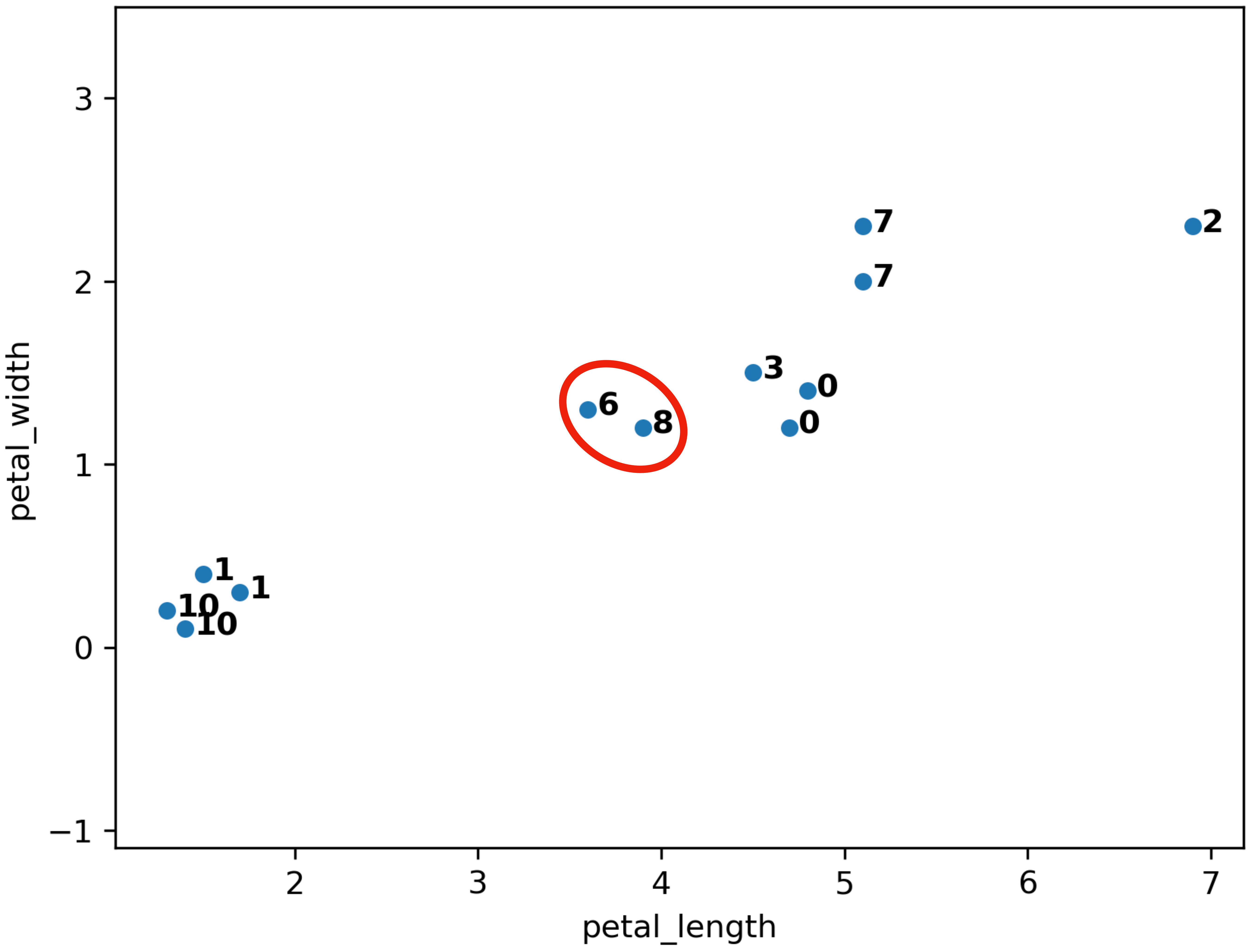

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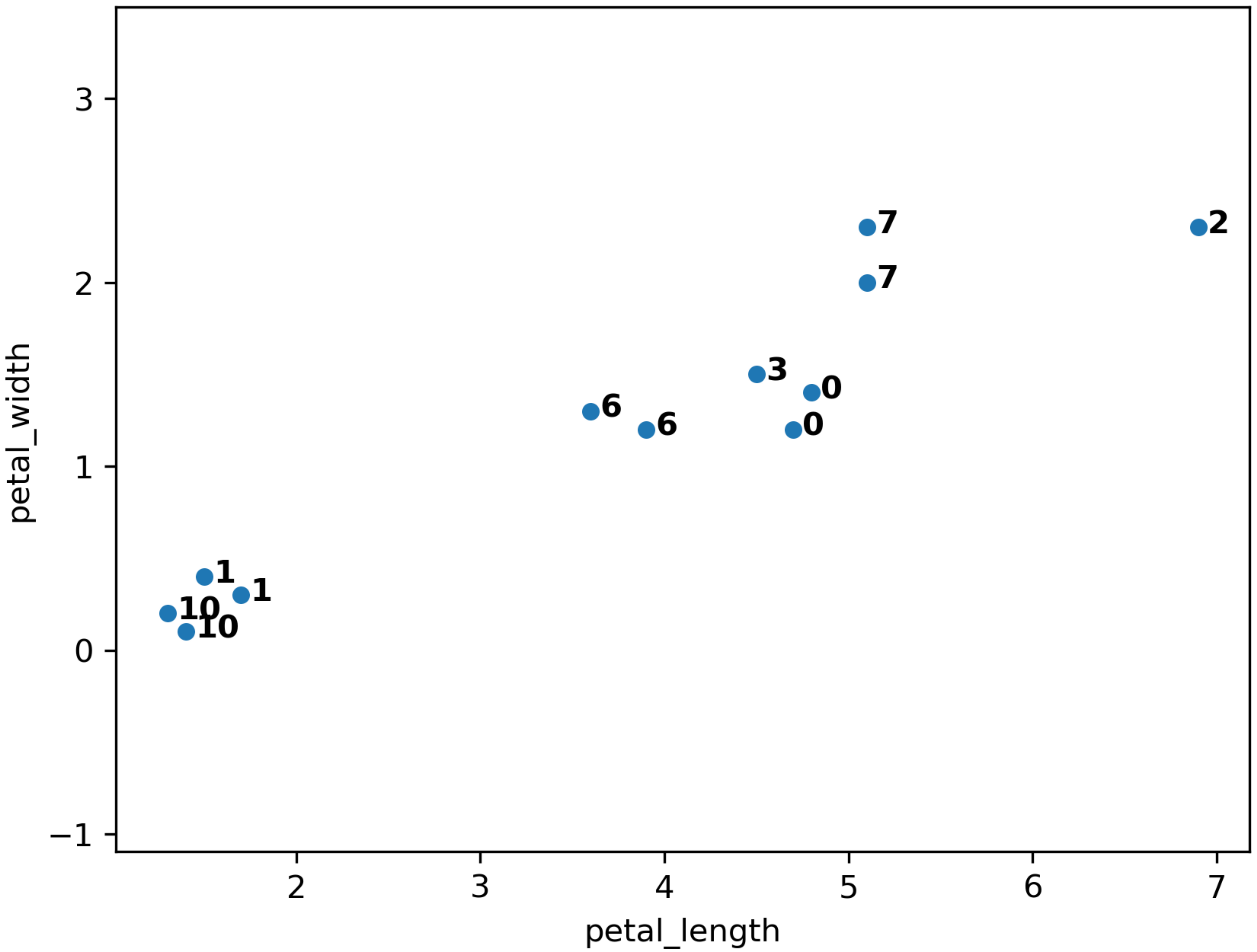

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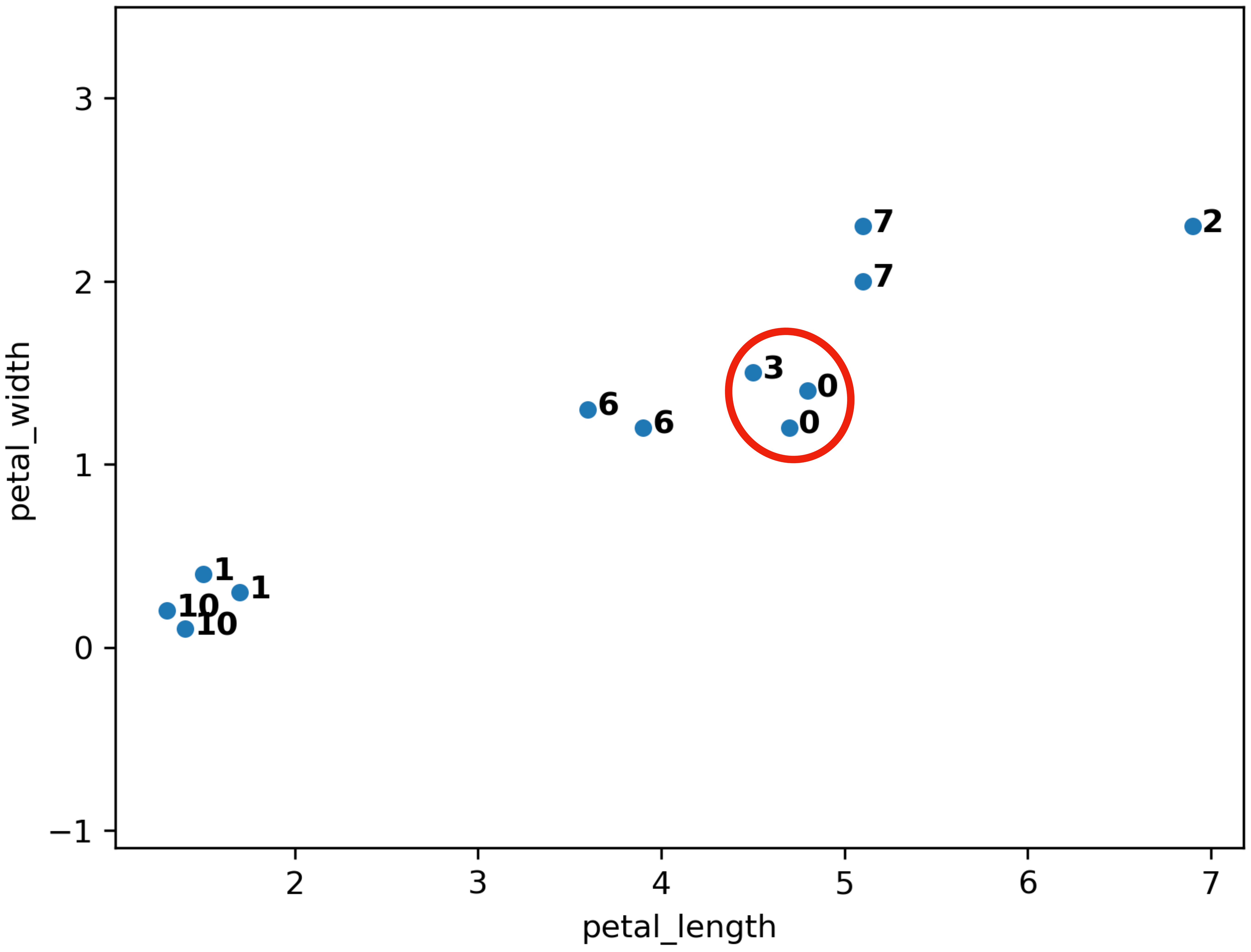

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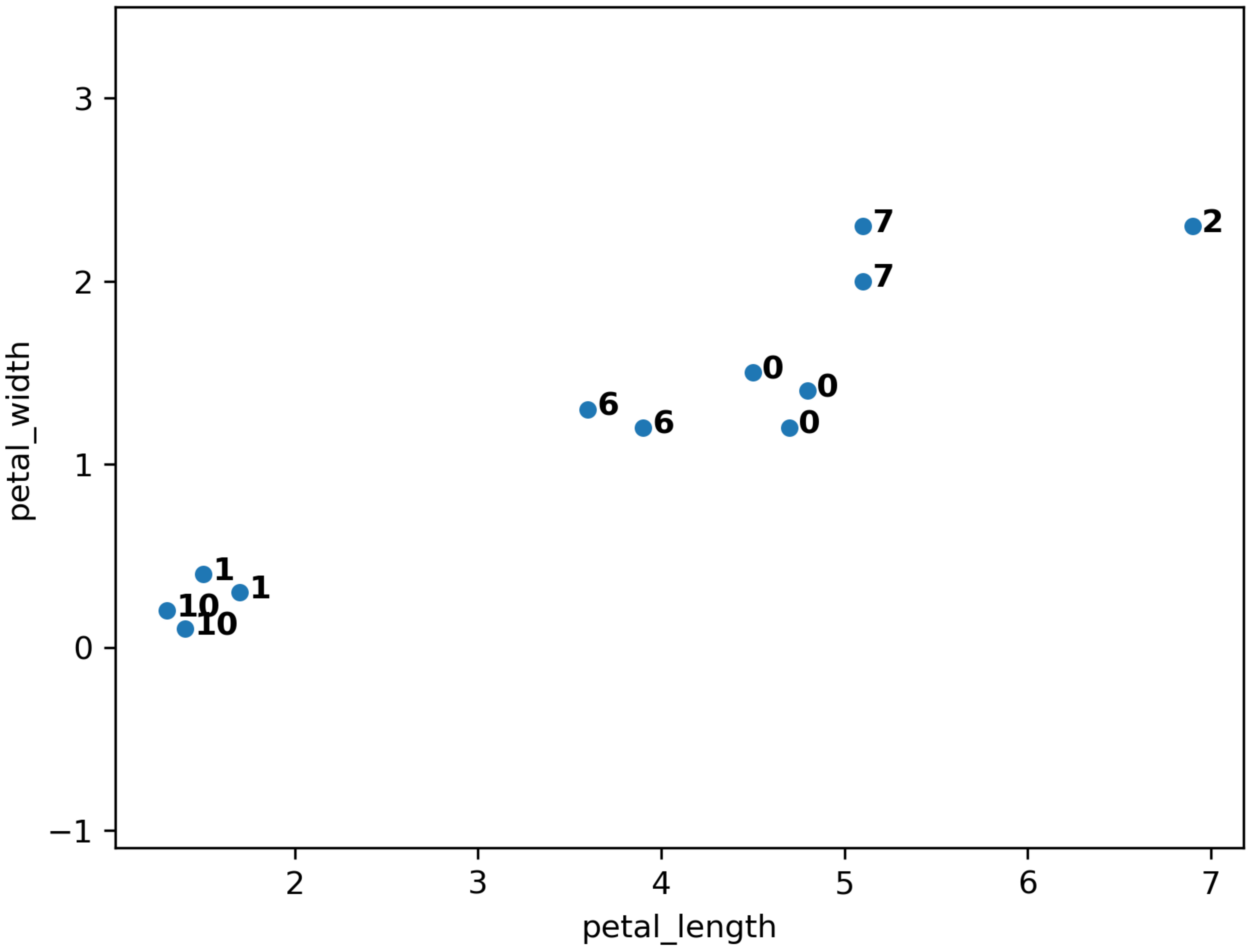

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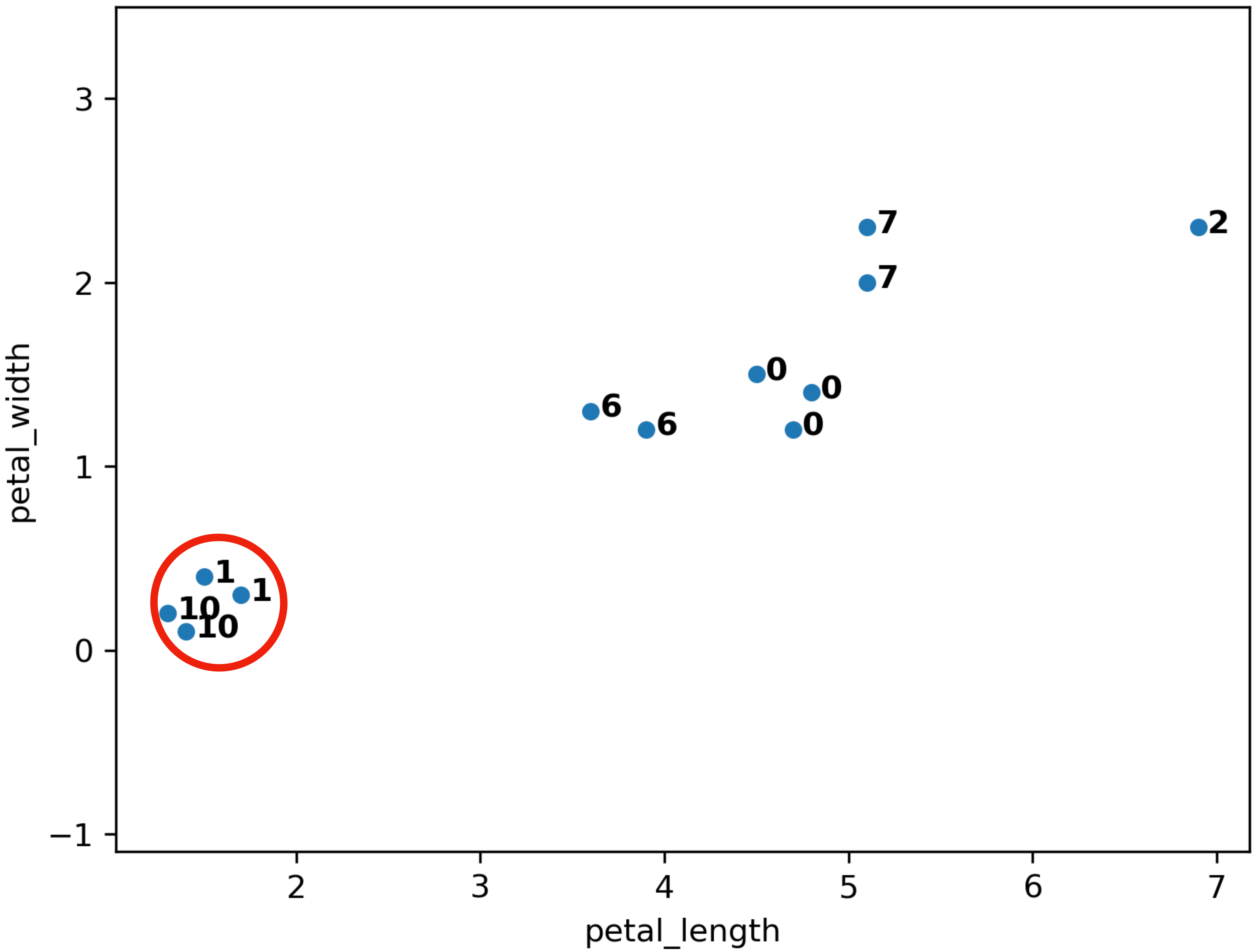

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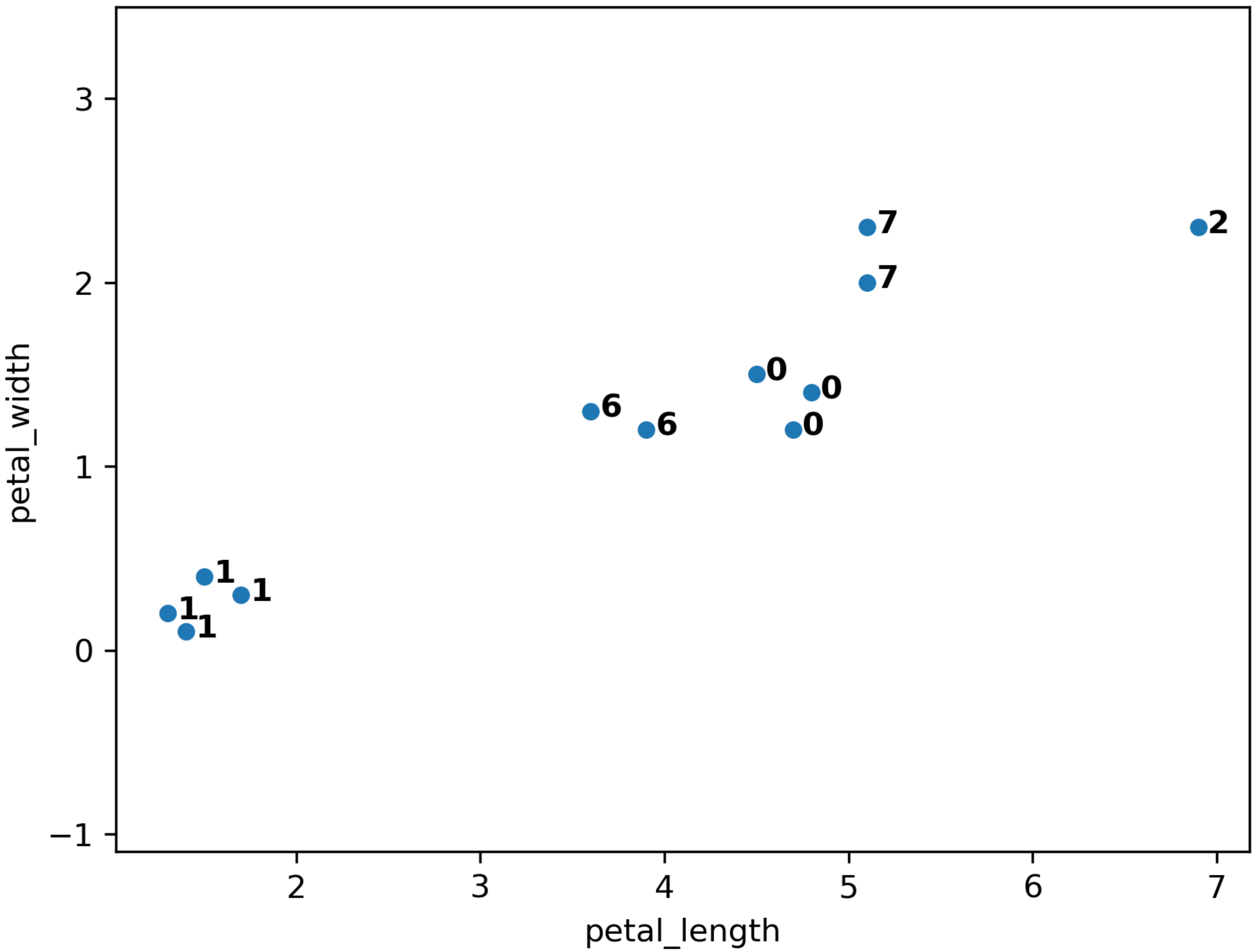

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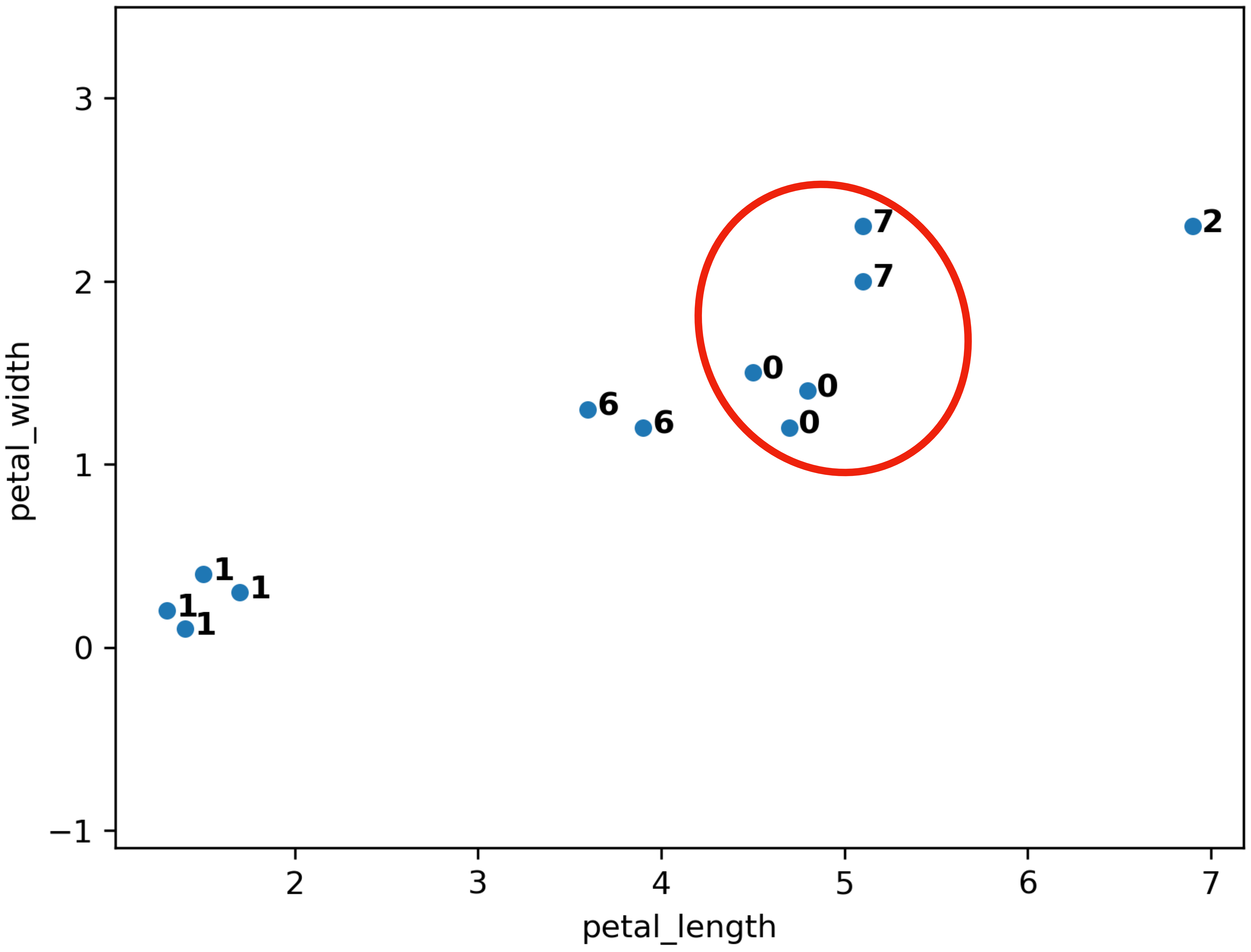

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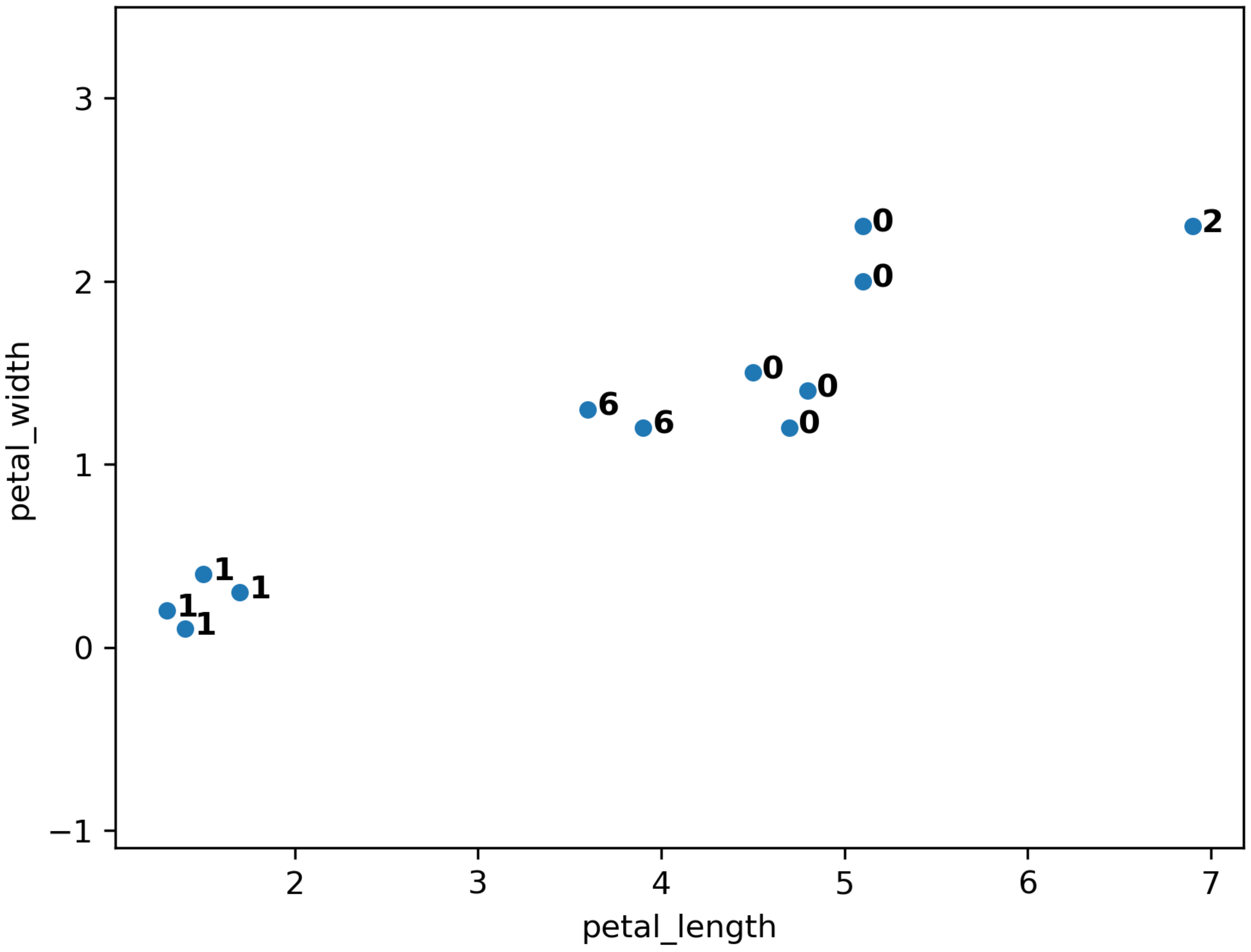

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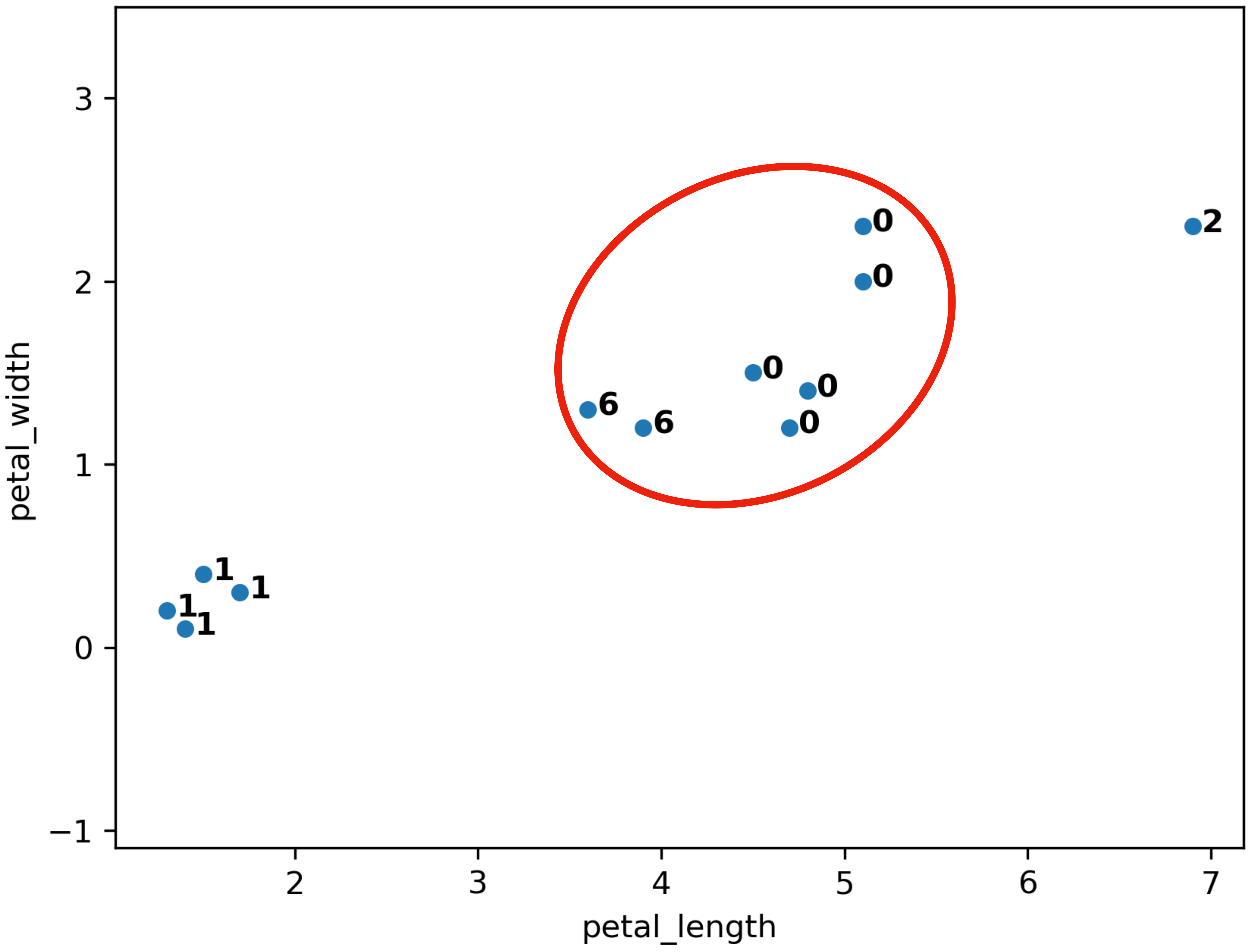

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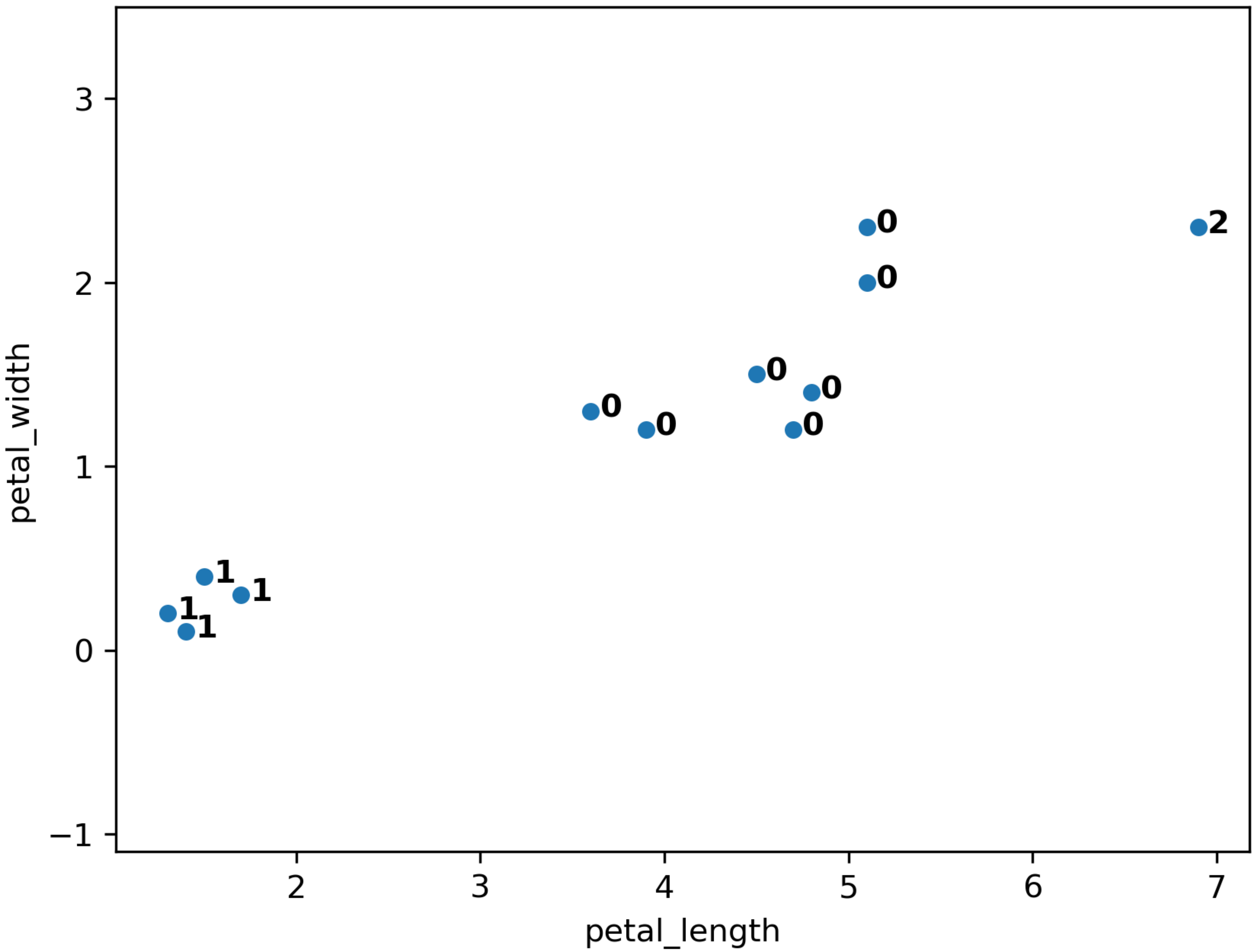

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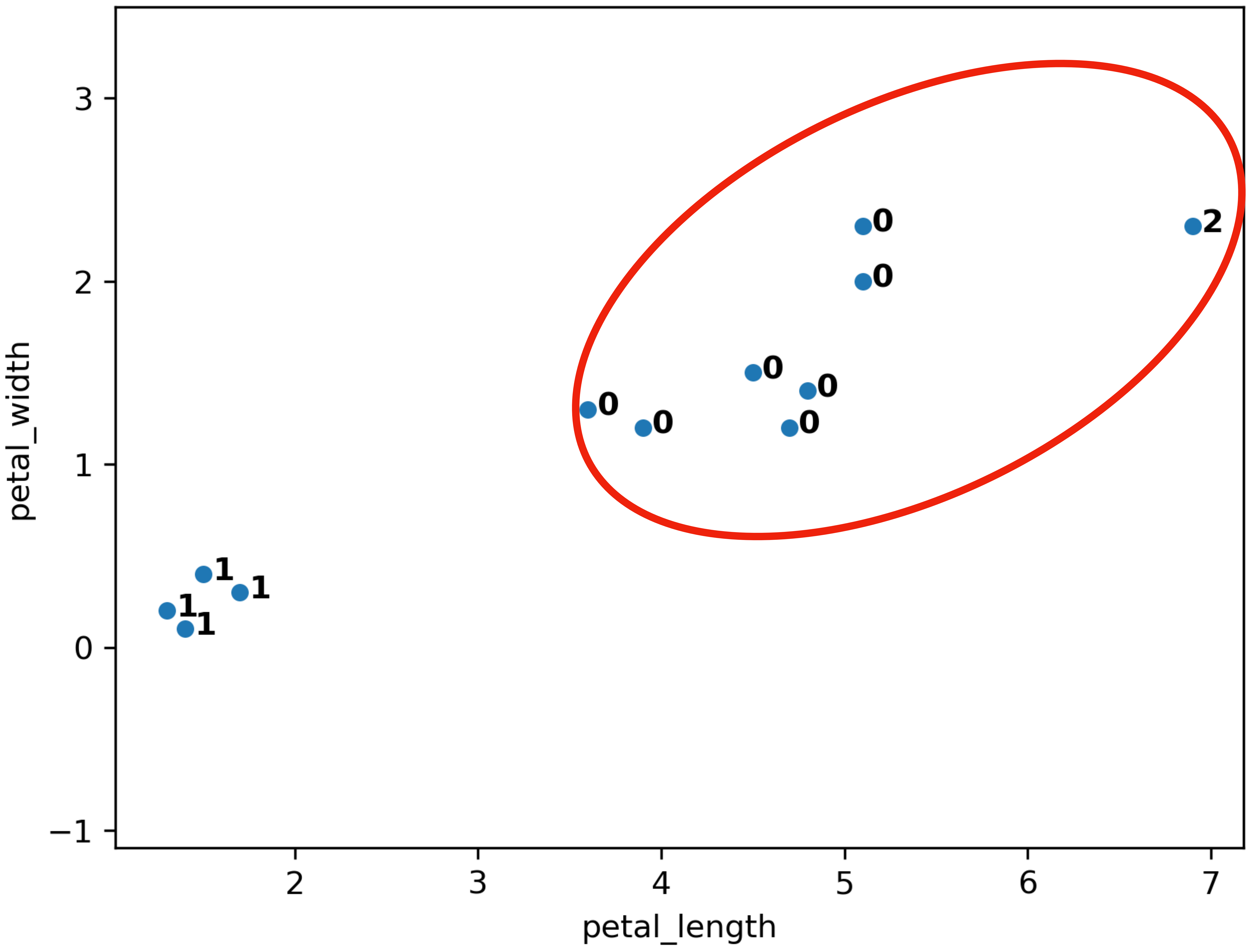

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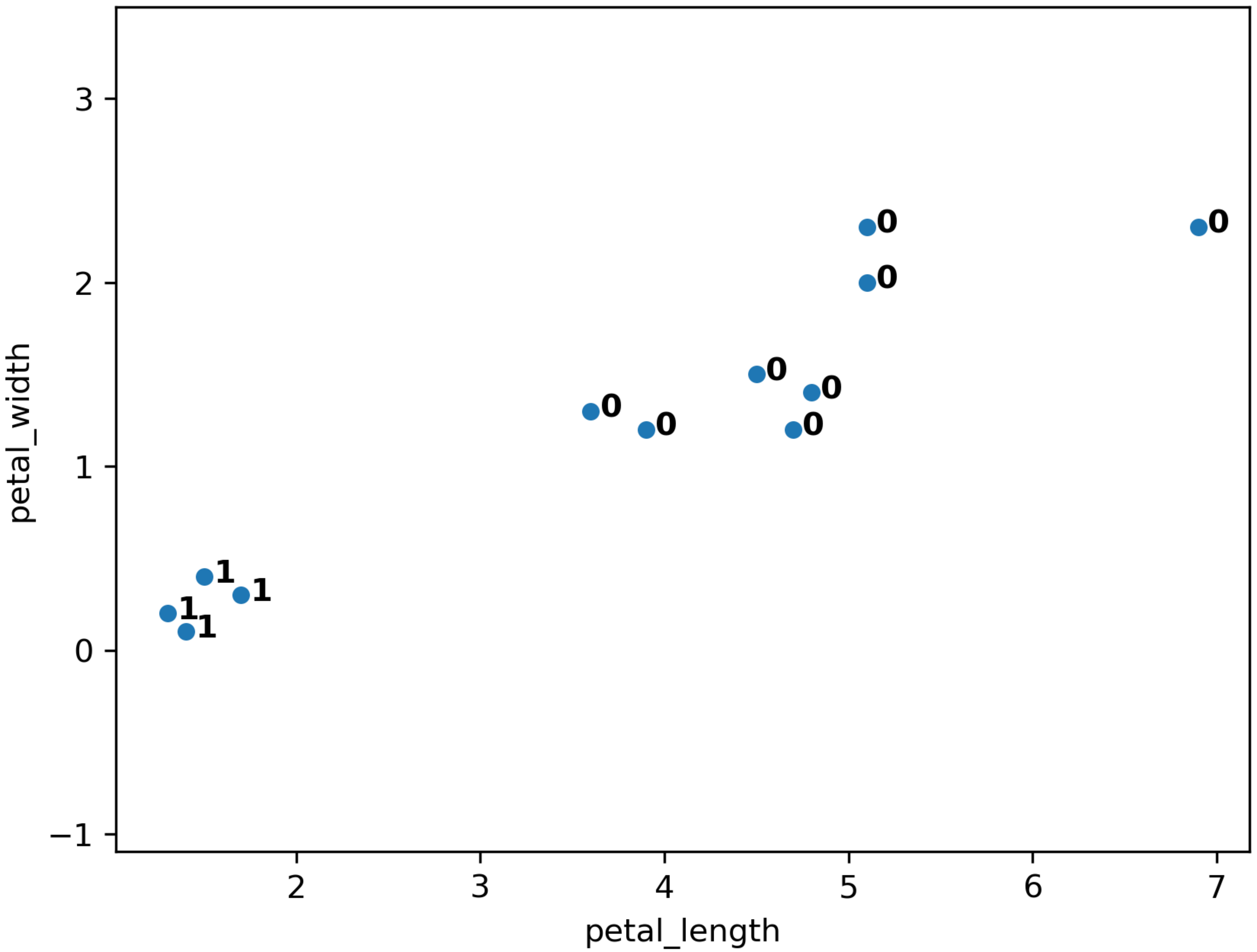


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Hypothesis testing
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SVD & PCA
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